

CompTIA Server+

Deploying and Managing Server Storage

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Introduction

Server+

NFS

Linux Filesystem

NTFS

SMB

Shared Storage

Welcome to the **Deploying and Managing Server Storage** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Different operating systems have different ways of configuring storage. Linux operating systems use a different filing system when compared to Windows operating systems.

In this module, the configuration of storage for different server operating systems will be investigated and implemented.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Configuring Storage in Linux
- Exercise 2 - Installing and Configuring Shared Storage in Linux using NFS
- Exercise 3 - Configuring Storage in Windows

- Exercise 4 - Configuring Shared Storage in Windows using SMB

After completing this module, you should be able to:

- Partition a Storage Device in Linux
- Format a Partition with a Filesystem in Linux
- Mount Storage in Linux
- Configure the Permissions on the NFS Directory
- Install and Configure an NFS Server
- Install and Configure an NFS Client
- Partition a Storage Device
- Format the Storage Device with the NTFS Filesystem
- Create a Network Share for Storage
- Connect a Windows 10 Client to the Shared Storage
- Map a Network Drive to the Shared Storage

Exam Objectives

The following exam objectives are covered in this module:

1.2 Given a scenario, deploy and manage storage

- Shared storage

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)
- **PLABALMA** - (Alma Linux 8.7 - Stand-alone Linux Workstation)
- **PLABUBUNTU** - (Linux Ubuntu - Stand-alone Linux Server)

Click **Next** to proceed to the first exercise.

Exercise 1 - Configuring Storage in Linux

Storage on a Linux operating system is configured by creating a partition, formatting it, and mounting it to a mount point.

In this exercise, the creation of a partition and mounting it will be explored on a Linux operating system

Learning Outcomes

After completing this exercise, you should be able to:

- Partition a Storage Device in Linux
- Format a Partition with a Filesystem in Linux
- Mount Storage in Linux

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABUBUNTU** - (Linux Ubuntu - Stand-alone Linux Server)

Task 1 - Partition a Storage Device in Linux

In this task, you will partition storage on the PLABUBUNTU Linux device.

Step 1

Connect to **PLABUBUNTU**.

Click the **Terminal** icon in the left pane.

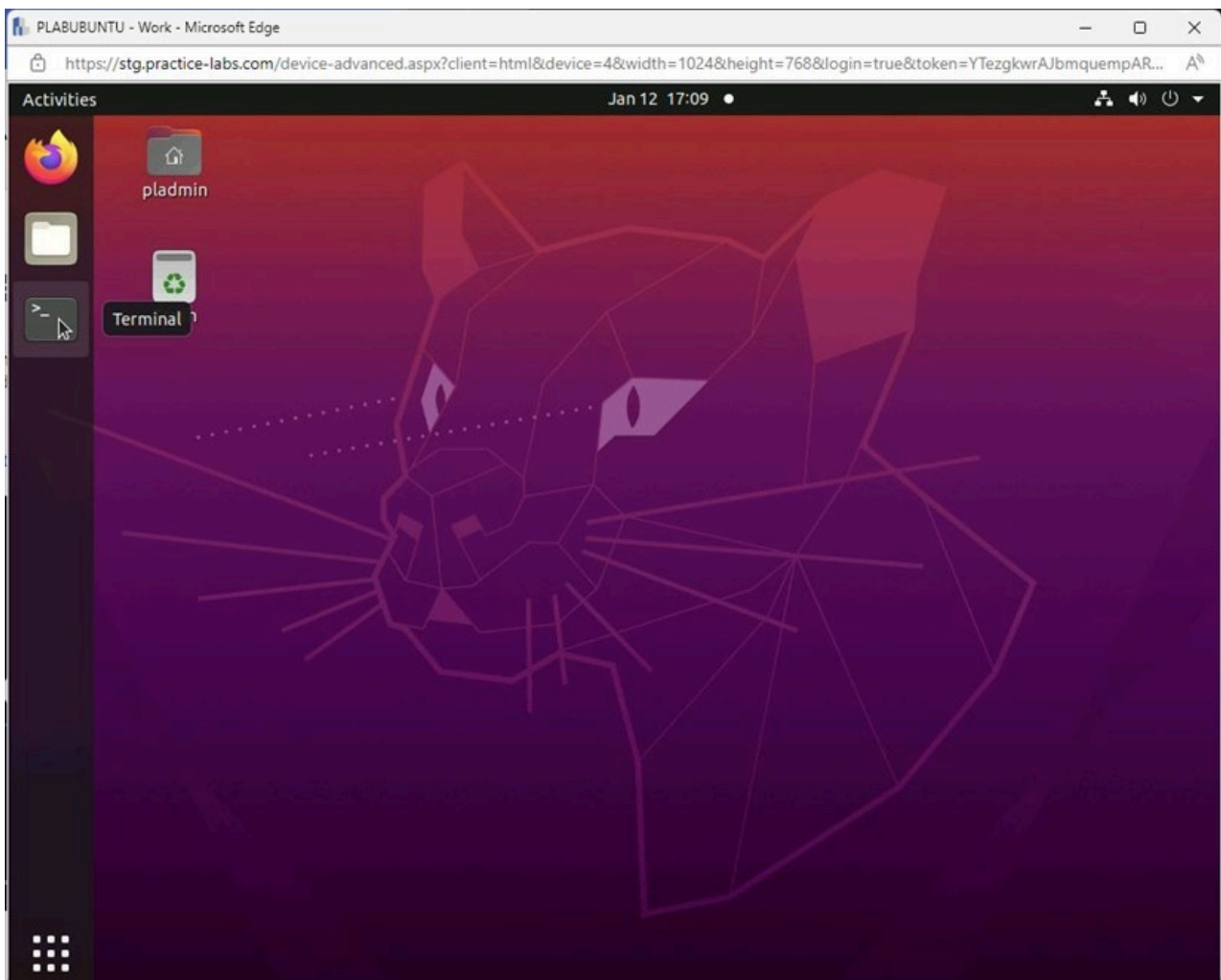
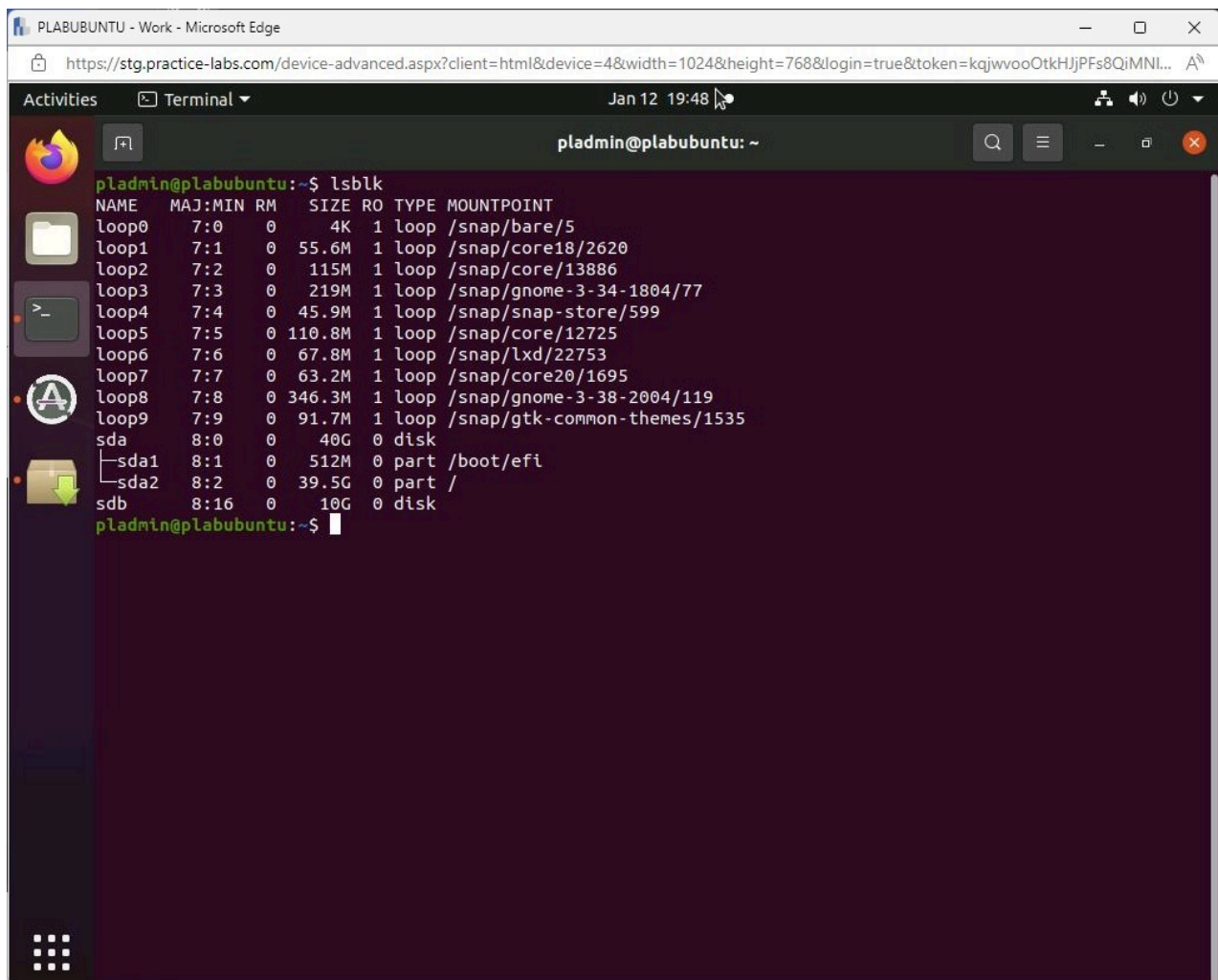


Figure 1.1 Screenshot of PLABUBUNTU: Displaying clicking the Terminal icon in the left pane.

Step 2

In the **Terminal** window, type the following and press **Enter**:

```
lsblk
```



The screenshot shows a web browser window titled "PLABUBUNTU - Work - Microsoft Edge" with the URL <https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=kqjwvooOtkHjPPfs8QiMNL...>. The browser displays a terminal window for a user named "pladmin" on a system named "plabubuntu". The terminal shows the command "lsblk" being executed, which lists the following partitions:

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
loop0	7:0	0	4K	1	loop	/snap/bare/5
loop1	7:1	0	55.6M	1	loop	/snap/core18/2620
loop2	7:2	0	115M	1	loop	/snap/core/13886
loop3	7:3	0	219M	1	loop	/snap/gnome-3-34-1804/77
loop4	7:4	0	45.9M	1	loop	/snap/snap-store/599
loop5	7:5	0	110.8M	1	loop	/snap/core/12725
loop6	7:6	0	67.8M	1	loop	/snap/lxd/22753
loop7	7:7	0	63.2M	1	loop	/snap/core20/1695
loop8	7:8	0	346.3M	1	loop	/snap/gnome-3-38-2004/119
loop9	7:9	0	91.7M	1	loop	/snap/gtk-common-themes/1535
sda	8:0	0	40G	0	disk	
├─sda1	8:1	0	512M	0	part	/boot/efi
└─sda2	8:2	0	39.5G	0	part	/
sdb	8:16	0	10G	0	disk	

Figure 1.2 Screenshot of PLABUBUNTU: Displaying typing the required command and pressing enter.

Note: The **lsblk** command is used to list all of the available partitions on the device.

Step 3

Locate the **sdb** partition in the output.

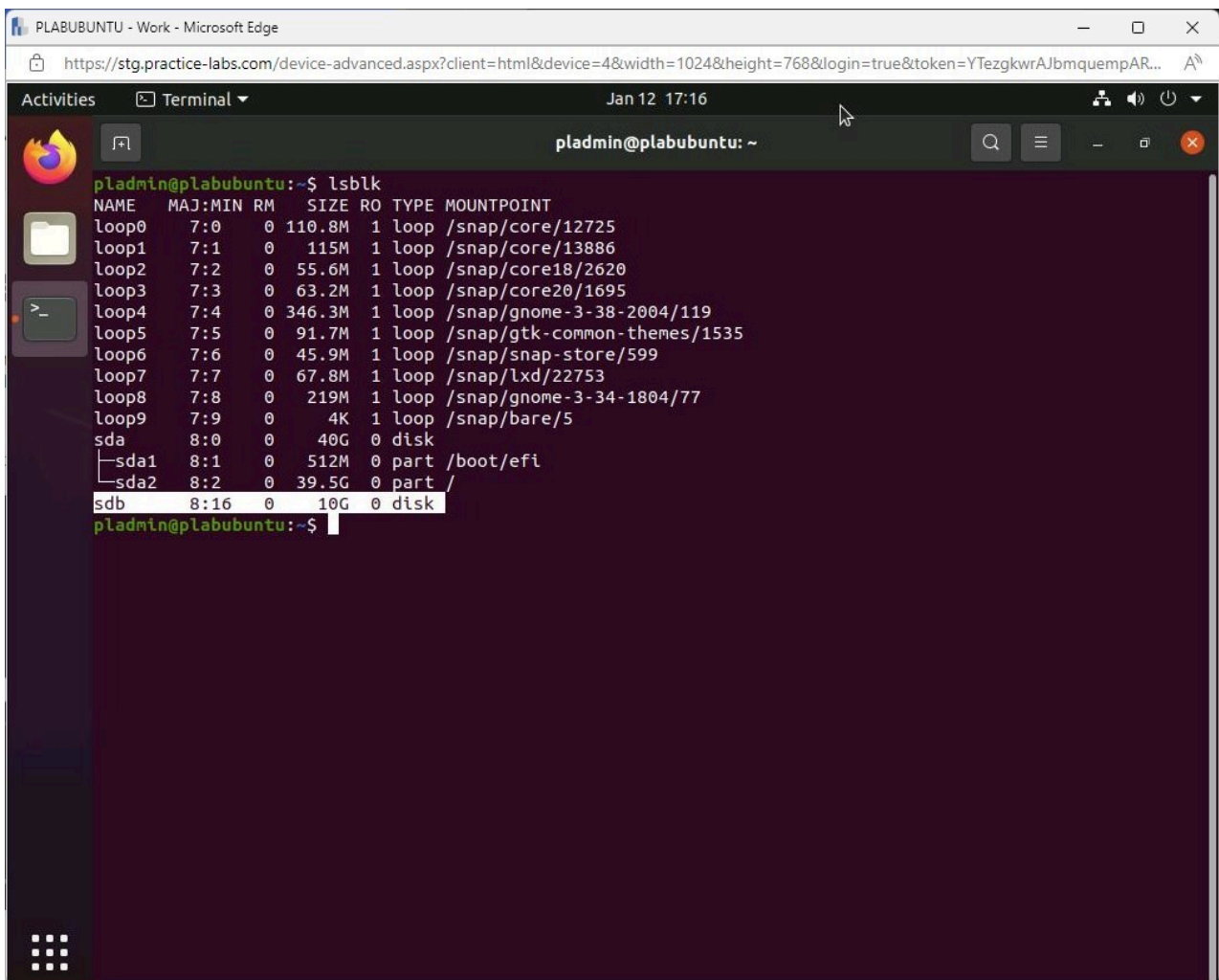


Figure 1.3 Screenshot of PLABUBUNTU: Displaying locating the sdb partition in the terminal window.

Step 4

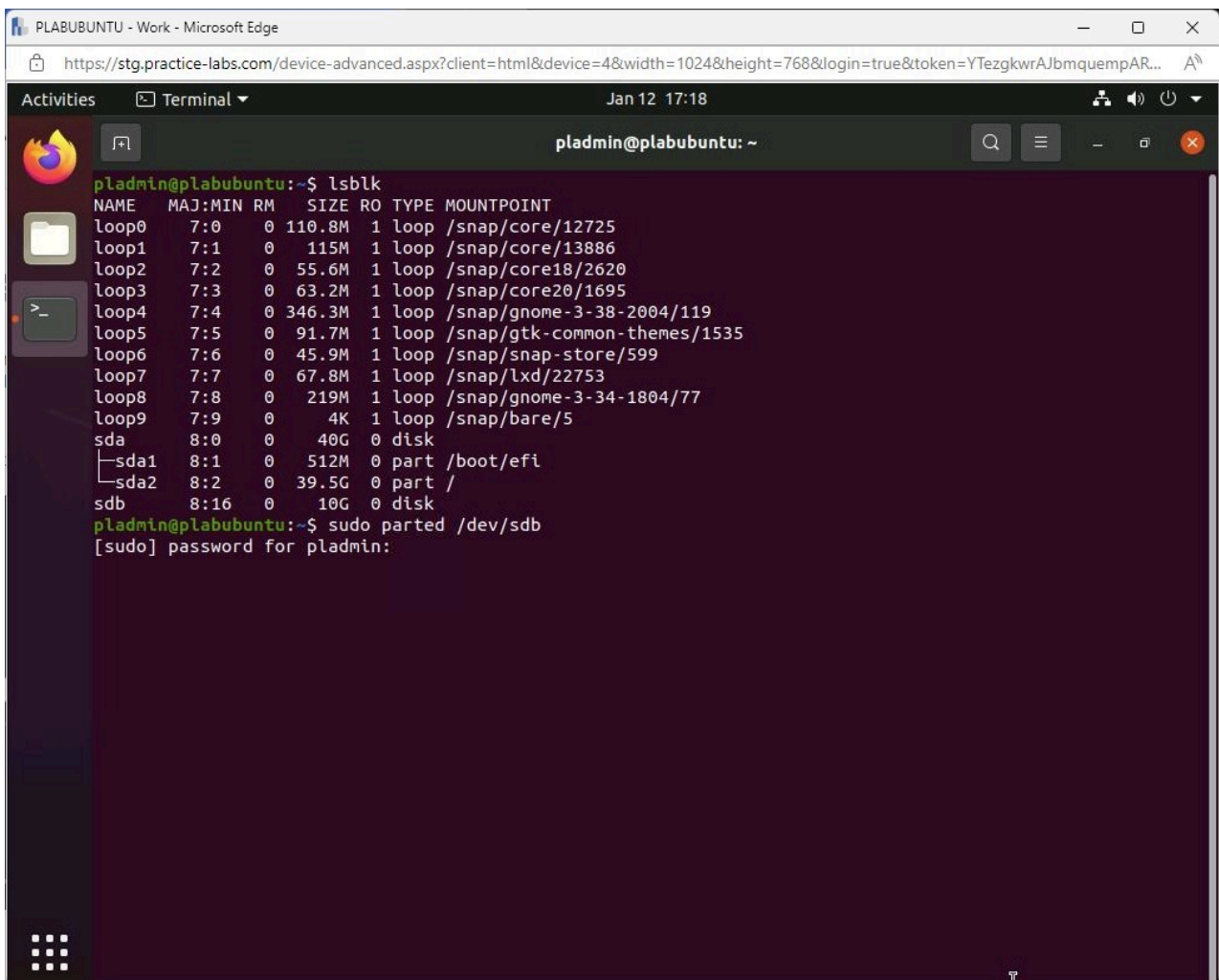
In the **Terminal** window, type the following and press **Enter**:

```
sudo parted /dev/sdb
```

Enter the following password:

Passw0rd

Press **Enter**.



The screenshot shows a terminal window titled 'PLABUBUNTU - Work - Microsoft Edge'. The browser address bar shows a URL from 'stg.practice-labs.com'. The terminal window has a title bar 'Jan 12 17:18' and a search bar. The terminal prompt is 'pladmin@plabubuntu: ~'. The user has entered the command 'lsblk', which displays a table of block devices. The table has columns: NAME, MAJ:MIN, RM, SIZE, RO, TYPE, and MOUNTPOINT. The devices listed are loop0 through loop9, sda, sda1, sda2, and sdb. The user has then entered the command 'sudo parted /dev/sdb', and the prompt has changed to '[sudo] password for pladmin:'.

```
pladmin@plabubuntu:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
loop0       7:0      0 110.8M 1 loop /snap/core/12725
loop1       7:1      0   115M 1 loop /snap/core/13886
loop2       7:2      0   55.6M 1 loop /snap/core18/2620
loop3       7:3      0   63.2M 1 loop /snap/core20/1695
loop4       7:4      0 346.3M 1 loop /snap/gnome-3-38-2004/119
loop5       7:5      0   91.7M 1 loop /snap/gtk-common-themes/1535
loop6       7:6      0   45.9M 1 loop /snap/snap-store/599
loop7       7:7      0   67.8M 1 loop /snap/lxd/22753
loop8       7:8      0  219M 1 loop /snap/gnome-3-34-1804/77
loop9       7:9      0     4K 1 loop /snap/bare/5
sda         8:0      0   40G   0 disk
├─sda1      8:1      0   512M   0 part /boot/efi
└─sda2      8:2      0  39.5G   0 part /
sdb         8:16     0   10G   0 disk
pladmin@plabubuntu:~$ sudo parted /dev/sdb
[sudo] password for pladmin:
```

Figure 1.4 Screenshot of PLABUBUNTU: Displaying entering the required command and password and pressing enter.

Note: The *parted* command is used to partition a specific partition on the device in this case, the *sdb* partition was selected.

Step 5

Next, type the following command and press **Enter**:

```
mklabel gpt
```


The screenshot shows a web browser window titled 'PLABUBUNTU - Work - Microsoft Edge' with the URL <https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=YTezgkwrAJbmquempAR...>. The browser displays a terminal window for 'pladmin@plabubuntu: ~'. The terminal shows the output of the 'lsblk' command, listing various loop and disk devices with their sizes, types, and mount points. Below this, the user has entered 'sudo parted /dev/sdb' and provided their password. The terminal now shows the GNU Parted 3.3 welcome message and the prompt '(parted) mklabel gpt'.

```
pladmin@plabubuntu:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINT
loop0       7:0      0 110.8M 1 loop  /snap/core/12725
loop1       7:1      0   115M 1 loop  /snap/core/13886
loop2       7:2      0   55.6M 1 loop  /snap/core18/2620
loop3       7:3      0   63.2M 1 loop  /snap/core20/1695
loop4       7:4      0 346.3M 1 loop  /snap/gnome-3-38-2004/119
loop5       7:5      0   91.7M 1 loop  /snap/gtk-common-themes/1535
loop6       7:6      0   45.9M 1 loop  /snap/snap-store/599
loop7       7:7      0   67.8M 1 loop  /snap/lxd/22753
loop8       7:8      0   219M 1 loop  /snap/gnome-3-34-1804/77
loop9       7:9      0     4K 1 loop  /snap/bare/5
sda         8:0      0   40G   0 disk
├─sda1      8:1      0   512M 0 part  /boot/efi
└─sda2      8:2      0  39.5G 0 part  /
sdb         8:16     0   10G   0 disk

pladmin@plabubuntu:~$ sudo parted /dev/sdb
[sudo] password for pladmin:
GNU Parted 3.3
Using /dev/sdb
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) mklabel gpt
```

Figure 1.5 Screenshot of PLABUBUNTU: Displaying entering the required command and pressing enter.

Step 6

Now, type the following command and press **Enter**:

```
mkpart primary ext4 1MB 10240MB
```


The screenshot shows a web browser window titled 'PLABUBUNTU - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=YTezgkwrAJbmquempAR...'. The browser displays a terminal window for 'pladmin@plabubuntu: ~'. The terminal shows the command 'lsblk' being executed, resulting in a table of block devices. Below this, the command 'sudo parted /dev/sdb' is entered, and the user is prompted for a password. The terminal output includes the following table:

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
loop0	7:0	0	110.8M	1	loop	/snap/core/12725
loop1	7:1	0	115M	1	loop	/snap/core/13886
loop2	7:2	0	55.6M	1	loop	/snap/core18/2620
loop3	7:3	0	63.2M	1	loop	/snap/core20/1695
loop4	7:4	0	346.3M	1	loop	/snap/gnome-3-38-2004/119
loop5	7:5	0	91.7M	1	loop	/snap/gtk-common-themes/1535
loop6	7:6	0	45.9M	1	loop	/snap/snap-store/599
loop7	7:7	0	67.8M	1	loop	/snap/lxd/22753
loop8	7:8	0	219M	1	loop	/snap/gnome-3-34-1804/77
loop9	7:9	0	4K	1	loop	/snap/bare/5
sda	8:0	0	40G	0	disk	
└─sda1	8:1	0	512M	0	part	/boot/efi
└─sda2	8:2	0	39.5G	0	part	/
sdb	8:16	0	10G	0	disk	

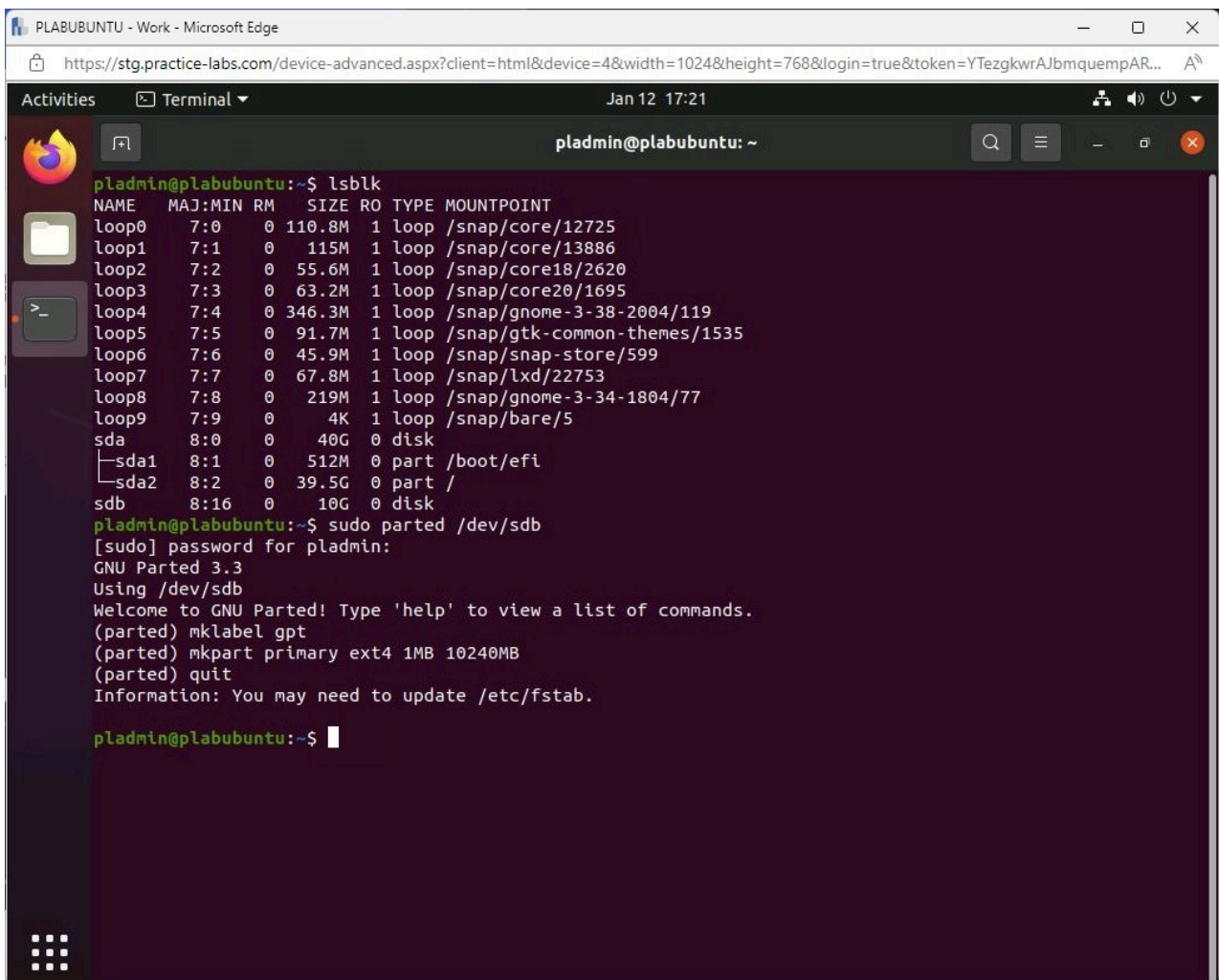
Below the table, the terminal shows the command 'sudo parted /dev/sdb' being executed, and the user is prompted for a password. The output of the 'parted' command is shown, including the GNU Parted version and the command 'mklabel gpt'.

Figure 1.6 Screenshot of PLABUBUNTU: Displaying entering the required command and pressing enter.

Step 7

Type the following command and press **Enter**:

```
quit
```



The screenshot shows a web browser window titled "PLABUBUNTU - Work - Microsoft Edge" with the URL <https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=YTezgkwrAJbmquempAR...>. The browser displays a terminal window for a user named "pladmin" on a system named "plabubuntu". The terminal shows the output of the `lsblk` command, listing various loop and disk devices. Subsequently, the user runs `sudo parted /dev/sdb`, and the terminal shows the GNU Parted 3.3 interface, including a password prompt and several commands entered: `mklabel gpt`, `mkpart primary ext4 1MB 10240MB`, and `quit`. The terminal also displays a message about updating `/etc/fstab`.

```
pladmin@plabubuntu:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
loop0       7:0      0 110.8M 1 loop /snap/core/12725
loop1       7:1      0   115M 1 loop /snap/core/13886
loop2       7:2      0   55.6M 1 loop /snap/core18/2620
loop3       7:3      0   63.2M 1 loop /snap/core20/1695
loop4       7:4      0 346.3M 1 loop /snap/gnome-3-38-2004/119
loop5       7:5      0   91.7M 1 loop /snap/gtk-common-themes/1535
loop6       7:6      0   45.9M 1 loop /snap/snap-store/599
loop7       7:7      0   67.8M 1 loop /snap/lxd/22753
loop8       7:8      0   219M 1 loop /snap/gnome-3-34-1804/77
loop9       7:9      0     4K 1 loop /snap/bare/5
sda         8:0      0   40G   0 disk
├─sda1      8:1      0    512M 0 part /boot/efi
└─sda2      8:2      0   39.5G 0 part /
sdb         8:16     0    10G   0 disk

pladmin@plabubuntu:~$ sudo parted /dev/sdb
[sudo] password for pladmin:
GNU Parted 3.3
Using /dev/sdb
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) mklabel gpt
(parted) mkpart primary ext4 1MB 10240MB
(parted) quit
Information: You may need to update /etc/fstab.

pladmin@plabubuntu:~$
```

Figure 1.7 Screenshot of PLABUBUNTU: Displaying entering the required command and pressing enter.

Step 8

Type the following command to clear the **Terminal** window:

```
clear
```

Press **Enter**.

The screenshot shows a web browser window titled 'PLABUBUNTU - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=YTezgkwrAJbmquempAR...'. The browser displays a terminal window for 'pladmin@plabubuntu: ~' dated 'Jan 12 17:23'. The terminal shows the command 'lsblk' being executed, resulting in a table of disk and loop devices. Below this, the command 'sudo parted /dev/sdb' is entered, followed by the password prompt and the start of the GNU Parted 3.3 interface.

```
pladmin@plabubuntu:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
loop0       7:0      0 110.8M 1 loop /snap/core/12725
loop1       7:1      0   115M 1 loop /snap/core/13886
loop2       7:2      0   55.6M 1 loop /snap/core18/2620
loop3       7:3      0   63.2M 1 loop /snap/core20/1695
loop4       7:4      0 346.3M 1 loop /snap/gnome-3-38-2004/119
loop5       7:5      0   91.7M 1 loop /snap/gtk-common-themes/1535
loop6       7:6      0   45.9M 1 loop /snap/snap-store/599
loop7       7:7      0   67.8M 1 loop /snap/lxd/22753
loop8       7:8      0  219M 1 loop /snap/gnome-3-34-1804/77
loop9       7:9      0     4K 1 loop /snap/bare/5
sda         8:0      0   40G   0 disk
├─sda1      8:1      0   512M   0 part /boot/efi
└─sda2      8:2      0   39.5G   0 part /
sdb         8:16     0    10G   0 disk

pladmin@plabubuntu:~$ sudo parted /dev/sdb
[sudo] password for pladmin:
GNU Parted 3.3
Using /dev/sdb
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) mklabel gpt
(parted) mkpart primary ext4 1MB 10240MB
(parted) quit
Information: You may need to update /etc/fstab.

pladmin@plabubuntu:~$ clear
```

Figure 1.8 Screenshot of PLABUBUNTU: Displaying entering the required command and pressing enter.

Keep the **Terminal** window open.

Task 2 - Format a Partition with a Filesystem in Linux

In this task, you will format a partition with a filesystem on the PLABUBUNTU Linux device.

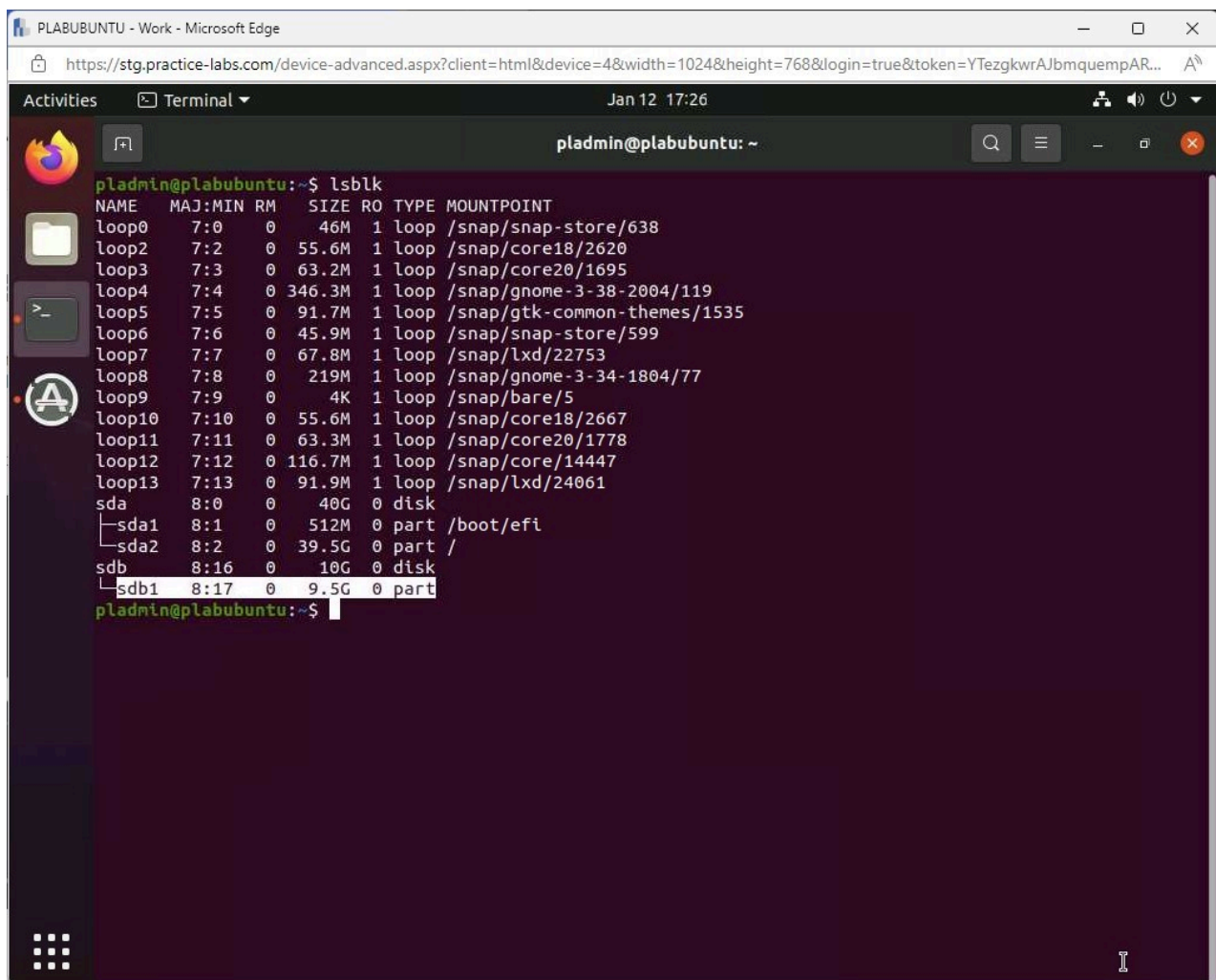
Step 1

Ensure you are connected to **PLABUBUNTU**, where the **Terminal** window is open.

In the **Terminal** window, type the following command to identify the device and partition to format:

```
lsblk
```

Press **Enter**.



The screenshot shows a web browser window titled 'PLABUBUNTU - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=48&width=1024&height=768&login=true&token=YTezgkwrAJbmquempAR...'. The browser displays a terminal window for 'pladmin@plabubuntu: ~' with the date 'Jan 12 17:26'. The terminal shows the command 'lsblk' being executed, resulting in a table of storage devices and partitions. The output is as follows:

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
loop0	7:0	0	46M	1	loop	/snap/snap-store/638
loop2	7:2	0	55.6M	1	loop	/snap/core18/2620
loop3	7:3	0	63.2M	1	loop	/snap/core20/1695
loop4	7:4	0	346.3M	1	loop	/snap/gnome-3-38-2004/119
loop5	7:5	0	91.7M	1	loop	/snap/gtk-common-themes/1535
loop6	7:6	0	45.9M	1	loop	/snap/snap-store/599
loop7	7:7	0	67.8M	1	loop	/snap/lxd/22753
loop8	7:8	0	219M	1	loop	/snap/gnome-3-34-1804/77
loop9	7:9	0	4K	1	loop	/snap/bare/5
loop10	7:10	0	55.6M	1	loop	/snap/core18/2667
loop11	7:11	0	63.3M	1	loop	/snap/core20/1778
loop12	7:12	0	116.7M	1	loop	/snap/core/14447
loop13	7:13	0	91.9M	1	loop	/snap/lxd/24061
sda	8:0	0	40G	0	disk	
├─sda1	8:1	0	512M	0	part	/boot/efi
└─sda2	8:2	0	39.5G	0	part	/
sdb	8:16	0	10G	0	disk	
└─sdb1	8:17	0	9.5G	0	part	

The terminal prompt is 'pladmin@plabubuntu:~\$'.

Figure 1.9 Screenshot of PLABUBUNTU: Displaying executing the command in the terminal window and pressing enter.

Note: Notice the storage device and partition is sdb1.

Step 2

Type the following command and press **Enter**:

```
sudo mkfs -t ext4 /dev/sdb1
```



```

pladmin@plabubuntu:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
loop0       7:0    0   46M  1 loop /snap/snap-store/638
loop2       7:2    0  55.6M  1 loop /snap/core18/2620
loop3       7:3    0  63.2M  1 loop /snap/core20/1695
loop4       7:4    0 346.3M  1 loop /snap/gnome-3-38-2004/119
loop5       7:5    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop6       7:6    0  45.9M  1 loop /snap/snap-store/599
loop7       7:7    0  67.8M  1 loop /snap/lxd/22753
loop8       7:8    0  219M  1 loop /snap/gnome-3-34-1804/77
loop9       7:9    0    4K  1 loop /snap/bare/5
loop10      7:10   0  55.6M  1 loop /snap/core18/2667
loop11      7:11   0  63.3M  1 loop /snap/core20/1778
loop12      7:12   0 116.7M  1 loop /snap/core/14447
loop13      7:13   0  91.9M  1 loop /snap/lxd/24061
sda         8:0    0   40G  0 disk
├─sda1      8:1    0   512M  0 part /boot/efi
└─sda2      8:2    0   39.5G  0 part /
sdb         8:16   0   10G  0 disk
└─sdb1      8:17   0    9.5G  0 part

pladmin@plabubuntu:~$ sudo mkfs -t ext4 /dev/sdb1
mkfs2fs 1.45.5 (07-Jan-2020)
Discarding device blocks: done
Creating filesystem with 2499840 4k blocks and 625856 inodes
Filesystem UUID: f7c52b5f-dc7d-4fc9-a3de-3f1b3b7234d1
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

pladmin@plabubuntu:~$

```

Figure 1.10 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Note: The **mkfs** command formats a partition with a specific file system. In this case, the partition is formatted with the **ext4** file system.

Note: Type the **Clear** command and press **Enter** before proceeding to the next task.

Keep the **Terminal** window open.

Task 3 - Mount Storage in Linux

In this task, you will mount the storage on the PLABUBUNTU Linux device.

Step 1

In **PLABUBUNTU**, the **Terminal** window is open.

In the **Terminal** window, type the following command to create a new directory:

```
sudo mkdir /srv/nfs
```

Press **Enter**.

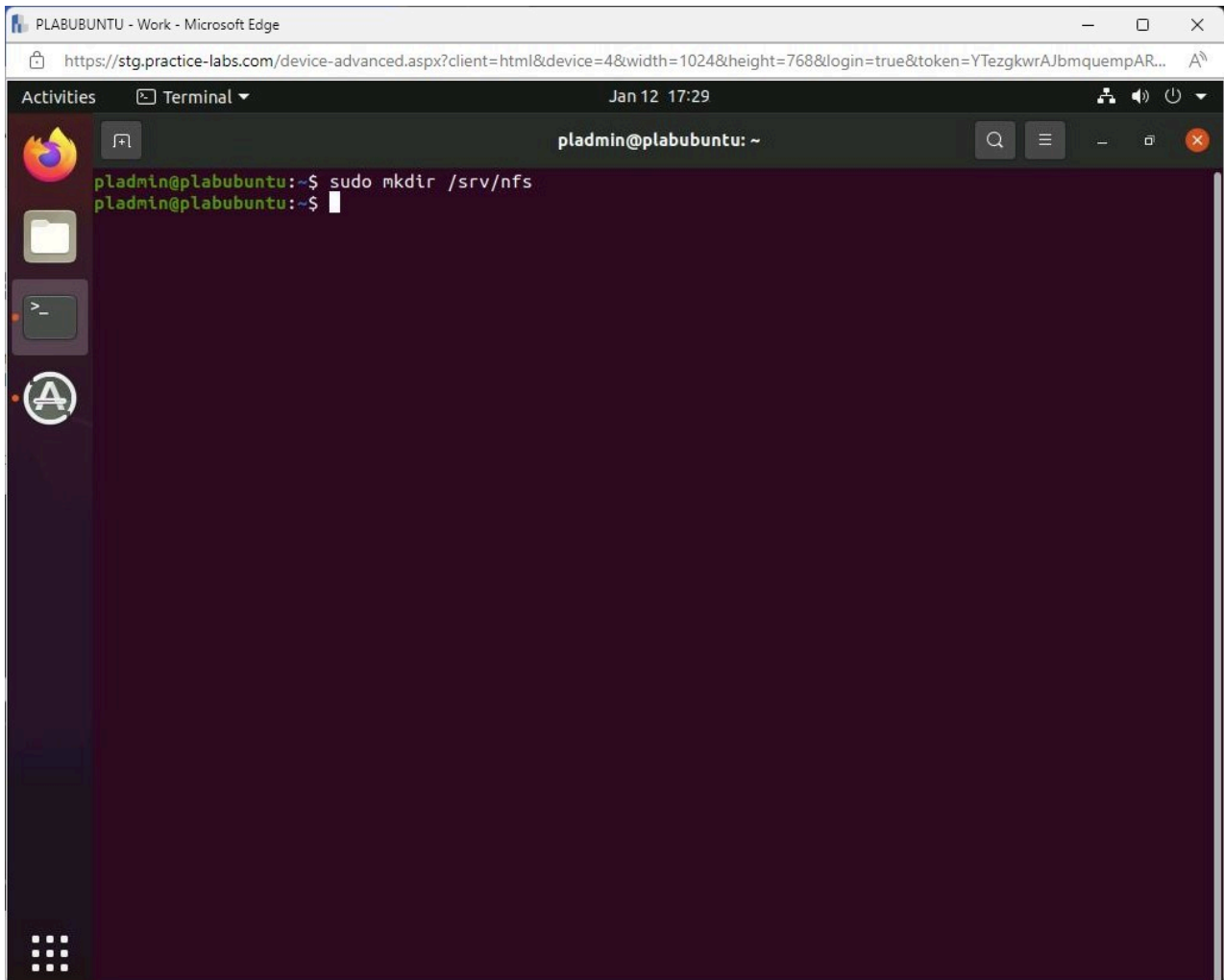


Figure 1.11 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Step 2

Next, type the following command to mount the storage:

```
sudo mount /dev/sdb1 /srv/nfs
```

Press **Enter**.

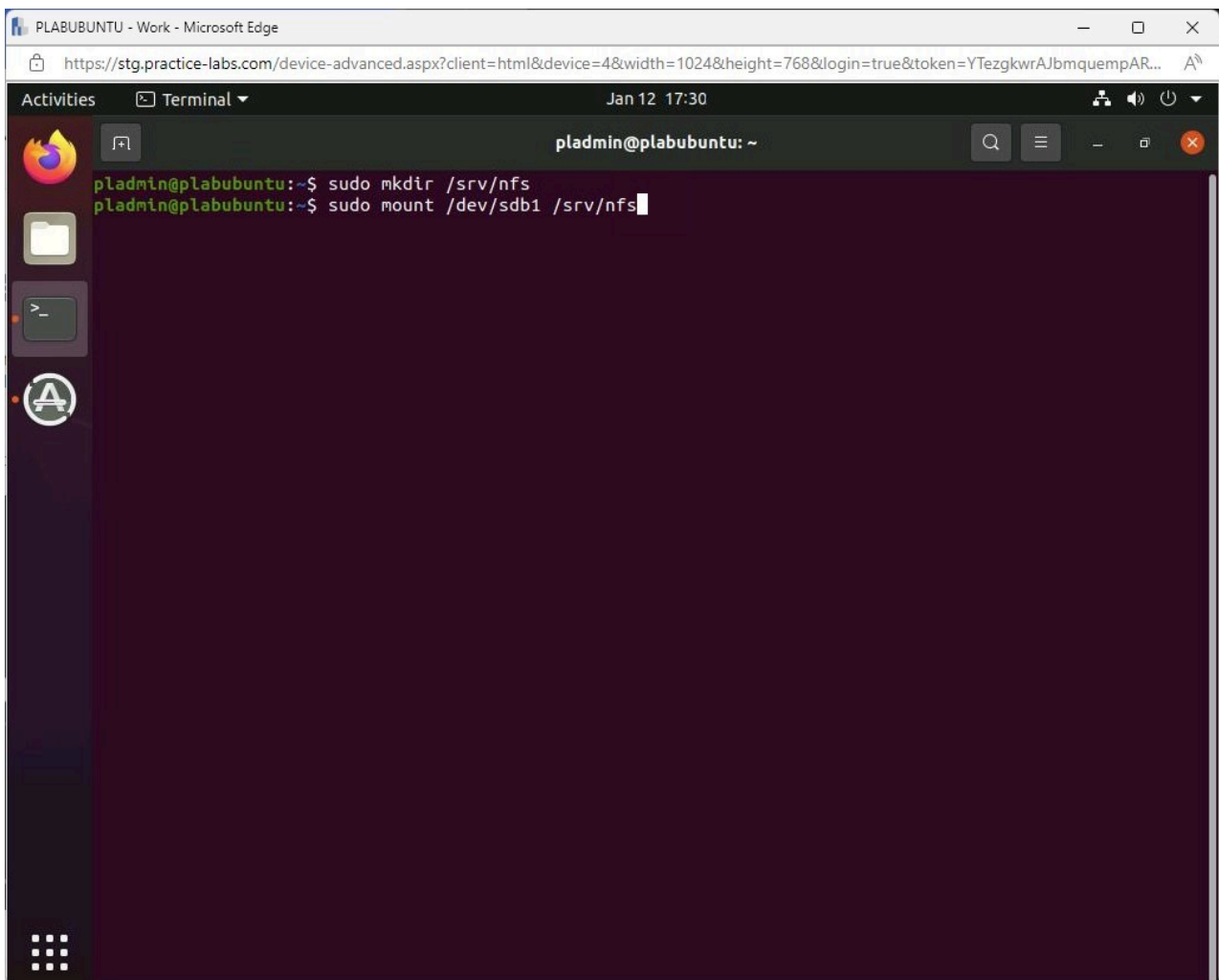


Figure 1.12 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

***Note:** The **mount** command is used to mount the previously created partition to the created directory, enabling the system to access the partition.*

Step 3

Next, verify the mount point by typing the following command:

```
lsblk -f
```

Press **Enter**.

The screenshot shows a web browser window titled 'PLABUBUNTU - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=YTezgkwrAJbmquempAR...'. The browser displays a terminal window for 'pladmin@plabubuntu: ~' with the date 'Jan 12 17:30'. The terminal shows the following commands and output:

```
pladmin@plabubuntu:~$ sudo mkdir /srv/nfs
pladmin@plabubuntu:~$ sudo mount /dev/sdb1 /srv/nfs
pladmin@plabubuntu:~$ lsblk -f
```

NAME	FSTYPE	LABEL	UUID	FSAVAIL	FSUSE%	MOUNTPOINT
loop0	squashfs			0	100%	/snap/snap-store/638
loop2	squashfs			0	100%	/snap/core18/2620
loop3	squashfs			0	100%	/snap/core20/1695
loop4	squashfs			0	100%	/snap/gnome-3-38-2004/119
loop5	squashfs			0	100%	/snap/gtk-common-themes/1535
loop6	squashfs			0	100%	/snap/snap-store/599
loop7	squashfs			0	100%	/snap/lxd/22753
loop8	squashfs			0	100%	/snap/gnome-3-34-1804/77
loop9	squashfs			0	100%	/snap/bare/5
loop10	squashfs			0	100%	/snap/core18/2667
loop11	squashfs			0	100%	/snap/core20/1778
loop12	squashfs			0	100%	/snap/core/14447
loop13	squashfs			0	100%	/snap/lxd/24061
sda						
├sda1	vfat		2B47-E9FD	505.8M	1%	/boot/efi
└sda2	ext4		3c9f64f2-040e-476d-99f8-fbcb2a0b7614	25.3G	29%	/
sdb						
└sdb1	ext4		f7c52b5f-dc7d-4fc9-a3de-3f1b3b7234d1	8.8G	0%	/srv/nfs

```
pladmin@plabubuntu:~$
```

Figure 1.13 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Note: Type the **Clear** command and press **Enter** before proceeding to the next Exercise.

Keep the **Terminal** window open.

Exercise 2 - Installing and Configuring Shared Storage in Linux using NFS

To enable access to a remote share on a network using linux devices, a Network File System (NFS) server needs to be implemented to facilitate the network share.

In this exercise, a Network File System (NFS) Server and client will be configured to access a network file share.

Learning Outcomes

After completing this exercise, you should be able to:

- Configure the Permissions on the NFS Directory
- Install and Configure an NFS Server
- Install and Configure an NFS Client

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABALMA** - (Alma Linux 8.7 - Stand-alone Linux Workstation)
- **PLABUBUNTU** - (Linux Ubuntu - Stand-alone Linux Server)

Task 1 - Configure the Permissions on the NFS Directory

In this task, you will configure the permissions on the NFS directory.

Step 1

Ensure you are connected to **PLABUBUNTU**, and the **Terminal** window is open.

In the **Terminal** window, type the following command to set the permissions:

```
sudo chmod o+w /srv/nfs
```

Press **Enter**.

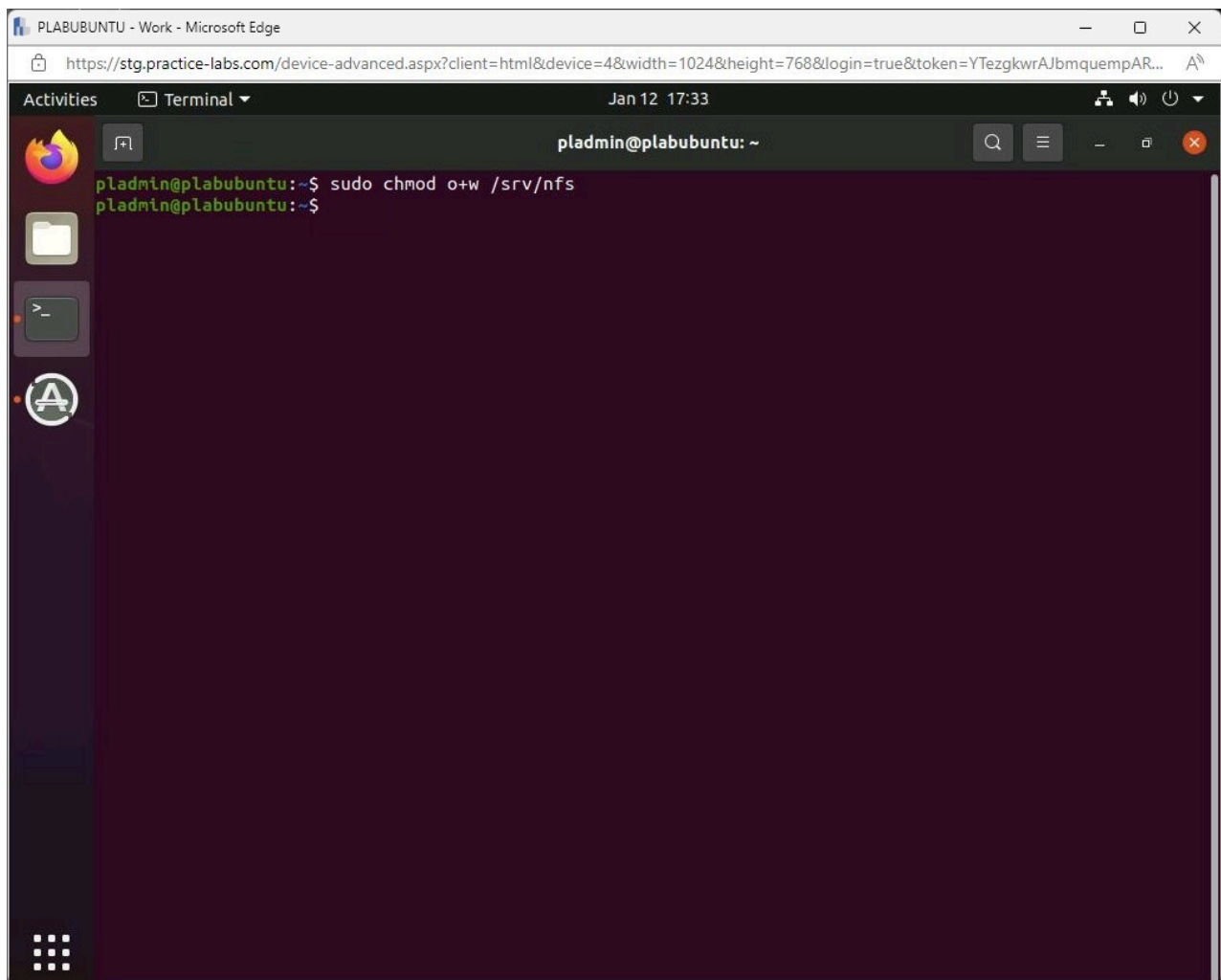


Figure 2.1 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

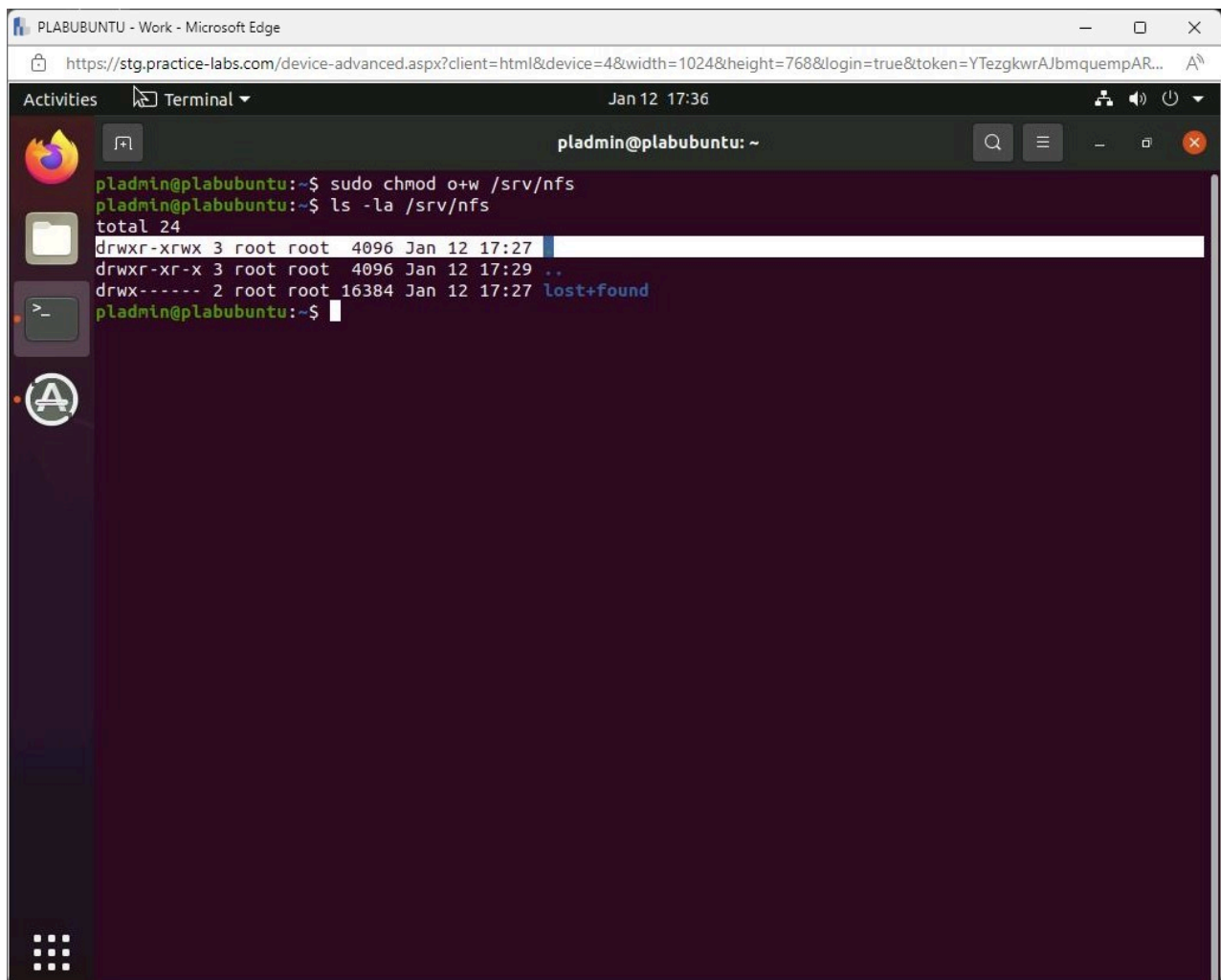
Note: The **chmod** command is used to change the file permissions of a file or directory. In this example, the **chmod** command was used to change the directory's permissions to write for the Others group.

Step 2

Next, type the following command to verify the permissions are set:

```
ls -la /srv/nfs
```

Press **Enter**.



The screenshot shows a terminal window titled 'pladmin@plabubuntu: ~' with a search bar and window controls. The terminal output is as follows:

```
pladmin@plabubuntu:~$ sudo chmod o+w /srv/nfs
pladmin@plabubuntu:~$ ls -la /srv/nfs
total 24
drwxr-xrwx 3 root root 4096 Jan 12 17:27
drwxr-xr-x 3 root root 4096 Jan 12 17:29 ..
drwx----- 2 root root 16384 Jan 12 17:27 lost+found
pladmin@plabubuntu:~$
```

Figure 2.2 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Note: Notice the permissions are set to **drwxr-xrwx root root 4096 Jan 12 17:27**. Please note your date and timestamp will vary from the one shown on the screenshot.

Keep the **Terminal** window open.

Task 2 - Install and Configure an NFS Server

In this task, you will install and configure an NFS server.

Step 1

In **PLABUBUNTU**, the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt update
```

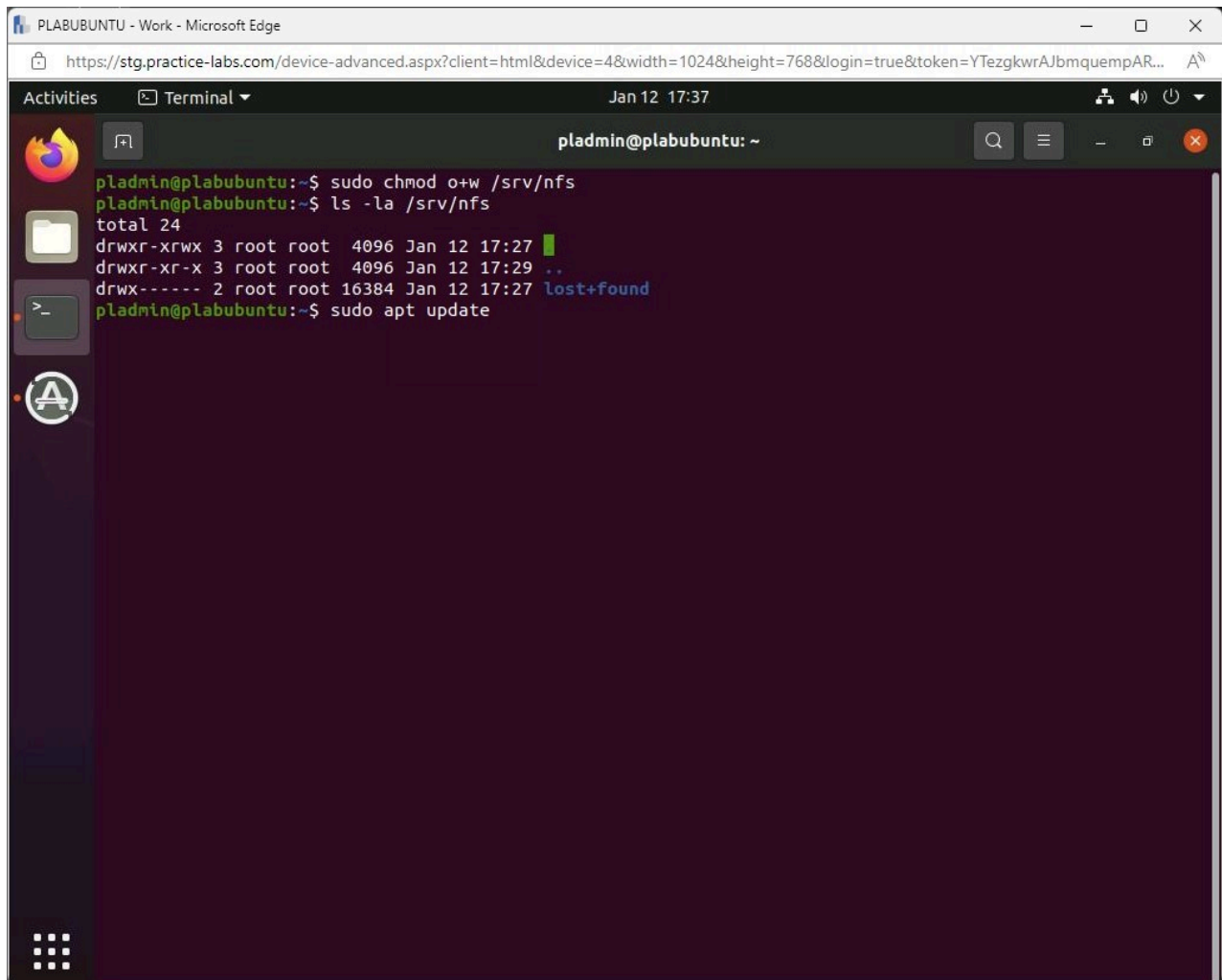


Figure 2.3 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Note: This command might take a while to complete.

Step 2

Next, type the following command to install an NFS server:

```
sudo apt install nfs-kernel-server -y
```

Press **Enter**.

The screenshot shows a web browser window titled 'PLABUBUNTU - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=kqjwvooOtkHjPPFs8QiMNL...'. The browser displays a terminal window with the prompt 'pladmin@plabubuntu: ~'. The terminal output shows the execution of 'sudo apt install nfs-kernel-server', which includes downloading various packages from the Ubuntu archive, building the dependency tree, and reading state information. The final output indicates that 92 packages can be upgraded and suggests running 'apt list --upgradable' to see them. The prompt 'pladmin@plabubuntu:~\$' is visible at the bottom of the terminal window.

```

Get:29 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [604 B]
Get:30 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [45.7 kB]
Get:31 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 DEP-11 Metadata [7,996 B]
Get:32 http://archive.ubuntu.com/ubuntu focal-backports/main DEP-11 48x48 Icons [6,984 B]
Get:33 http://archive.ubuntu.com/ubuntu focal-backports/main DEP-11 64x64 Icons [10.5 kB]
Get:34 http://archive.ubuntu.com/ubuntu focal-backports/main DEP-11 64x64@2 Icons [29 B]
Get:35 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 c-n-f Metadata [1,420 B]
Get:36 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [24.9 kB]
Get:37 http://archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [16.3 kB]
Get:38 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 DEP-11 Metadata [30.5 kB]
Get:39 http://archive.ubuntu.com/ubuntu focal-backports/universe DEP-11 64x64@2 Icons [29 B]
Get:40 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [880 B]
Get:41 http://archive.ubuntu.com/ubuntu focal-security/main amd64 Packages [1,929 kB]
Get:42 http://archive.ubuntu.com/ubuntu focal-security/main Translation-en [316 kB]
Get:43 http://archive.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metadata [59.9 kB]
Get:44 http://archive.ubuntu.com/ubuntu focal-security/main DEP-11 48x48 Icons [18.9 kB]
Get:45 http://archive.ubuntu.com/ubuntu focal-security/main DEP-11 64x64 Icons [36.0 kB]
Get:46 http://archive.ubuntu.com/ubuntu focal-security/main DEP-11 64x64@2 Icons [29 B]
Get:47 http://archive.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [11.7 kB]
Get:48 http://archive.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [1,419 kB]
Get:49 http://archive.ubuntu.com/ubuntu focal-security/restricted Translation-en [200 kB]
Get:50 http://archive.ubuntu.com/ubuntu focal-security/restricted amd64 c-n-f Metadata [616 B]
Get:51 http://archive.ubuntu.com/ubuntu focal-security/universe amd64 Packages [783 kB]
Get:52 http://archive.ubuntu.com/ubuntu focal-security/universe Translation-en [152 kB]
Get:53 http://archive.ubuntu.com/ubuntu focal-security/universe amd64 DEP-11 Metadata [93.8 kB]
Get:54 http://archive.ubuntu.com/ubuntu focal-security/universe DEP-11 64x64@2 Icons [29 B]
Get:55 http://archive.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [16.9 kB]
Get:56 http://archive.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [22.2 kB]
Get:57 http://archive.ubuntu.com/ubuntu focal-security/multiverse Translation-en [5,464 B]
Get:58 http://archive.ubuntu.com/ubuntu focal-security/multiverse amd64 DEP-11 Metadata [940 B]
Get:59 http://archive.ubuntu.com/ubuntu focal-security/multiverse DEP-11 64x64@2 Icons [29 B]
Get:60 http://archive.ubuntu.com/ubuntu focal-security/multiverse amd64 c-n-f Metadata [516 B]
Fetched 13.0 MB in 3s (4,033 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
92 packages can be upgraded. Run 'apt list --upgradable' to see them.
pladmin@plabubuntu:~$ sudo apt install nfs-kernel-server -y

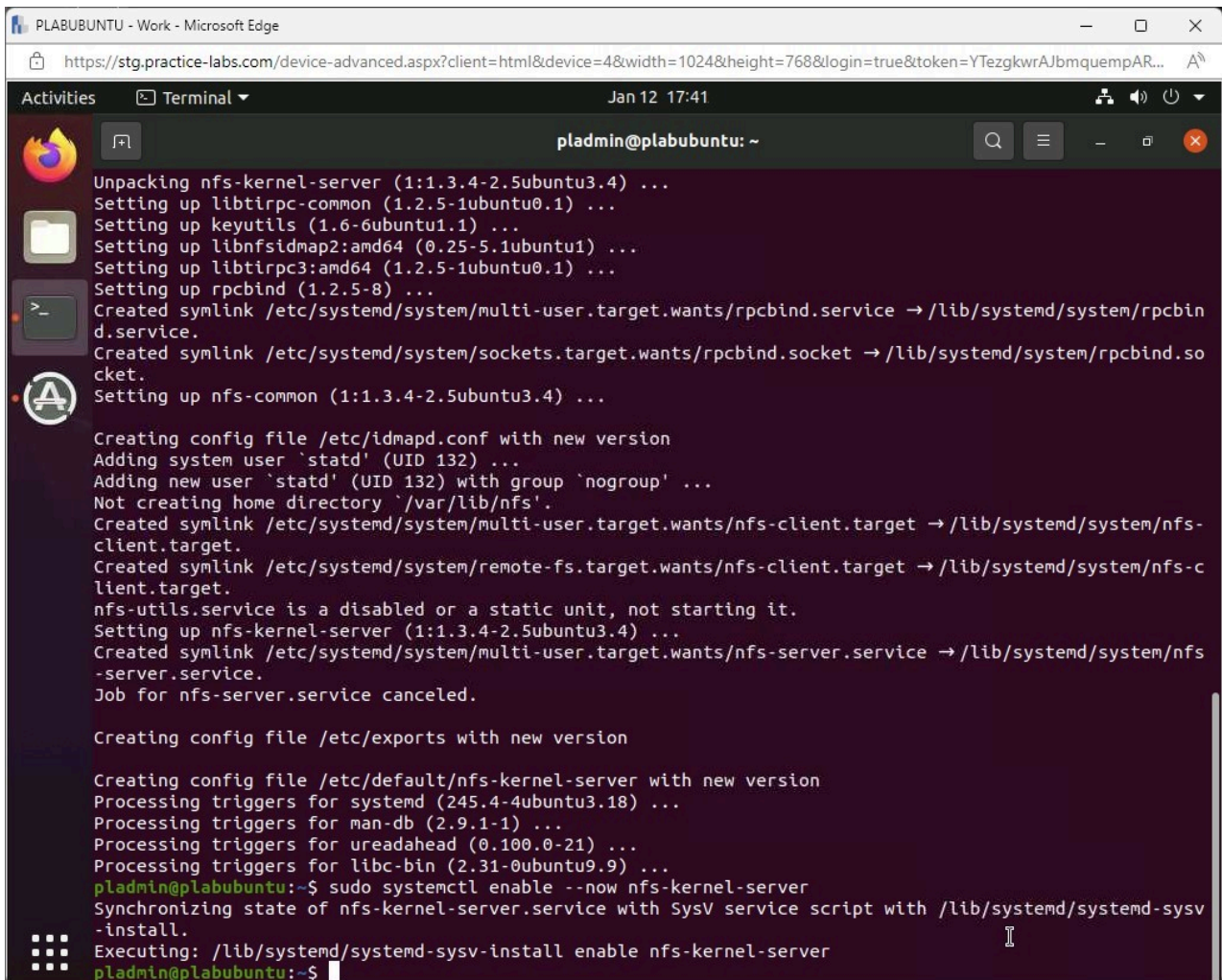
```

Figure 2.4 Screenshot of PLABUBUNTU: Displaying executing the command in the terminal window and pressing enter.

Step 3

Type the following command and press **Enter**:

```
sudo systemctl enable --now nfs-kernel-server
```

```
PLABUBUNTU - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=YTezgkwrAJbmquempAR...

Activities Terminal Jan 12 17:41
pladmin@plabubuntu: ~

Unpacking nfs-kernel-server (1:1.3.4-2.5ubuntu3.4) ...
Setting up libtirpc-common (1.2.5-1ubuntu0.1) ...
Setting up keyutils (1.6-6ubuntu1.1) ...
Setting up libnfsidmap2:amd64 (0.25-5.1ubuntu1) ...
Setting up libtirpc3:amd64 (1.2.5-1ubuntu0.1) ...
Setting up rpcbind (1.2.5-8) ...
Created symlink /etc/systemd/system/multi-user.target.wants/rpcbind.service → /lib/systemd/system/rpcbind.service.
Created symlink /etc/systemd/system/sockets.target.wants/rpcbind.socket → /lib/systemd/system/rpcbind.socket.
Setting up nfs-common (1:1.3.4-2.5ubuntu3.4) ...

Creating config file /etc/idmapd.conf with new version
Adding system user `statd' (UID 132) ...
Adding new user `statd' (UID 132) with group `nogroup' ...
Not creating home directory `/var/lib/nfs'.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
Created symlink /etc/systemd/system/remote-fs.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
nfs-utils.service is a disabled or a static unit, not starting it.
Setting up nfs-kernel-server (1:1.3.4-2.5ubuntu3.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /lib/systemd/system/nfs-server.service.
Job for nfs-server.service canceled.

Creating config file /etc/exports with new version

Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for systemd (245.4-4ubuntu3.18) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ureadahead (0.100.0-21) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
pladmin@plabubuntu:~$ sudo systemctl enable --now nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nfs-kernel-server
pladmin@plabubuntu:~$
```

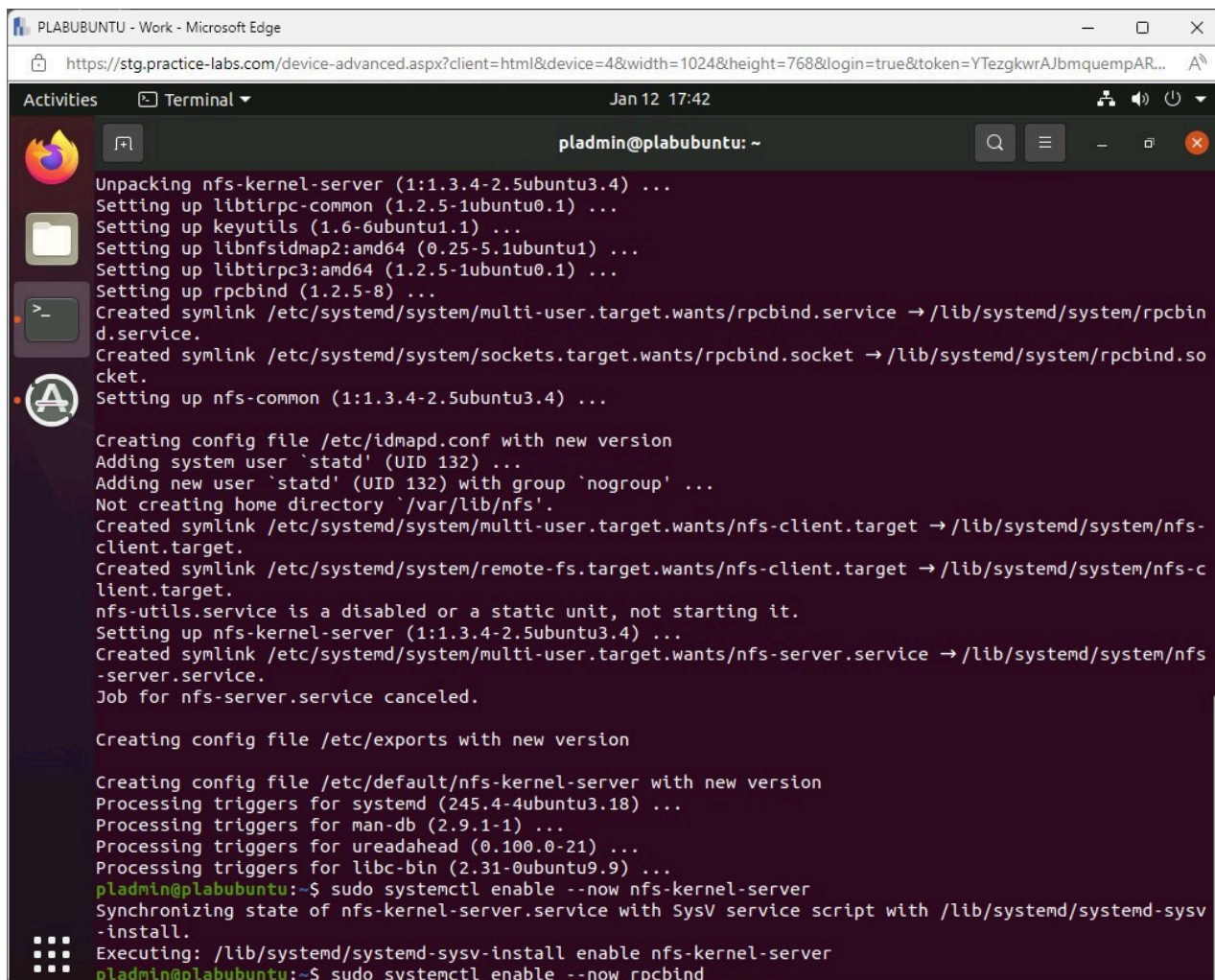
Figure 2.5 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Note: The **systemctl** command is used to start, stop and enable services on a linux device. In this case, the command enabled the NFS service to start automatically on the system's startup.

Step 4

Type the following command and press **Enter**:

```
sudo systemctl enable --now rpcbind
```



```
PLABUBUNTU - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=YTezgkwrAJbmquempAR...

Activities Terminal Jan 12 17:42
pladmin@plabubuntu: ~

Unpacking nfs-kernel-server (1:1.3.4-2.5ubuntu3.4) ...
Setting up libtirpc-common (1.2.5-1ubuntu0.1) ...
Setting up keyutils (1.6-6ubuntu1.1) ...
Setting up libnfsidmap2:amd64 (0.25-5.1ubuntu1) ...
Setting up libtirpc3:amd64 (1.2.5-1ubuntu0.1) ...
Setting up rpcbind (1.2.5-8) ...
Created symlink /etc/systemd/system/multi-user.target.wants/rpcbind.service → /lib/systemd/system/rpcbind.service.
Created symlink /etc/systemd/system/sockets.target.wants/rpcbind.socket → /lib/systemd/system/rpcbind.socket.
Setting up nfs-common (1:1.3.4-2.5ubuntu3.4) ...

Creating config file /etc/idmapd.conf with new version
Adding system user `statd' (UID 132) ...
Adding new user `statd' (UID 132) with group `nogroup' ...
Not creating home directory `/var/lib/nfs'.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
Created symlink /etc/systemd/system/remote-fs.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
nfs-utils.service is a disabled or a static unit, not starting it.
Setting up nfs-kernel-server (1:1.3.4-2.5ubuntu3.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /lib/systemd/system/nfs-server.service.
Job for nfs-server.service canceled.

Creating config file /etc/exports with new version

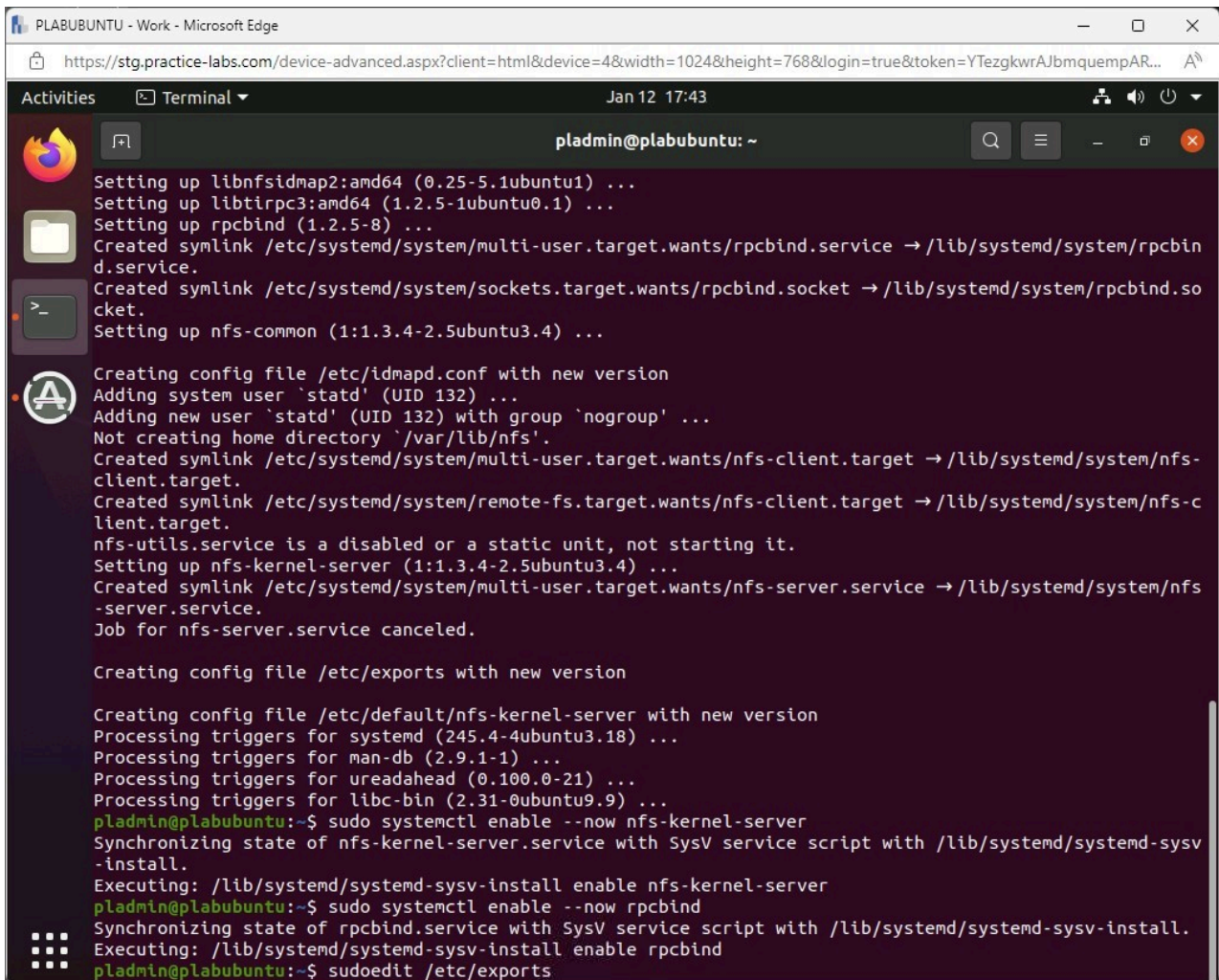
Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for systemd (245.4-4ubuntu3.18) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ureadahead (0.100.0-21) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
pladmin@plabubuntu:~$ sudo systemctl enable --now nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nfs-kernel-server
pladmin@plabubuntu:~$ sudo systemctl enable --now rpcbind
```

Figure 2.6 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Step 5

Next, type the following command and press **Enter**:

```
sudoedit /etc/exports
```



```

PLABUBUNTU - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=YTezgkwrAJbmquempAR...

Activities Terminal Jan 12 17:43
pladmin@plabubuntu: ~

Setting up libnfsidmap2:amd64 (0.25-5.1ubuntu1) ...
Setting up libtirpc3:amd64 (1.2.5-1ubuntu0.1) ...
Setting up rpcbind (1.2.5-8) ...
Created symlink /etc/systemd/system/multi-user.target.wants/rpcbind.service → /lib/systemd/system/rpcbind.service.
Created symlink /etc/systemd/system/sockets.target.wants/rpcbind.socket → /lib/systemd/system/rpcbind.socket.
Setting up nfs-common (1:1.3.4-2.5ubuntu3.4) ...

Creating config file /etc/idmapd.conf with new version
Adding system user `statd' (UID 132) ...
Adding new user `statd' (UID 132) with group `nogroup' ...
Not creating home directory `/var/lib/nfs'.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
Created symlink /etc/systemd/system/remote-fs.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
nfs-utils.service is a disabled or a static unit, not starting it.
Setting up nfs-kernel-server (1:1.3.4-2.5ubuntu3.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /lib/systemd/system/nfs-server.service.
Job for nfs-server.service canceled.

Creating config file /etc/exports with new version

Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for systemd (245.4-4ubuntu3.18) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ureadahead (0.100.0-21) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
pladmin@plabubuntu:~$ sudo systemctl enable --now nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nfs-kernel-server
pladmin@plabubuntu:~$ sudo systemctl enable --now rpcbind
Synchronizing state of rpcbind.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable rpcbind
pladmin@plabubuntu:~$ sudoedit /etc/exports

```

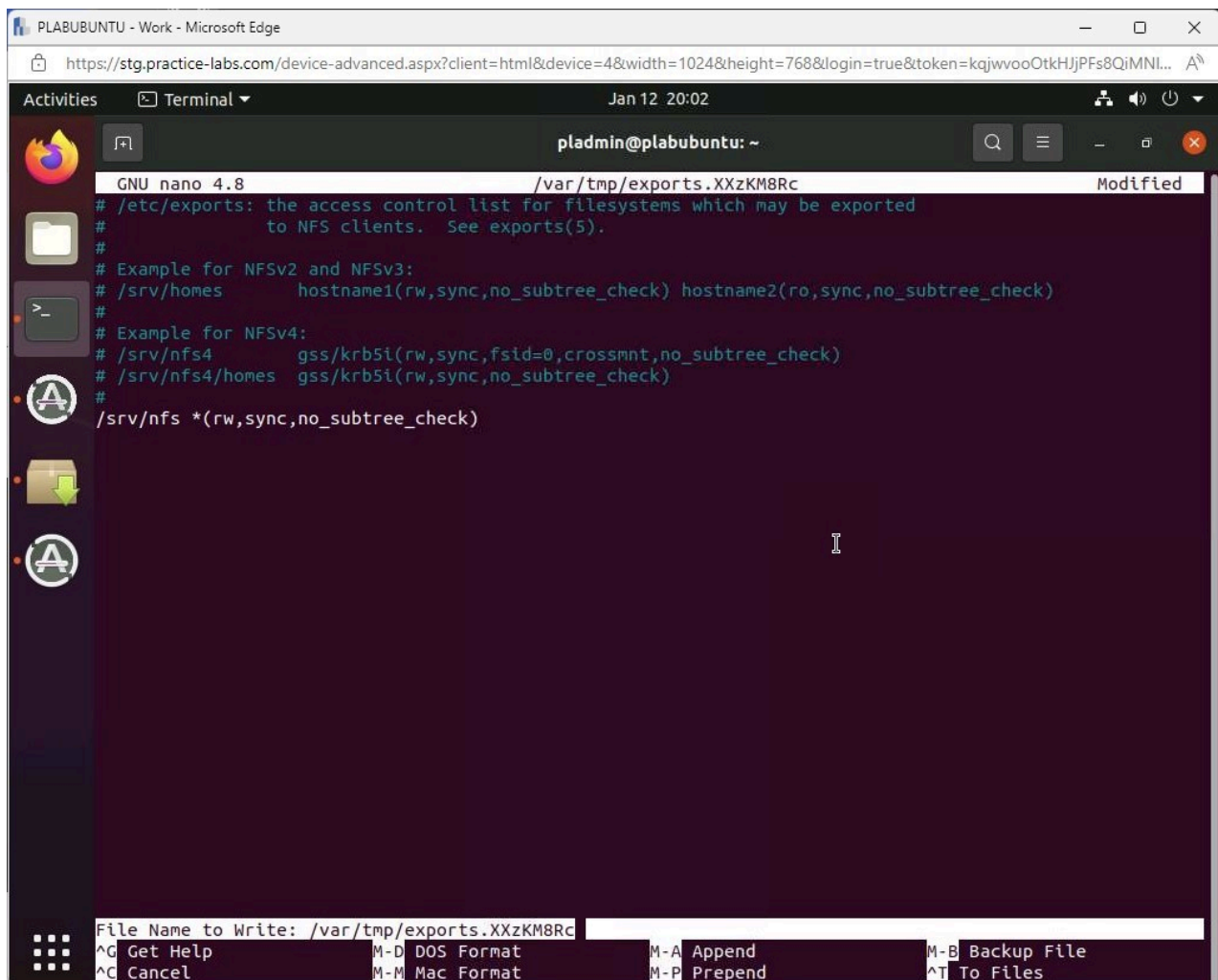
Figure 2.7 Screenshot of PLABUBUNTU: Displaying executing the command in the terminal window and pressing enter.

Step 6

Move the cursor to the end of the **/etc/exports** file (after the #) and type the following:

```
/srv/nfs *(rw,sync,no_subtree_check)
```

Press **Ctrl+O** and then press **Enter**.



```
GNU nano 4.8 /var/tmp/exports.XXzKM8Rc Modified
# /etc/exports: the access control list for filesystems which may be exported
# to NFS clients. See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4 gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes gss/krb5i(rw,sync,no_subtree_check)
#
/srv/nfs *(rw,sync,no_subtree_check)
```

File Name to Write: /var/tmp/exports.XXzKM8Rc

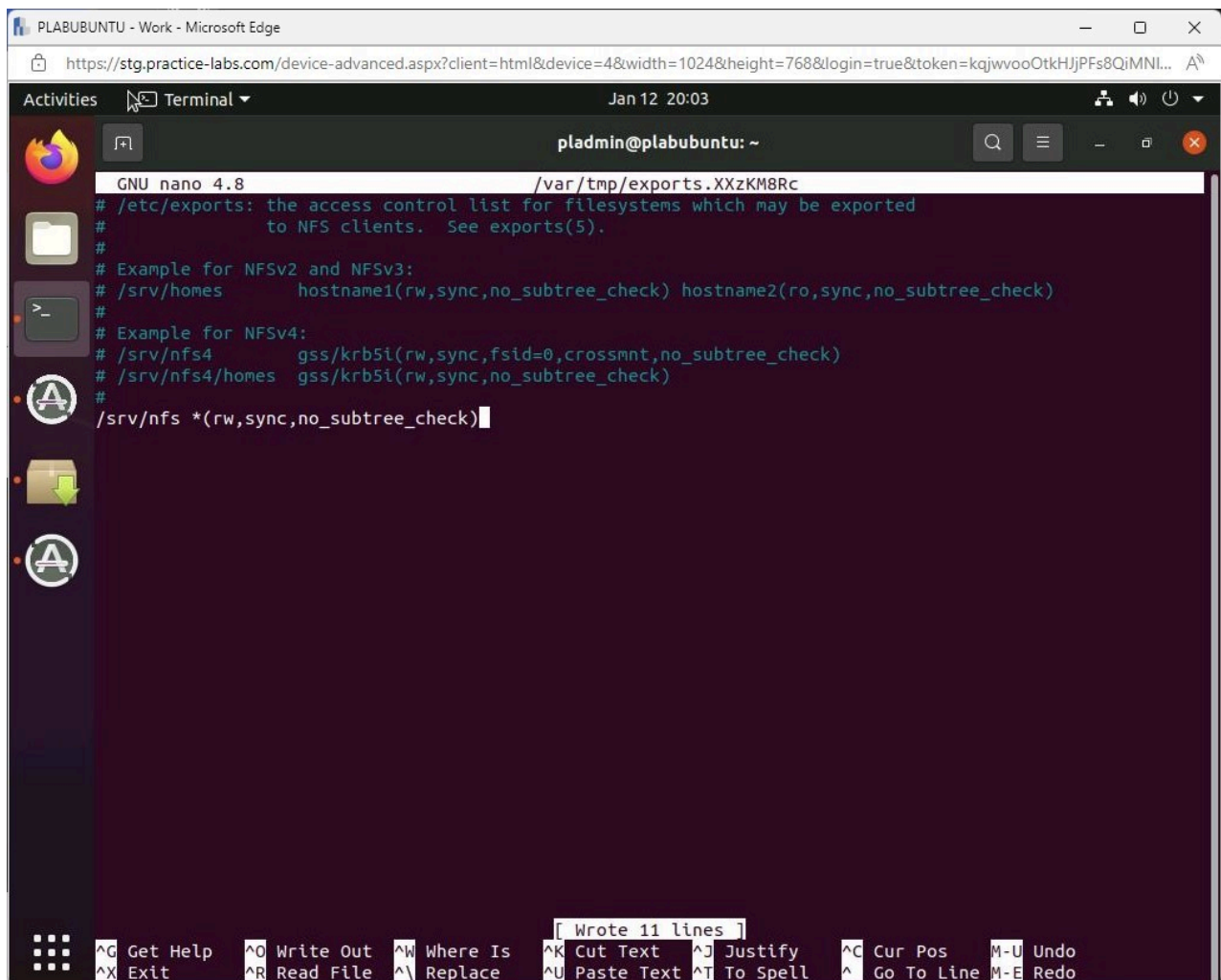
^G Get Help M-D DOS Format M-A Append M-B Backup File
^C Cancel M-M Mac Format M-P Prepend ^T To Files

Figure 2.8 Screenshot of PLABUBUNTU: Displaying entering the required text in the nano editor.

Note: By editing the `/etc/exports` file, the recently created partition that was mounted to the directory will be mounted persistent when the device starts up.

Step 7

Press the **Ctrl+X** keys to exit the **nano** editor.



```
PLABUBUNTU - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=kqjwvooOtkHJjPFs8QiMNL...

Activities Terminal Jan 12 20:03
pladmin@plabubuntu: ~
GNU nano 4.8 /var/tmp/exports.XXzKM8Rc
# /etc/exports: the access control list for filesystems which may be exported
# to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4 gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes gss/krb5i(rw,sync,no_subtree_check)
#
/srv/nfs *(rw,sync,no_subtree_check)

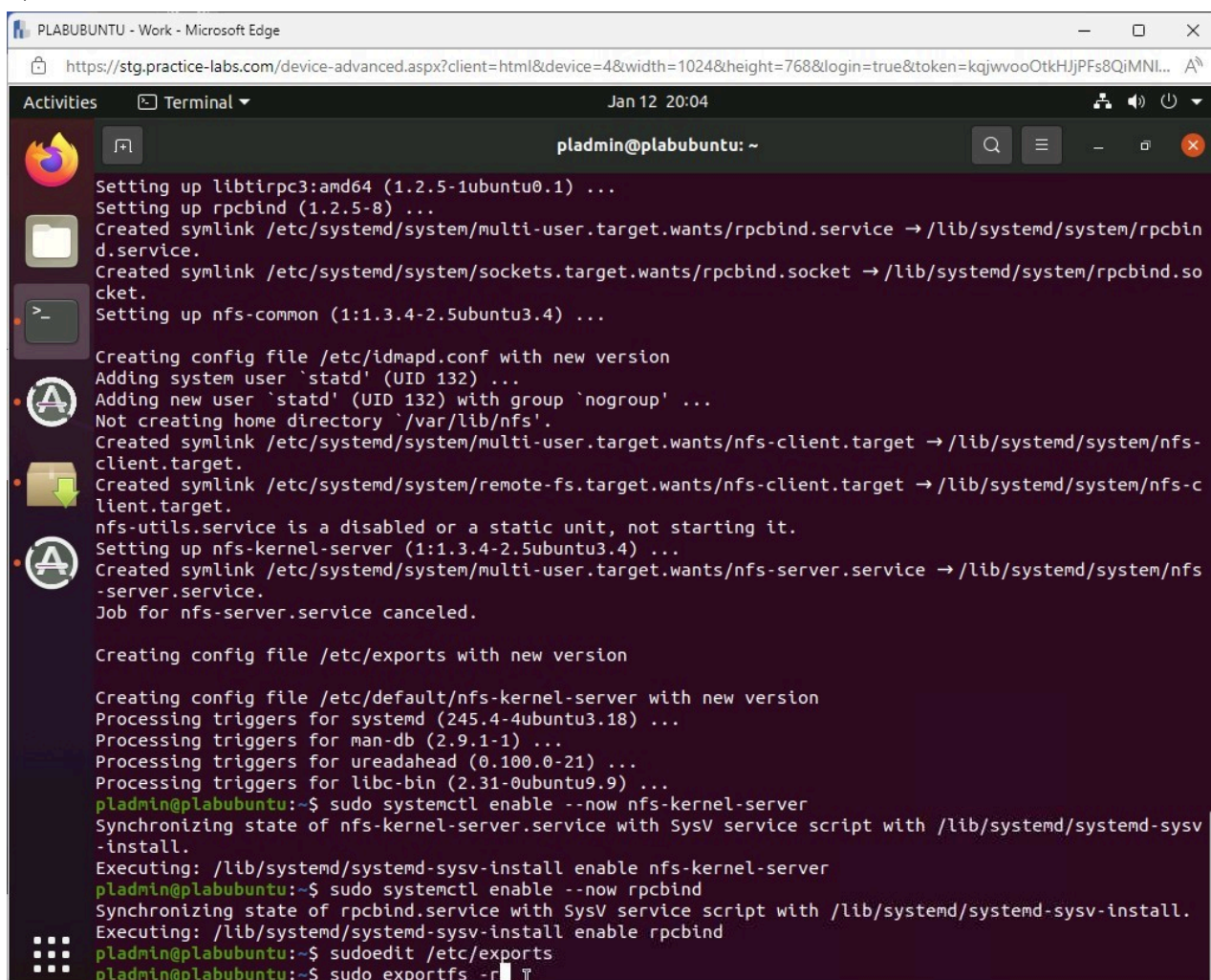
[ Wrote 11 lines ]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line M-E Redo
```

Figure 2.9 Screenshot of PLABUBUNTU: Displaying pressing the required keys to exit the nano editor.

Step 8

In the **Terminal** window, type the following command and press **Enter**:

```
sudo exportfs -r
```



```
PLABUBUNTU - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=kqjwvooOtkHjPFs8QiMNL...

Activities Terminal Jan 12 20:04 pladmin@plabubuntu: ~

Setting up libtirpc3:amd64 (1.2.5-1ubuntu0.1) ...
Setting up rpcbind (1.2.5-8) ...
Created symlink /etc/systemd/system/multi-user.target.wants/rpcbind.service → /lib/systemd/system/rpcbind.service.
Created symlink /etc/systemd/system/sockets.target.wants/rpcbind.socket → /lib/systemd/system/rpcbind.socket.
Setting up nfs-common (1:1.3.4-2.5ubuntu3.4) ...

Creating config file /etc/idmapd.conf with new version
Adding system user `statd' (UID 132) ...
Adding new user `statd' (UID 132) with group `nogroup' ...
Not creating home directory `/var/lib/nfs'.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
Created symlink /etc/systemd/system/remote-fs.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
nfs-utils.service is a disabled or a static unit, not starting it.
Setting up nfs-kernel-server (1:1.3.4-2.5ubuntu3.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /lib/systemd/system/nfs-server.service.
Job for nfs-server.service canceled.

Creating config file /etc/exports with new version

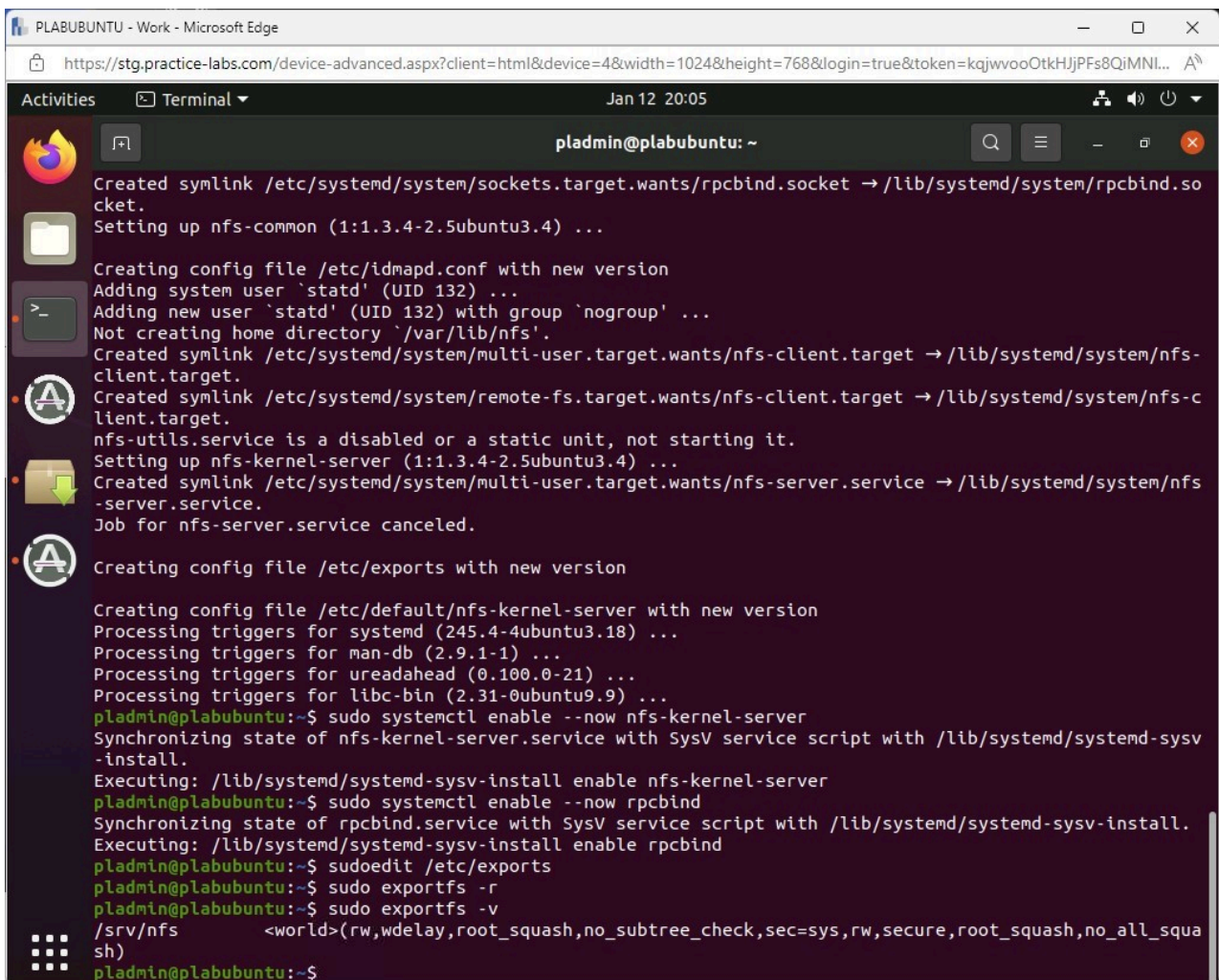
Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for systemd (245.4-4ubuntu3.18) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ureadahead (0.100.0-21) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
pladmin@plabubuntu:~$ sudo systemctl enable --now nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nfs-kernel-server
pladmin@plabubuntu:~$ sudo systemctl enable --now rpcbind
Synchronizing state of rpcbind.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable rpcbind
pladmin@plabubuntu:~$ sudoedit /etc/exports
pladmin@plabubuntu:~$ sudo exportfs -v
```

Figure 2.10 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Step 9

Next, type the following command and press **Enter**:

```
sudo exportfs -v
```

The screenshot shows a terminal window titled "PLABUBUNTU - Work - Microsoft Edge" with the URL <https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=kqjwvooOtkHjPPFs8QiMNL...>. The terminal output shows the following commands and their results:

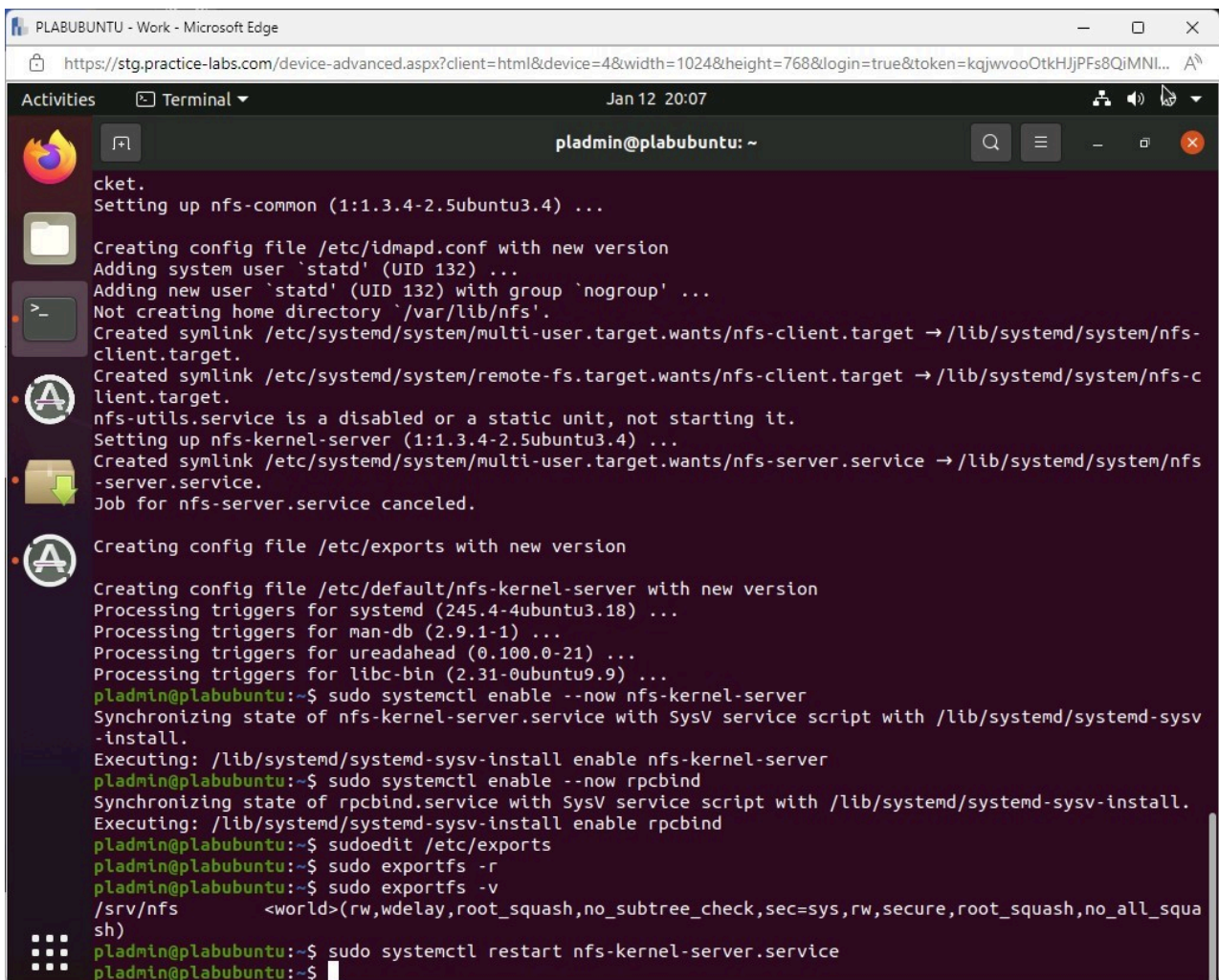
```
Created symlink /etc/systemd/system/sockets.target.wants/rpcbind.socket → /lib/systemd/system/rpcbind.socket.
Setting up nfs-common (1:1.3.4-2.5ubuntu3.4) ...
Creating config file /etc/idmapd.conf with new version
Adding system user `statd' (UID 132) ...
Adding new user `statd' (UID 132) with group `nogroup' ...
Not creating home directory `/var/lib/nfs'.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
Created symlink /etc/systemd/system/remote-fs.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
nfs-utils.service is a disabled or a static unit, not starting it.
Setting up nfs-kernel-server (1:1.3.4-2.5ubuntu3.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /lib/systemd/system/nfs-server.service.
Job for nfs-server.service canceled.
Creating config file /etc/exports with new version
Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for systemd (245.4-4ubuntu3.18) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ureadahead (0.100.0-21) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
pladmin@plabubuntu:~$ sudo systemctl enable --now nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nfs-kernel-server
pladmin@plabubuntu:~$ sudo systemctl enable --now rpcbind
Synchronizing state of rpcbind.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable rpcbind
pladmin@plabubuntu:~$ sudoedit /etc/exports
pladmin@plabubuntu:~$ sudo exportfs -r
pladmin@plabubuntu:~$ sudo exportfs -v
/srv/nfs          <world>(rw,wdelay,root_squash,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squash)
pladmin@plabubuntu:~$
```

Figure 2.11 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Step 10

Type the following command and press **Enter**:

```
sudo systemctl restart nfs-kernel-server.service
```



```
PLABUBUNTU - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=kqjwvooOtkHjPFs8QiMNI...

Activities Terminal Jan 12 20:07 pladmin@plabubuntu: ~

cknet.
Setting up nfs-common (1:1.3.4-2.5ubuntu3.4) ...
Creating config file /etc/idmapd.conf with new version
Adding system user `statd' (UID 132) ...
Adding new user `statd' (UID 132) with group `nogroup' ...
Not creating home directory `/var/lib/nfs'.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
Created symlink /etc/systemd/system/remote-fs.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
nfs-utils.service is a disabled or a static unit, not starting it.
Setting up nfs-kernel-server (1:1.3.4-2.5ubuntu3.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /lib/systemd/system/nfs-server.service.
Job for nfs-server.service canceled.

Creating config file /etc/exports with new version
Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for systemd (245.4-4ubuntu3.18) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ureadahead (0.100.0-21) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
pladmin@plabubuntu:~$ sudo systemctl enable --now nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nfs-kernel-server
pladmin@plabubuntu:~$ sudo systemctl enable --now rpcbind
Synchronizing state of rpcbind.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable rpcbind
pladmin@plabubuntu:~$ sudoedit /etc/exports
pladmin@plabubuntu:~$ sudo exportfs -r
pladmin@plabubuntu:~$ sudo exportfs -v
/srv/nfs <world>(rw,wdelay,root_squash,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squash)
pladmin@plabubuntu:~$ sudo systemctl restart nfs-kernel-server.service
pladmin@plabubuntu:~$
```

Figure 2.12 Screenshot of PLABUBUNTU: Displaying executing the required command in the terminal window and pressing enter.

Task 3 - Install and Configure an NFS Client

In this task, you will install and configure an NFS Client.

Step 1

Connect to **PLABALMA**.

Select **Activities** and then select **Terminal**.

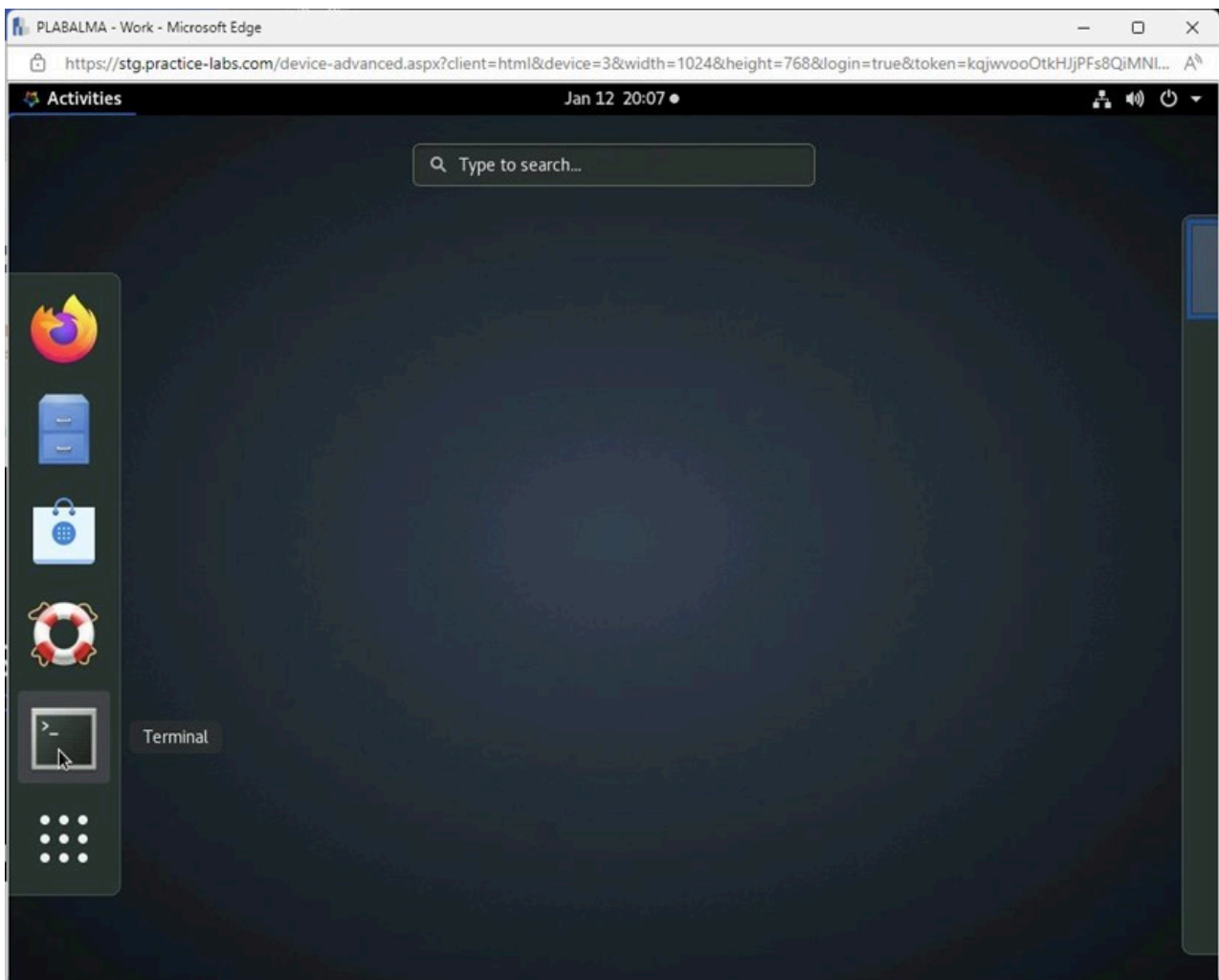


Figure 2.13 Screenshot of PLABALMA: Displaying selecting Activities and clicking the Terminal icon.

Step 2

In the **Terminal** window, type the following command to install an NFS client:

```
sudo dnf install nfs-utils
```

Press **Enter**.

When prompted, enter the following password and press **Enter**:

Passw0rd

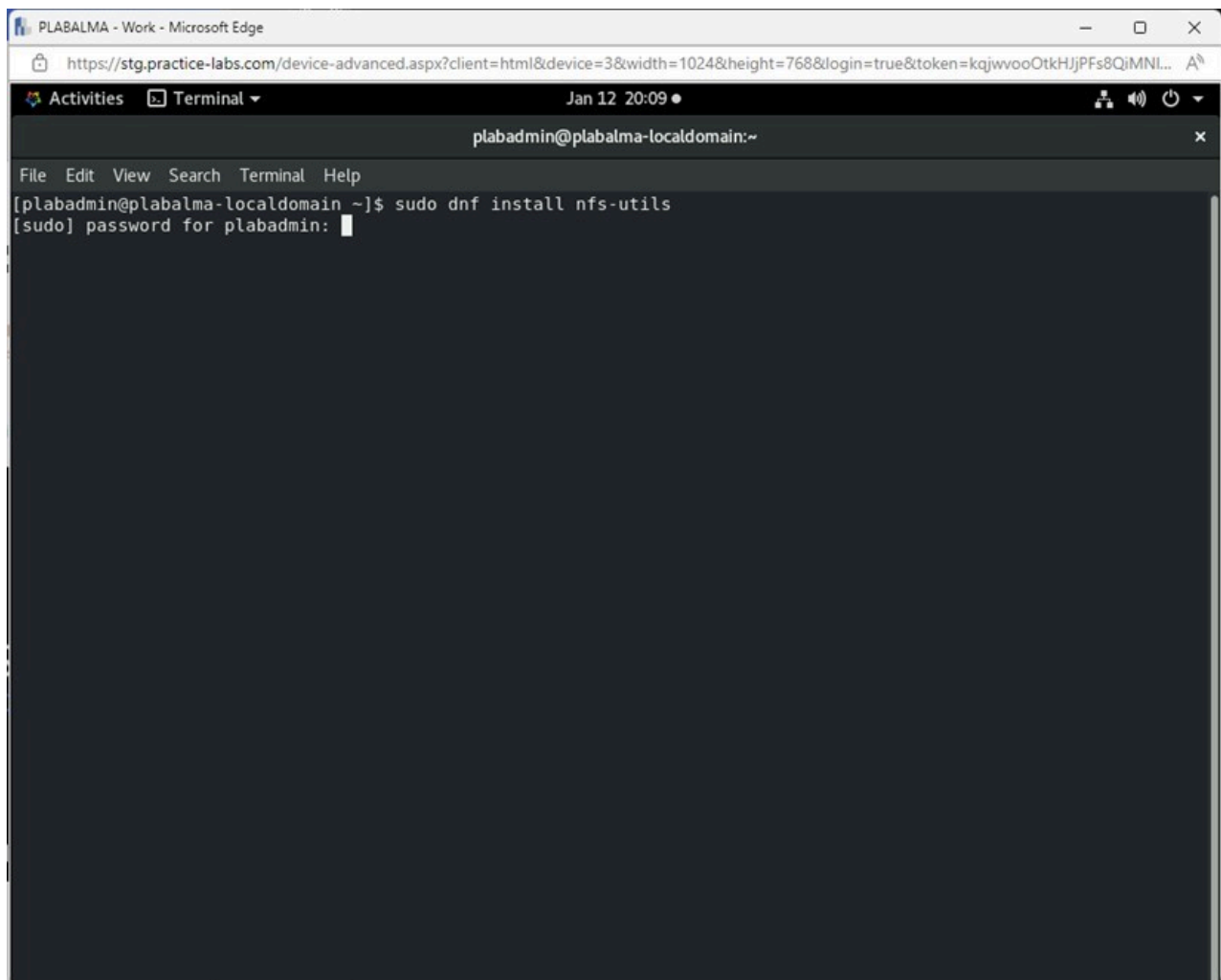
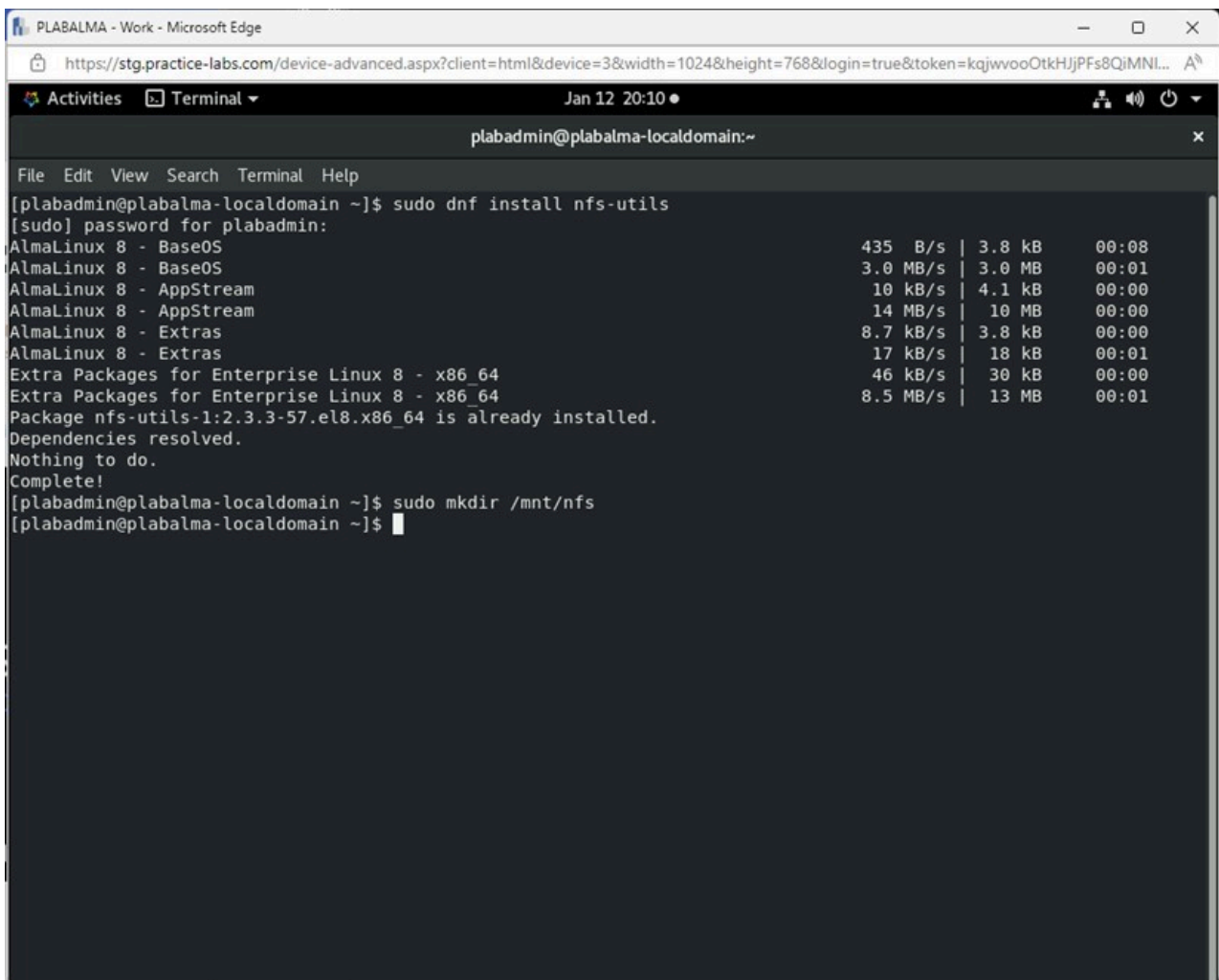


Figure 2.14 Screenshot of PLABALMA: Displaying entering the required command in the terminal window and pressing enter.

Step 3

Next, type the following command and press **Enter**:

```
sudo mkdir /mnt/nfs
```



The screenshot shows a terminal window titled "plabadmin@plabalma-localdomain:~" within a Microsoft Edge browser. The terminal displays the command `sudo dnf install nfs-utils` and its output, which includes a progress bar for installing various packages and their dependencies. The output shows that the package `nfs-utils-1:2.3.3-57.el8.x86_64` is already installed. The terminal also shows the command `sudo mkdir /mnt/nfs` being executed.

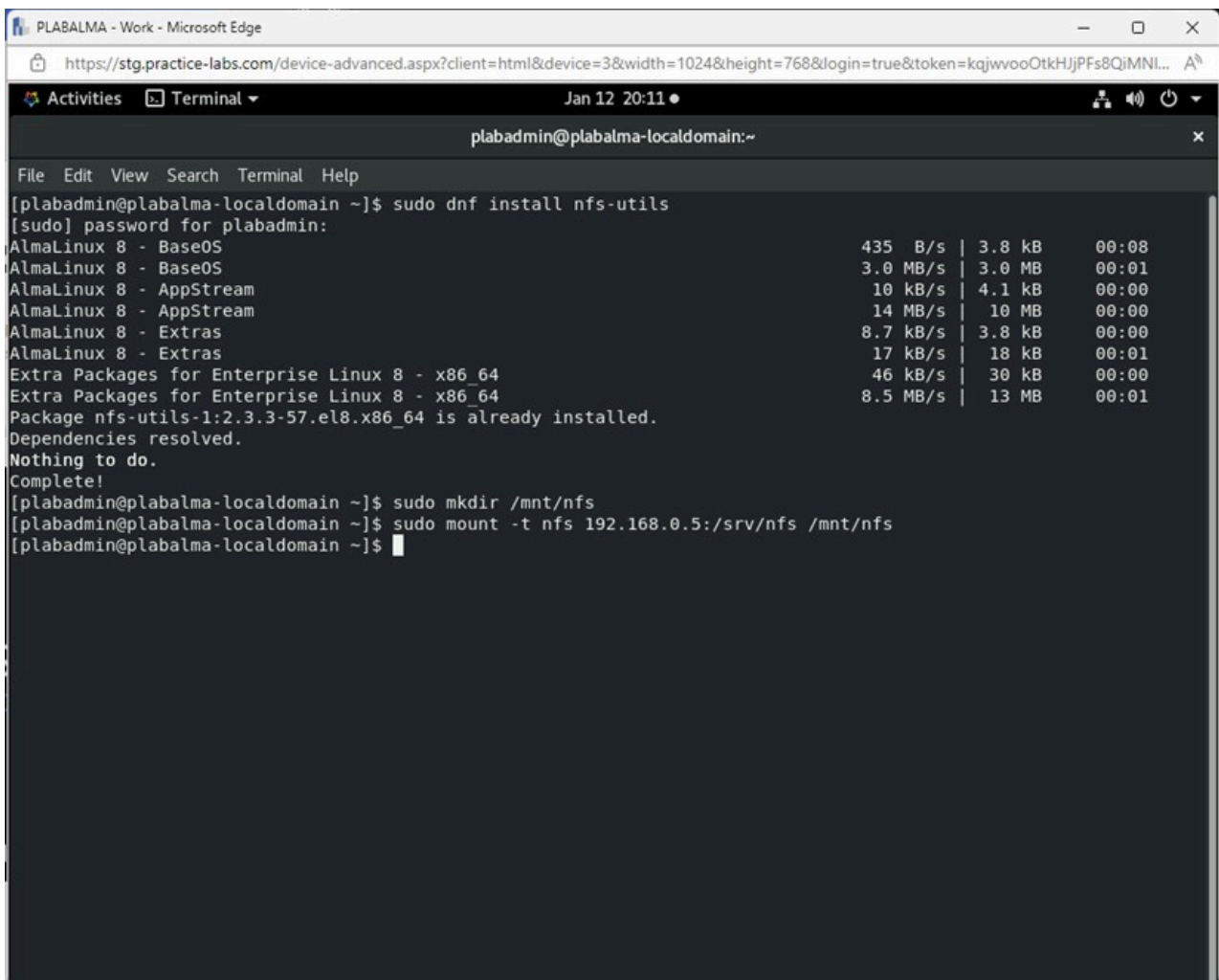
```
[plabadmin@plabalma-localdomain ~]$ sudo dnf install nfs-utils
[sudo] password for plabadmin:
AlmaLinux 8 - BaseOS                               435 B/s | 3.8 kB    00:08
AlmaLinux 8 - BaseOS                               3.0 MB/s | 3.0 MB    00:01
AlmaLinux 8 - AppStream                             10 kB/s | 4.1 kB    00:00
AlmaLinux 8 - AppStream                             14 MB/s | 10 MB    00:00
AlmaLinux 8 - Extras                                8.7 kB/s | 3.8 kB    00:00
AlmaLinux 8 - Extras                                17 kB/s | 18 kB    00:01
Extra Packages for Enterprise Linux 8 - x86_64      46 kB/s | 30 kB    00:00
Extra Packages for Enterprise Linux 8 - x86_64      8.5 MB/s | 13 MB    00:01
Package nfs-utils-1:2.3.3-57.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[plabadmin@plabalma-localdomain ~]$ sudo mkdir /mnt/nfs
[plabadmin@plabalma-localdomain ~]$
```

Figure 2.15 Screenshot of PLABALMA: Displaying entering the required command in the terminal window and pressing enter.

Step 4

Type the following command and press **Enter**:

```
sudo mount -t nfs 192.168.0.5:/srv/nfs /mnt/nfs
```

The screenshot shows a terminal window titled "plabadmin@plabalma-localdomain:~" within a Microsoft Edge browser. The terminal displays the command `sudo dnf install nfs-utils` and its output, which includes a progress bar for installing various packages. The output shows that `nfs-utils-1:2.3.3-57.el8.x86_64` is already installed. The terminal also shows the commands `sudo mkdir /mnt/nfs` and `sudo mount -t nfs 192.168.0.5:/srv/nfs /mnt/nfs`.

```
plabadmin@plabalma-localdomain:~$ sudo dnf install nfs-utils
[sudo] password for plabadmin:
AlmaLinux 8 - BaseOS                               435 B/s | 3.8 kB    00:08
AlmaLinux 8 - BaseOS                               3.0 MB/s | 3.0 MB    00:01
AlmaLinux 8 - AppStream                             10 kB/s | 4.1 kB    00:00
AlmaLinux 8 - AppStream                             14 MB/s | 10 MB    00:00
AlmaLinux 8 - Extras                                8.7 kB/s | 3.8 kB    00:00
AlmaLinux 8 - Extras                                17 kB/s | 18 kB    00:01
Extra Packages for Enterprise Linux 8 - x86_64      46 kB/s | 30 kB    00:00
Extra Packages for Enterprise Linux 8 - x86_64      8.5 MB/s | 13 MB    00:01
Package nfs-utils-1:2.3.3-57.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[plabadmin@plabalma-localdomain ~]$ sudo mkdir /mnt/nfs
[plabadmin@plabalma-localdomain ~]$ sudo mount -t nfs 192.168.0.5:/srv/nfs /mnt/nfs
[plabadmin@plabalma-localdomain ~]$
```

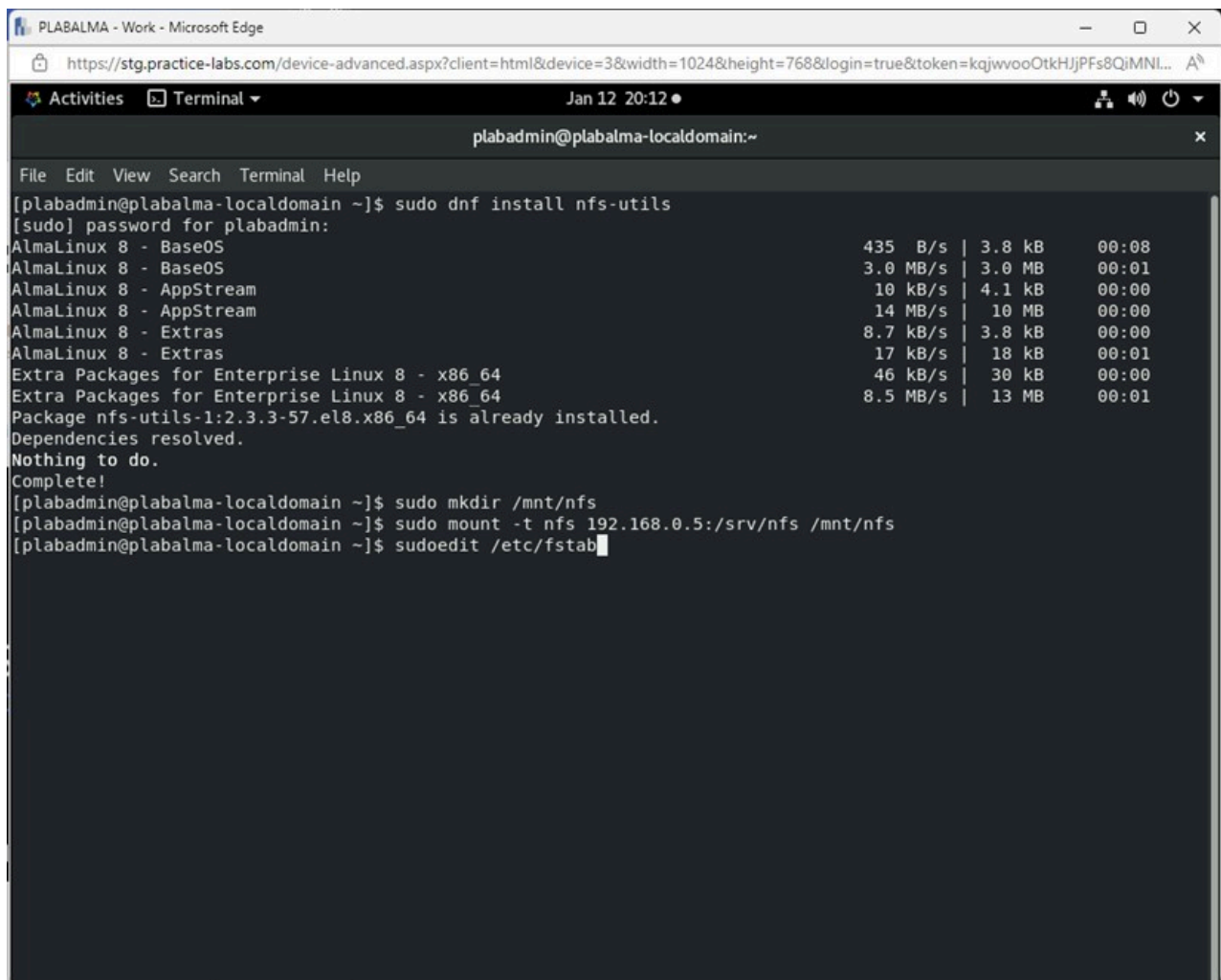
Figure 2.16 Screenshot of PLABALMA: Displaying entering the required command in the terminal window and pressing enter.

Note: The **mount** command is used to mount the remote partition to the host that was created on the **PLABUBUNTU** device.

Step 5

Type the following command and press **Enter**:

```
sudoedit /etc/fstab
```



The screenshot shows a terminal window titled "plabadmin@plabalma-localdomain:~" within a Microsoft Edge browser. The terminal displays the command `sudo dnf install nfs-utils` and its output, which includes a progress bar for installing various packages. The output shows that `nfs-utils-1:2.3.3-57.el8.x86_64` is already installed. The terminal then shows the command `sudo mkdir /mnt/nfs`, followed by `sudo mount -t nfs 192.168.0.5:/srv/nfs /mnt/nfs`, and finally `sudoedit /etc/fstab`.

```
File Edit View Search Terminal Help
[plabadmin@plabalma-localdomain ~]$ sudo dnf install nfs-utils
[sudo] password for plabadmin:
AlmaLinux 8 - BaseOS                               435 B/s | 3.8 kB    00:08
AlmaLinux 8 - BaseOS                               3.0 MB/s | 3.0 MB    00:01
AlmaLinux 8 - AppStream                             10 kB/s | 4.1 kB    00:00
AlmaLinux 8 - AppStream                             14 MB/s | 10 MB    00:00
AlmaLinux 8 - Extras                                8.7 kB/s | 3.8 kB    00:00
AlmaLinux 8 - Extras                                17 kB/s | 18 kB    00:01
Extra Packages for Enterprise Linux 8 - x86_64       46 kB/s | 30 kB    00:00
Extra Packages for Enterprise Linux 8 - x86_64       8.5 MB/s | 13 MB    00:01
Package nfs-utils-1:2.3.3-57.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[plabadmin@plabalma-localdomain ~]$ sudo mkdir /mnt/nfs
[plabadmin@plabalma-localdomain ~]$ sudo mount -t nfs 192.168.0.5:/srv/nfs /mnt/nfs
[plabadmin@plabalma-localdomain ~]$ sudoedit /etc/fstab
```

Figure 2.17 Screenshot of PLABALMA: Displaying entering the required command in the terminal window and pressing enter.

Step 6

Press **i** to edit the file.

Move the cursor to the end of the file, press **Enter** and add the below line:

```
192.168.0.5:/srv/nfs/ /mnt/nfs nfs defaults rw,sync
0 0
```

```

#
# /etc/fstab
# Created by anaconda on Fri Jun 10 11:38:26 2022
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/almalinux-root /                    xfs     defaults        0 0
UUID=dfc1eacf-e8df-42b9-a9ab-9b28e5139340 /boot      xfs     defaults        0 0
UUID=D2D6-2E20 /boot/efi  vfat    umask=0077,shortname=winnt 0 2
/dev/mapper/almalinux-swap none        swap    defaults        0 0
192.168.05:/srv/nfs/ /mnt/nfs  nfs     defaults        rw,sync 0 0

```

Figure 2.18 Screenshot of PLABALMA: Displaying entering the required line to the end of the file.

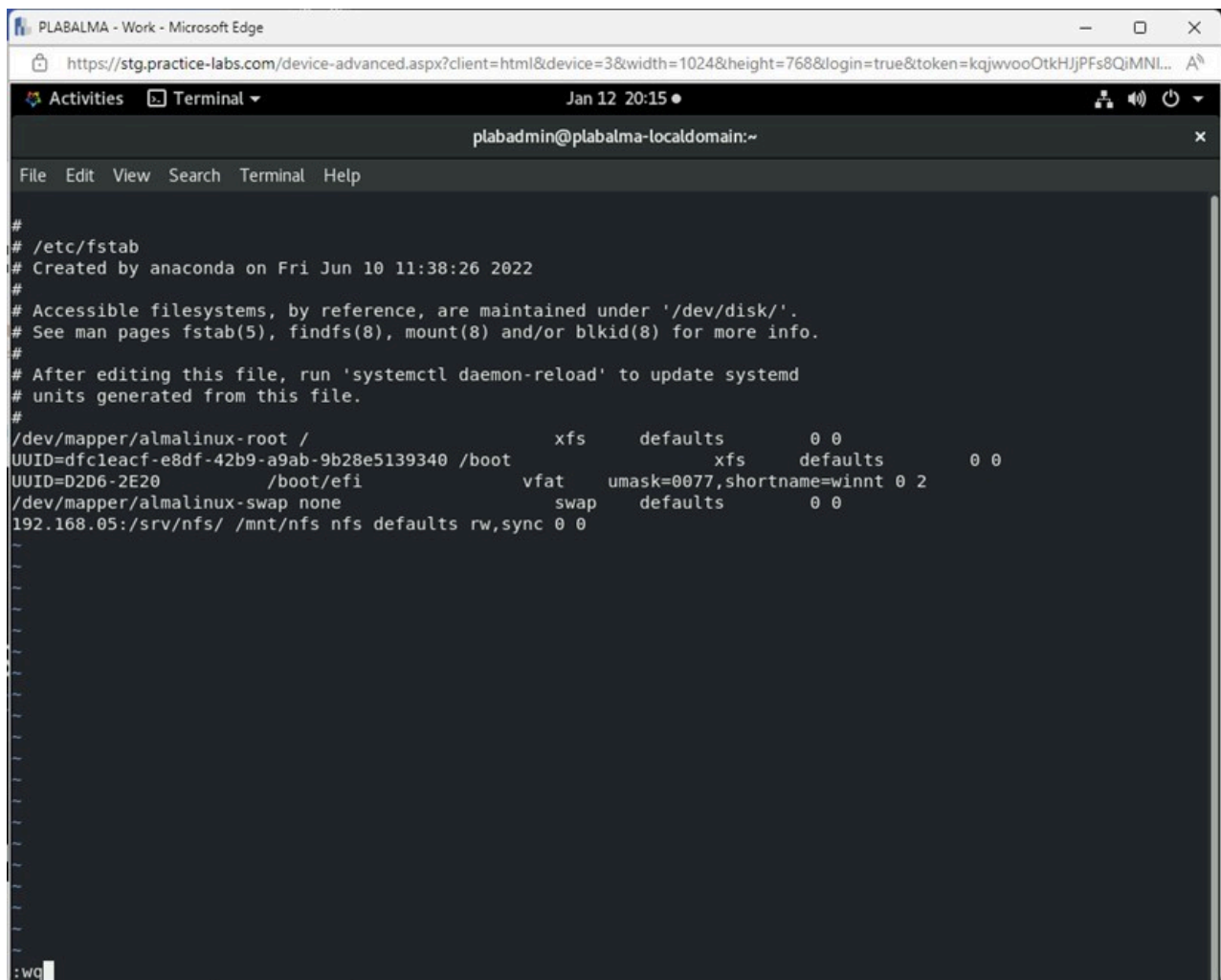
Note: Editing the *fstab* file will ensure the shared remote file system will be persistently mounted to the device.

Step 7

Press the **ESC** key.

Next, type the following and press **Enter**:

```
:wq
```



The screenshot shows a web browser window titled "PLABALMA - Work - Microsoft Edge" with the URL <https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=kqjwvooOtkHjPFs8QiMNL...>. Below the browser window is a terminal window titled "plabadmin@plabalma-localhost:~". The terminal displays the contents of the `/etc/fstab` file, which includes comments and mount entries for various filesystems.

```
#  
# /etc/fstab  
# Created by anaconda on Fri Jun 10 11:38:26 2022  
#  
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.  
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.  
#  
# After editing this file, run 'systemctl daemon-reload' to update systemd  
# units generated from this file.  
#  
/dev/mapper/almalinux-root / xfs defaults 0 0  
UUID=dfc1eacf-e8df-42b9-a9ab-9b28e5139340 /boot xfs defaults 0 0  
UUID=D2D6-2E20 /boot/efi vfat umask=0077,shortname=winnt 0 2  
/dev/mapper/almalinux-swap none swap defaults 0 0  
192.168.05:/srv/nfs/ /mnt/nfs nfs defaults rw,sync 0 0
```

Figure 2.19 Screenshot of PLABALMA: Displaying entering the required command in the terminal window and pressing enter.

Step 8

Test the mount by typing the following command:

```
df -h
```

Press **Enter**.

```

PLABALMA - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=kqjwvoOtkHjPFs8QiMNL...
Activities Terminal Jan 12 20:16
plabadmin@plabalma-localdomain:~
File Edit View Search Terminal Help
[plabadmin@plabalma-localdomain ~]$ sudo dnf install nfs-utils
[sudo] password for plabadmin:
AlmaLinux 8 - BaseOS 435 B/s | 3.8 kB 00:08
AlmaLinux 8 - BaseOS 3.0 MB/s | 3.0 MB 00:01
AlmaLinux 8 - AppStream 10 kB/s | 4.1 kB 00:00
AlmaLinux 8 - AppStream 14 MB/s | 10 MB 00:00
AlmaLinux 8 - Extras 8.7 kB/s | 3.8 kB 00:00
AlmaLinux 8 - Extras 17 kB/s | 18 kB 00:01
Extra Packages for Enterprise Linux 8 - x86_64 46 kB/s | 30 kB 00:00
Extra Packages for Enterprise Linux 8 - x86_64 8.5 MB/s | 13 MB 00:01
Package nfs-utils-1:2.3.3-57.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[plabadmin@plabalma-localdomain ~]$ sudo mkdir /mnt/nfs
[plabadmin@plabalma-localdomain ~]$ sudo mount -t nfs 192.168.0.5:/srv/nfs /mnt/nfs
[plabadmin@plabalma-localdomain ~]$ sudoedit /etc/fstab
[plabadmin@plabalma-localdomain ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        1.8G   0 1.8G   0% /dev
tmpfs           1.8G   0 1.8G   0% /dev/shm
tmpfs           1.8G 9.2M 1.8G   1% /run
tmpfs           1.8G   0 1.8G   0% /sys/fs/cgroup
/dev/mapper/almalinux-root 35G  5.5G  30G  16% /
/dev/sda2       1014M 335M  680M  33% /boot
/dev/sda1       599M  5.8M  594M   1% /boot/efi
tmpfs           368M  32K  368M   1% /run/user/1000
192.168.0.5:/srv/nfs  9.3G   0  8.8G   0% /mnt/nfs
[plabadmin@plabalma-localdomain ~]$

```

Figure 2.20 Screenshot of PLABALMA: Displaying entering the required command in the terminal window and pressing enter.

Note: The **df** command is used to display the different mounted partitions on the Alma linux device.

Step 9

Close the **Terminal** window.

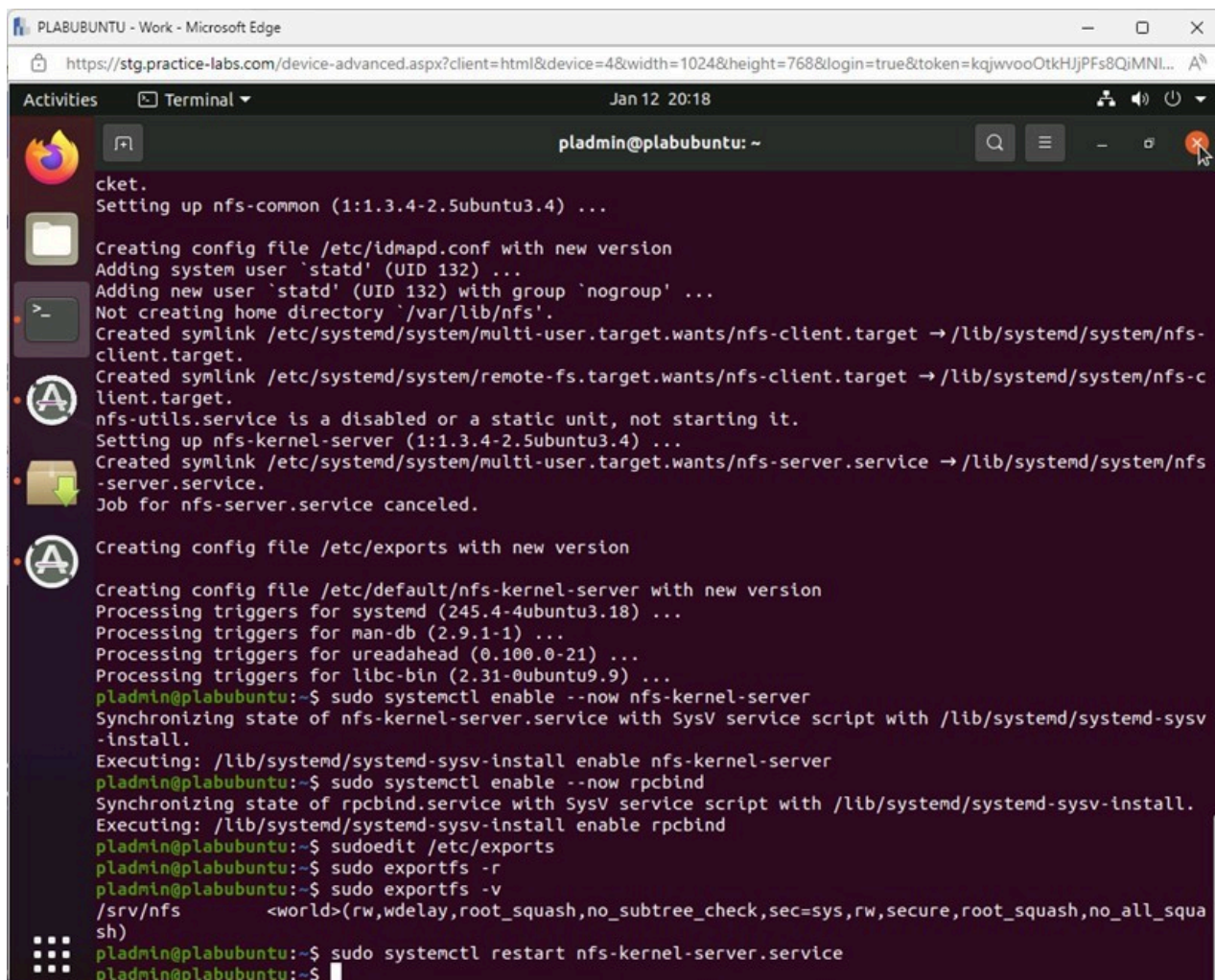
The screenshot shows a terminal window titled "PLABALMA - Work - Microsoft Edge" with the URL <https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=kqjwvoOtkHjPFs8QiMNL...>. The terminal prompt is `plabadmin@plabalma-localdomain:~`. The user runs `sudo dnf install nfs-utils`, which shows progress bars for various packages. After installation, the user runs `sudo mkdir /mnt/nfs`, `sudo mount -t nfs 192.168.0.5:/srv/nfs /mnt/nfs`, and `sudoedit /etc/fstab`. Finally, the user runs `df -h`, displaying the following output:

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	1.8G	0	1.8G	0%	/dev
tmpfs	1.8G	0	1.8G	0%	/dev/shm
tmpfs	1.8G	9.2M	1.8G	1%	/run
tmpfs	1.8G	0	1.8G	0%	/sys/fs/cgroup
/dev/mapper/almalinux-root	35G	5.5G	30G	16%	/
/dev/sda2	1014M	335M	680M	33%	/boot
/dev/sda1	599M	5.8M	594M	1%	/boot/efi
tmpfs	368M	32K	368M	1%	/run/user/1000
192.168.0.5:/srv/nfs	9.3G	0	8.8G	0%	/mnt/nfs

Figure 2.21 Screenshot of PLABALMA: Displaying closing the Terminal window.

Step 10

Connect to **PLABUBUNTU** and close the **Terminal** window.



The screenshot shows a terminal window titled 'PLABUBUNTU - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=4&width=1024&height=768&login=true&token=kqjwvooOtkHjPFs8QiMNL...'. The terminal output shows the following commands and their results:

```
ckct.
Setting up nfs-common (1:1.3.4-2.5ubuntu3.4) ...
Creating config file /etc/idmapd.conf with new version
Adding system user `statd' (UID 132) ...
Adding new user `statd' (UID 132) with group `nogroup' ...
Not creating home directory `/var/lib/nfs'.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
Created symlink /etc/systemd/system/remote-fs.target.wants/nfs-client.target → /lib/systemd/system/nfs-client.target.
nfs-utils.service is a disabled or a static unit, not starting it.
Setting up nfs-kernel-server (1:1.3.4-2.5ubuntu3.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /lib/systemd/system/nfs-server.service.
Job for nfs-server.service canceled.

Creating config file /etc/exports with new version
Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for systemd (245.4-4ubuntu3.18) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ureadahead (0.100.0-21) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
pladmin@plabubuntu:~$ sudo systemctl enable --now nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nfs-kernel-server
pladmin@plabubuntu:~$ sudo systemctl enable --now rpcbind
Synchronizing state of rpcbind.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable rpcbind
pladmin@plabubuntu:~$ sudoedit /etc/exports
pladmin@plabubuntu:~$ sudo exportfs -r
pladmin@plabubuntu:~$ sudo exportfs -v
/srv/nfs          <world>(rw,wdelay,root_squash,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squa
sh)
pladmin@plabubuntu:~$ sudo systemctl restart nfs-kernel-server.service
pladmin@plabubuntu:~$
```

Figure 2.22 Screenshot of PLABUBUNTU: Displaying closing the Terminal window.

Exercise 3 - Configuring Storage in Windows

Different methods can be used when configuring storage for Windows Operating systems, such as using the Command prompt (cmd) or the Graphical User Interface (GUI).

In this exercise, the Command Prompt will be used to configure storage on a Windows Operating system.

Learning Outcomes

After completing this exercise, you should be able to:

- Partition a Storage Device
- Format the Storage Device with the NTFS Filesystem

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)

Task 1 - Partition a Storage Device

In this task, you will partition a storage device using the Command Prompt.

Step 1

Connect to **PLABDM01**.

Minimize the **Server Manager** window.

Click the **Start** charm and type the following:

```
cmd
```

In the **Best match** pop-up menu, right-click on **Command Prompt** and select **Run as administrator**.

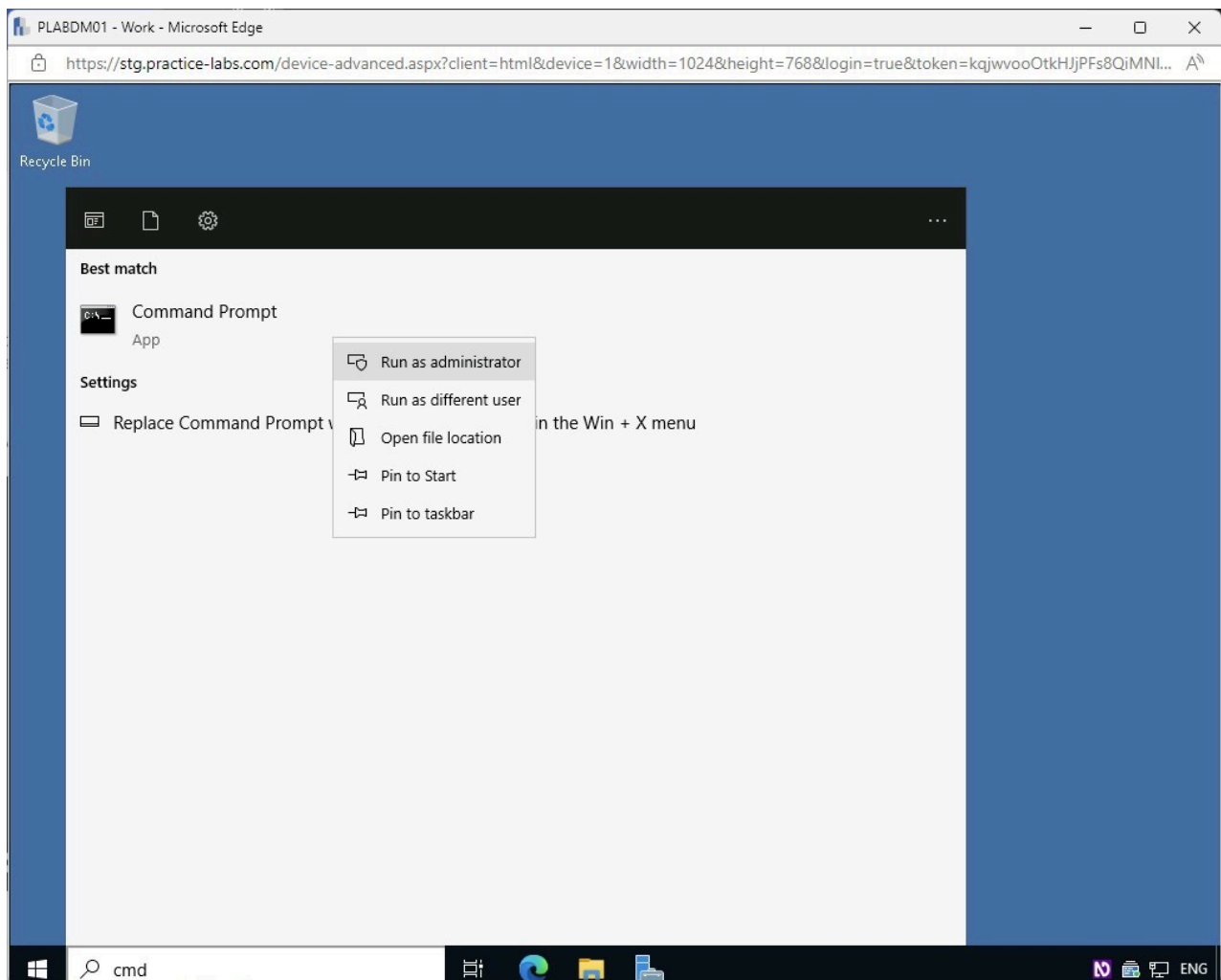


Figure 3.1 Screenshot of PLABDM01: Displaying clicking Start typing cmd, and selecting Run as administrator for Command Prompt.

Step 2

In the **Command Prompt** window, type the following command and press **Enter**:

```
diskpart
```

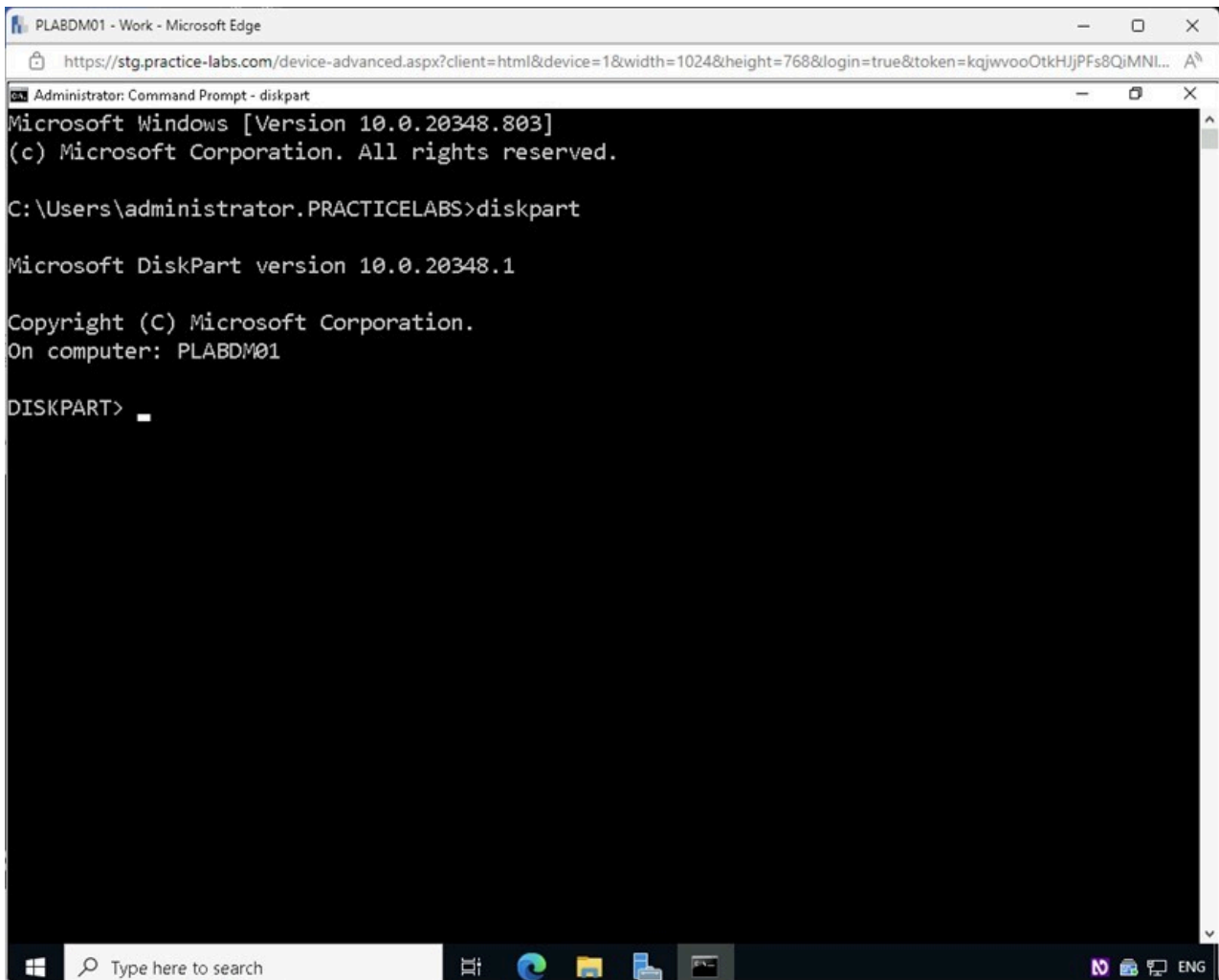


Figure 3.2 Screenshot of PLABDM01: Displaying typing the command in Command Prompt and pressing enter.

Note: Similar to a Linux operating system, the Command Prompt in Windows can be used to partition and format a Windows filing system. The **diskpart** command can be used to achieve this in the Command Prompt.

Step 3

In the **DISKPART** utility, type the following command and press **Enter**:

```
list disk
```



```
PLABDM01 - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=1&width=1024&height=768&login=true&token=kqjwvooOtkHjPPFs8QiMNL...
Select Administrator: Command Prompt - diskpart
Microsoft Windows [Version 10.0.20348.803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator.PRACTICELABS>diskpart

Microsoft DiskPart version 10.0.20348.1

Copyright (C) Microsoft Corporation.
On computer: PLABDM01

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online          80 GB     1024 KB
   Disk 1      Online         10 GB     1024 KB

DISKPART> _
```

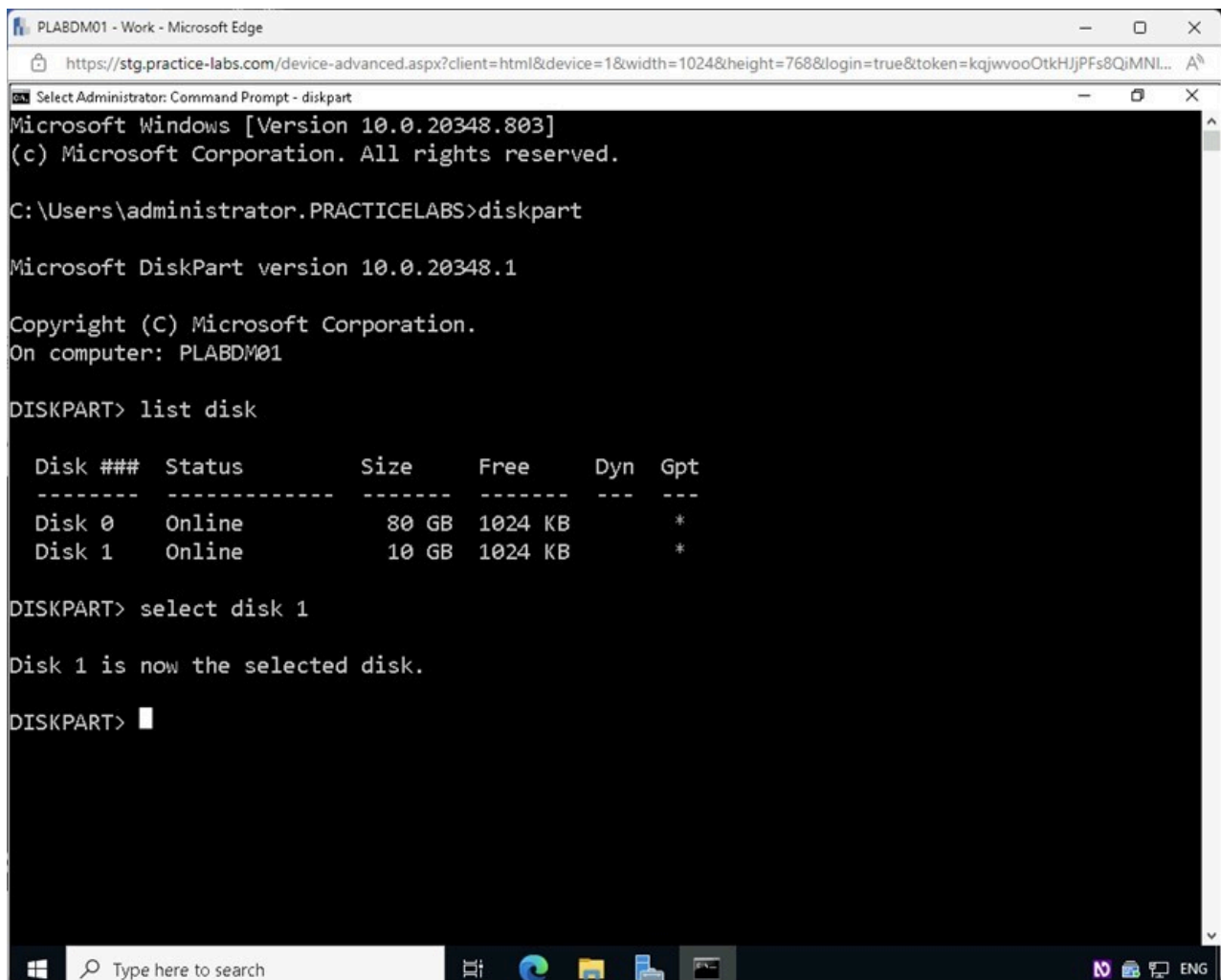
Figure 3.3 Screenshot of PLABDM01: Displaying typing the command in Command Prompt and pressing enter.

Note: Notice the **Disk 1** with the **Size of 10GB**.

Step 4

Next, type the following command and press **Enter**:

```
select disk 1
```



```
PLABDM01 - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=1&width=1024&height=768&login=true&token=kqjwvooOtkHjPFs8QiMNL...
Select Administrator: Command Prompt - diskpart
Microsoft Windows [Version 10.0.20348.803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator.PRACTICELABS>diskpart

Microsoft DiskPart version 10.0.20348.1

Copyright (C) Microsoft Corporation.
On computer: PLABDM01

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online          80 GB    1024 KB
   Disk 1      Online          10 GB    1024 KB

DISKPART> select disk 1

Disk 1 is now the selected disk.

DISKPART> █
```

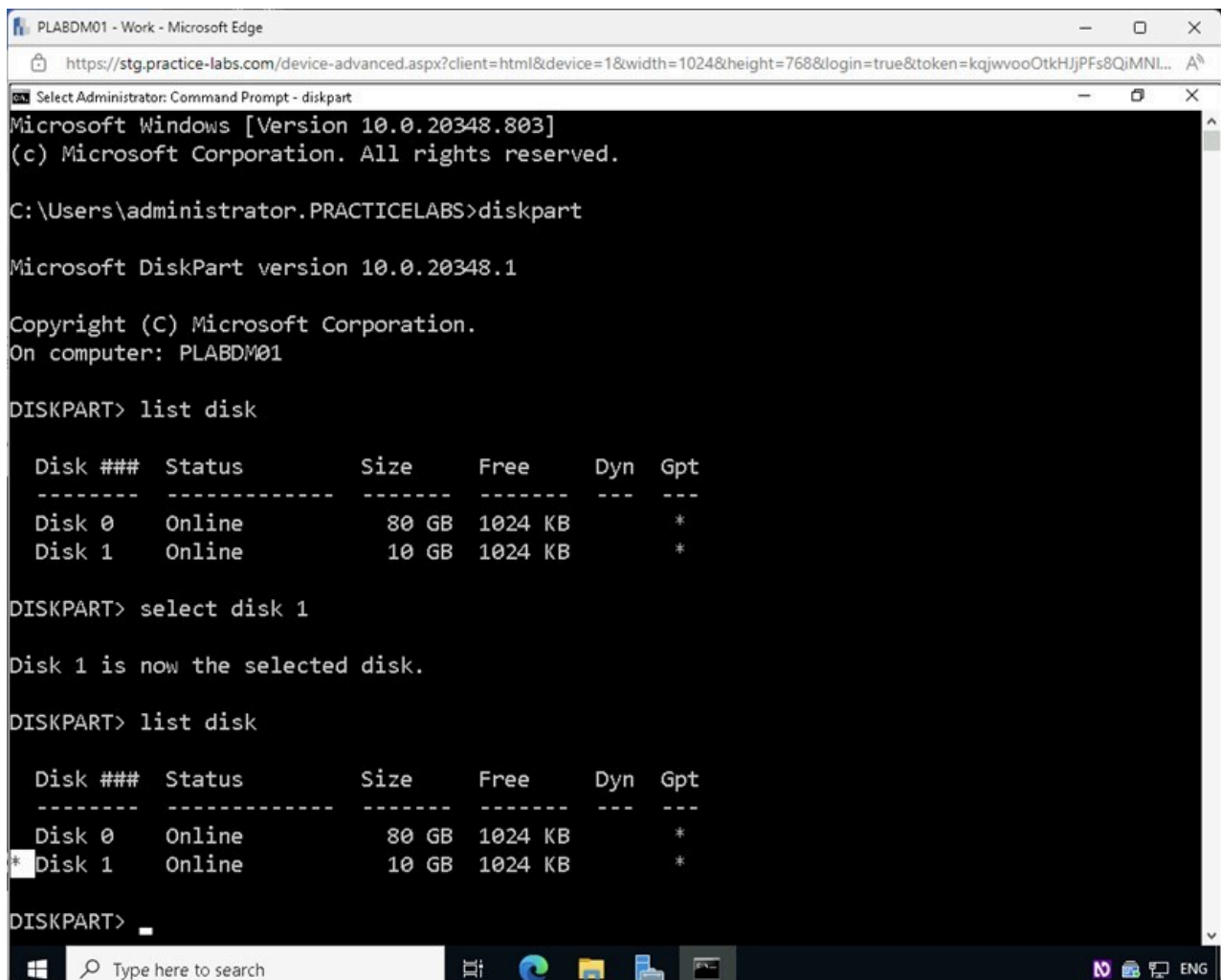
Figure 3.4 Screenshot of PLABDM01: Displaying typing the command in Command Prompt and pressing enter.

Note: *Disk 1* has a size of 10 GB. Hence, the number 1 is entered in the above command.

Step 5

Type the following command and press **Enter**:

```
list disk
```



```
PLABDM01 - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=1&width=1024&height=768&login=true&token=kqjwvooOtkHjPPFs8QiMNL...
Select Administrator: Command Prompt - diskpart
Microsoft Windows [Version 10.0.20348.803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator.PRACTICELABS>diskpart

Microsoft DiskPart version 10.0.20348.1

Copyright (C) Microsoft Corporation.
On computer: PLABDM01

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online          80 GB    1024 KB
   Disk 1      Online          10 GB    1024 KB

DISKPART> select disk 1

Disk 1 is now the selected disk.

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online          80 GB    1024 KB
   * Disk 1      Online          10 GB    1024 KB

DISKPART> 
```

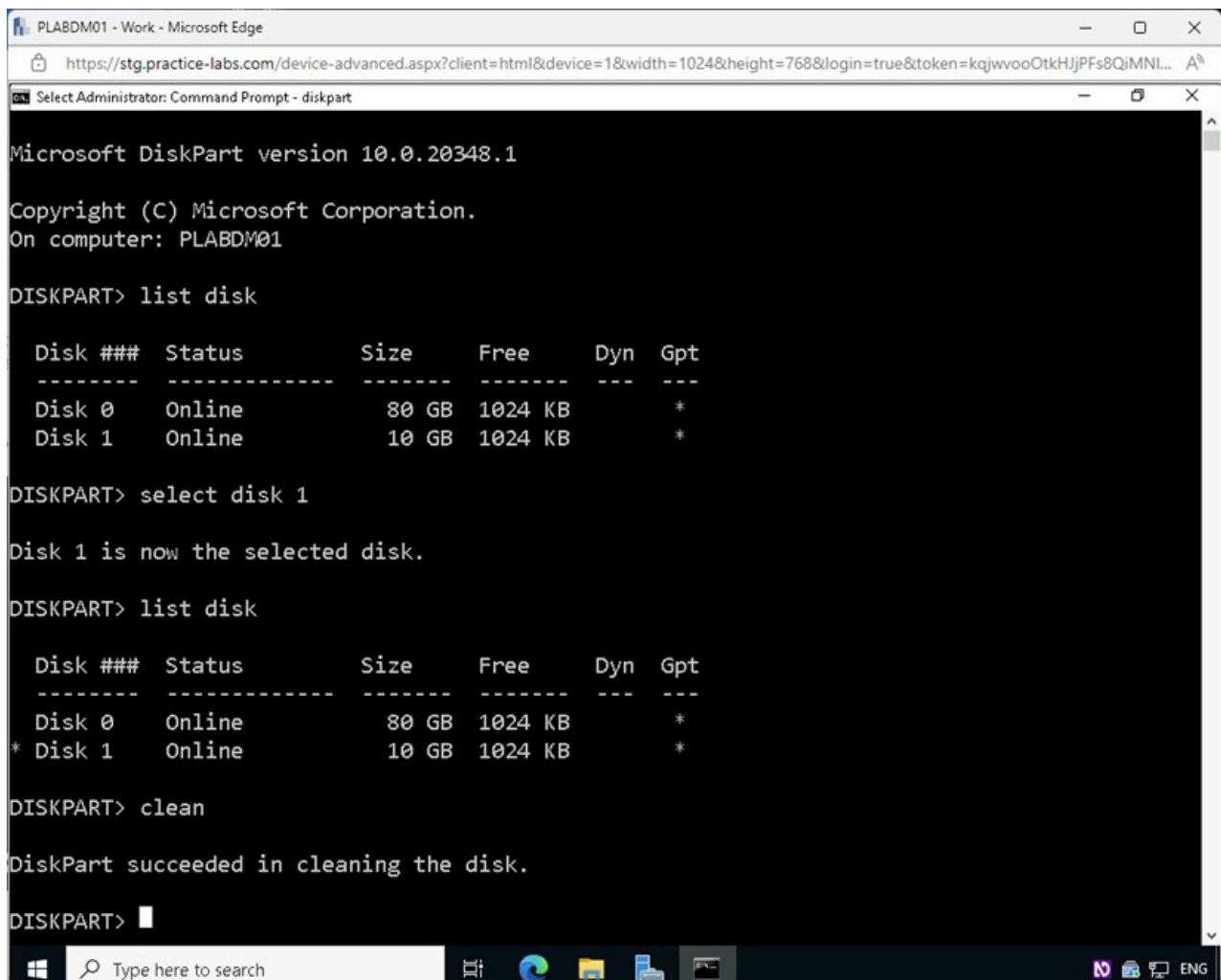
Figure 3.5 Screenshot of PLABDM01: Displaying typing the command in Command Prompt and pressing enter.

Note: Notice **Disk 1** has an asterisk (*) next to it.

Step 6

Type the following command and press **Enter**:

clean



```
Microsoft DiskPart version 10.0.20348.1

Copyright (C) Microsoft Corporation.
On computer: PLABDM01

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online          80 GB     1024 KB          *
   Disk 1      Online          10 GB     1024 KB          *

DISKPART> select disk 1

Disk 1 is now the selected disk.

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online          80 GB     1024 KB          *
* Disk 1      Online          10 GB     1024 KB          *

DISKPART> clean

DiskPart succeeded in cleaning the disk.

DISKPART> █
```

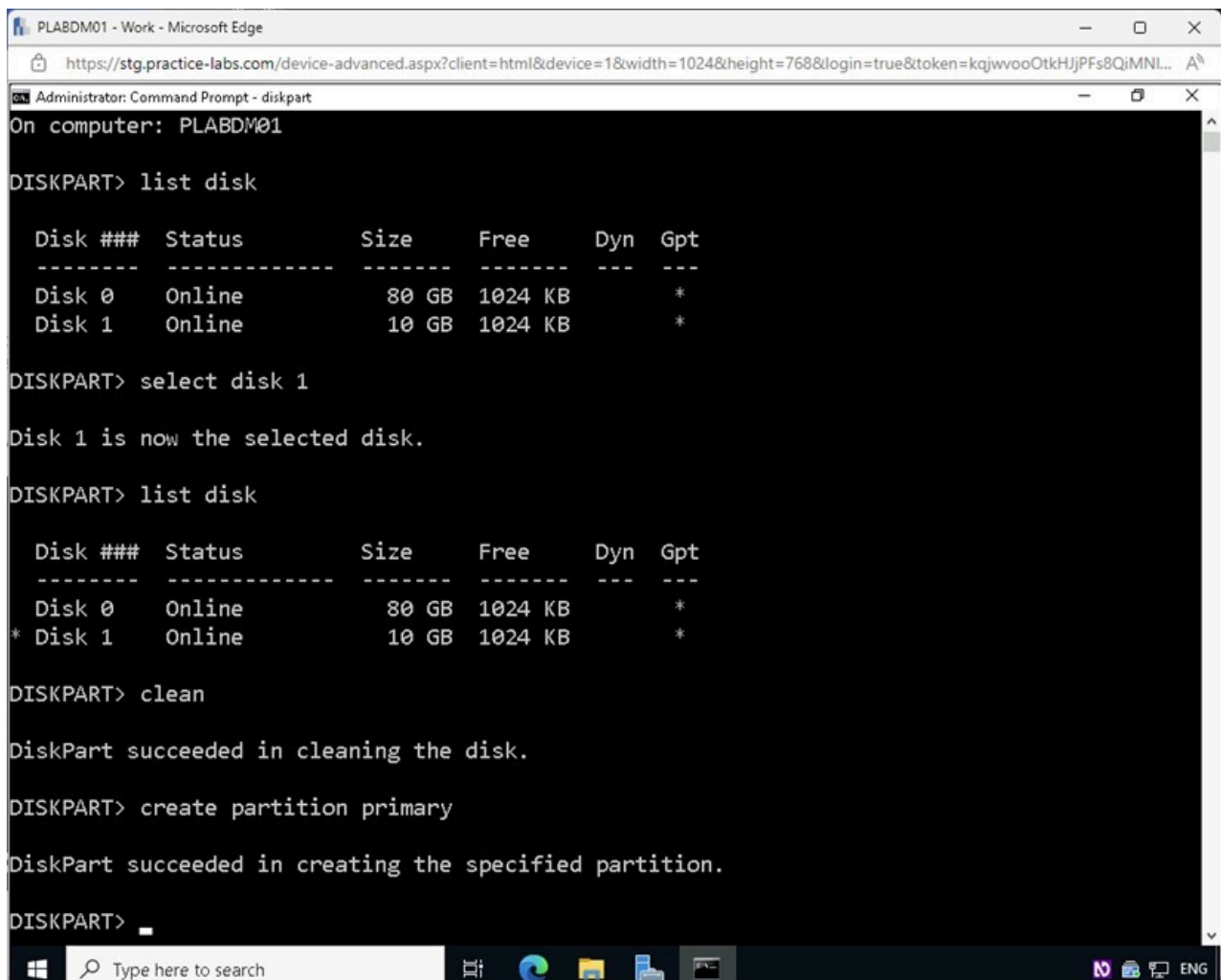
Figure 3.6 Screenshot of PLABDM01: Displaying typing the command in Command Prompt and pressing enter.

Step 7

Type the following command to create a primary partition:

```
create partition primary
```

Press **Enter**.



```
On computer: PLABDM01

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online            80 GB    1024 KB
   Disk 1      Online            10 GB    1024 KB

DISKPART> select disk 1

Disk 1 is now the selected disk.

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online            80 GB    1024 KB
* Disk 1      Online            10 GB    1024 KB

DISKPART> clean

DiskPart succeeded in cleaning the disk.

DISKPART> create partition primary

DiskPart succeeded in creating the specified partition.

DISKPART> _
```

Figure 3.7 Screenshot of PLABDM01: Displaying typing the command in Command Prompt and pressing enter.

Step 8

Type the following command to verify the new partition has been created:

```
list partition
```

Press **Enter**.


```

Administrator: Command Prompt - diskpart
Disk 0      Online      80 GB   1024 KB   *
Disk 1      Online      10 GB   1024 KB   *

DISKPART> select disk 1

Disk 1 is now the selected disk.

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online        80 GB     1024 KB           *
*  Disk 1      Online        10 GB     1024 KB           *

DISKPART> clean

DiskPart succeeded in cleaning the disk.

DISKPART> create partition primary

DiskPart succeeded in creating the specified partition.

DISKPART> list partition

   Partition ###  Type              Size      Offset
   -----  -
*  Partition 1    Primary           9 GB     1024 KB

DISKPART>

```

Figure 3.8 Screenshot of PLABDM01: Displaying typing the command in Command Prompt and pressing enter.

Note: Notice an asterisk (*) next to the partition you just created, confirming that it is selected.

Keep the **Command Prompt** window open.

Task 2 - Format the Storage Device with the NTFS Filesystem

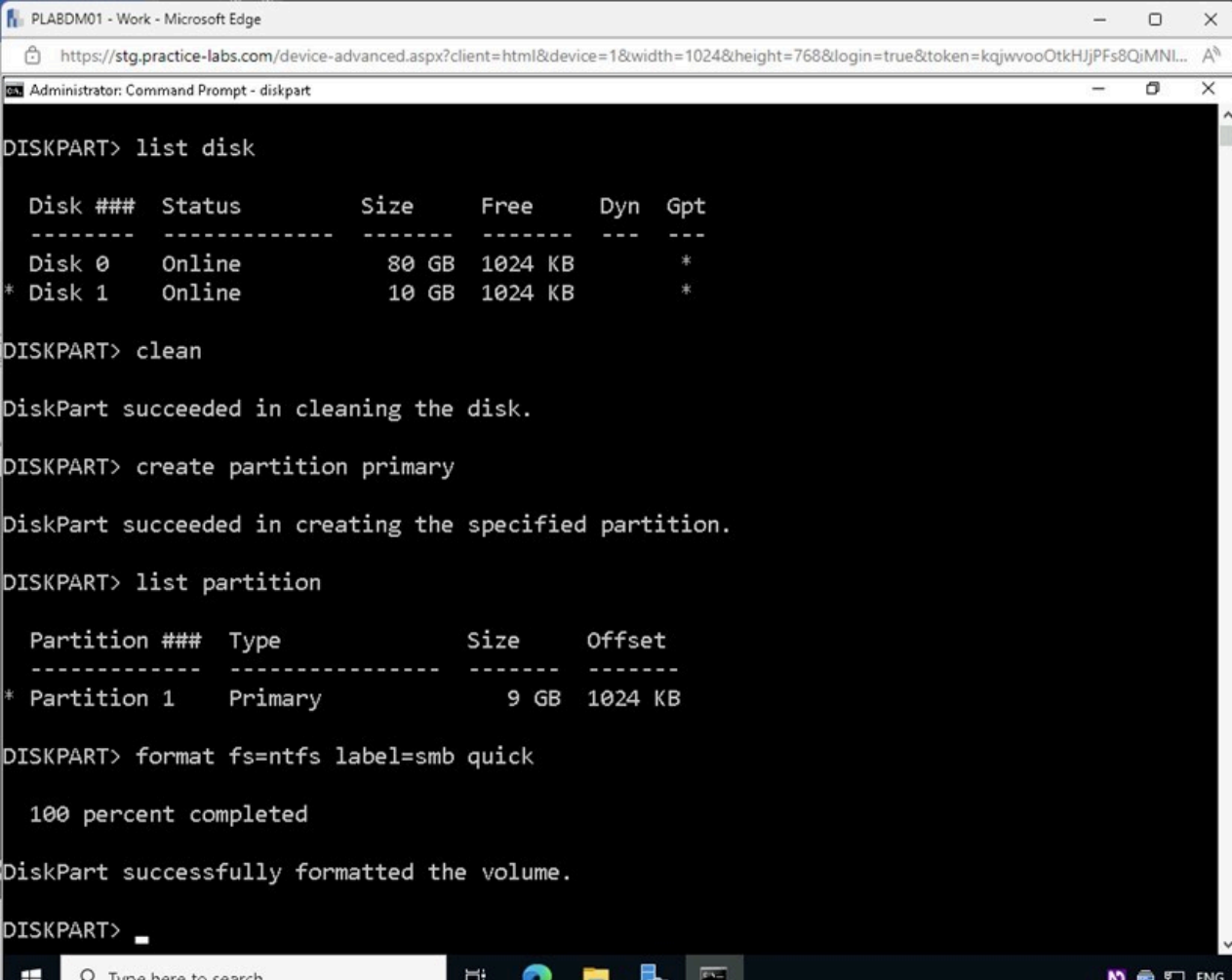
In this task, you will format the storage device with the NTFS filesystem using the command prompt.

Step 1

In **PLABDM01**, the **Command Prompt** window is open.

Type the following command and press **Enter**:

```
format fs=ntfs label=smb quick
```



The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt - diskpart". The window is open in a Microsoft Edge browser window titled "PLABDM01 - Work - Microsoft Edge". The address bar shows the URL "https://stg.practice-labs.com/device-advanced.aspx?client=html&device=1&width=1024&height=768&login=true&token=kqjwvooOtkHjPPFs8QiMNI...". The Command Prompt shows the following commands and output:

```
DISKPART> list disk

Disk ###  Status       Size       Free       Dyn  Gpt
-----  -
Disk 0    Online      30 GB      1024 KB
* Disk 1    Online      10 GB      1024 KB

DISKPART> clean

DiskPart succeeded in cleaning the disk.

DISKPART> create partition primary

DiskPart succeeded in creating the specified partition.

DISKPART> list partition

Partition ###  Type            Size       Offset
-----  -
* Partition 1   Primary         9 GB      1024 KB

DISKPART> format fs=ntfs label=smb quick

100 percent completed

DiskPart successfully formatted the volume.

DISKPART> _
```

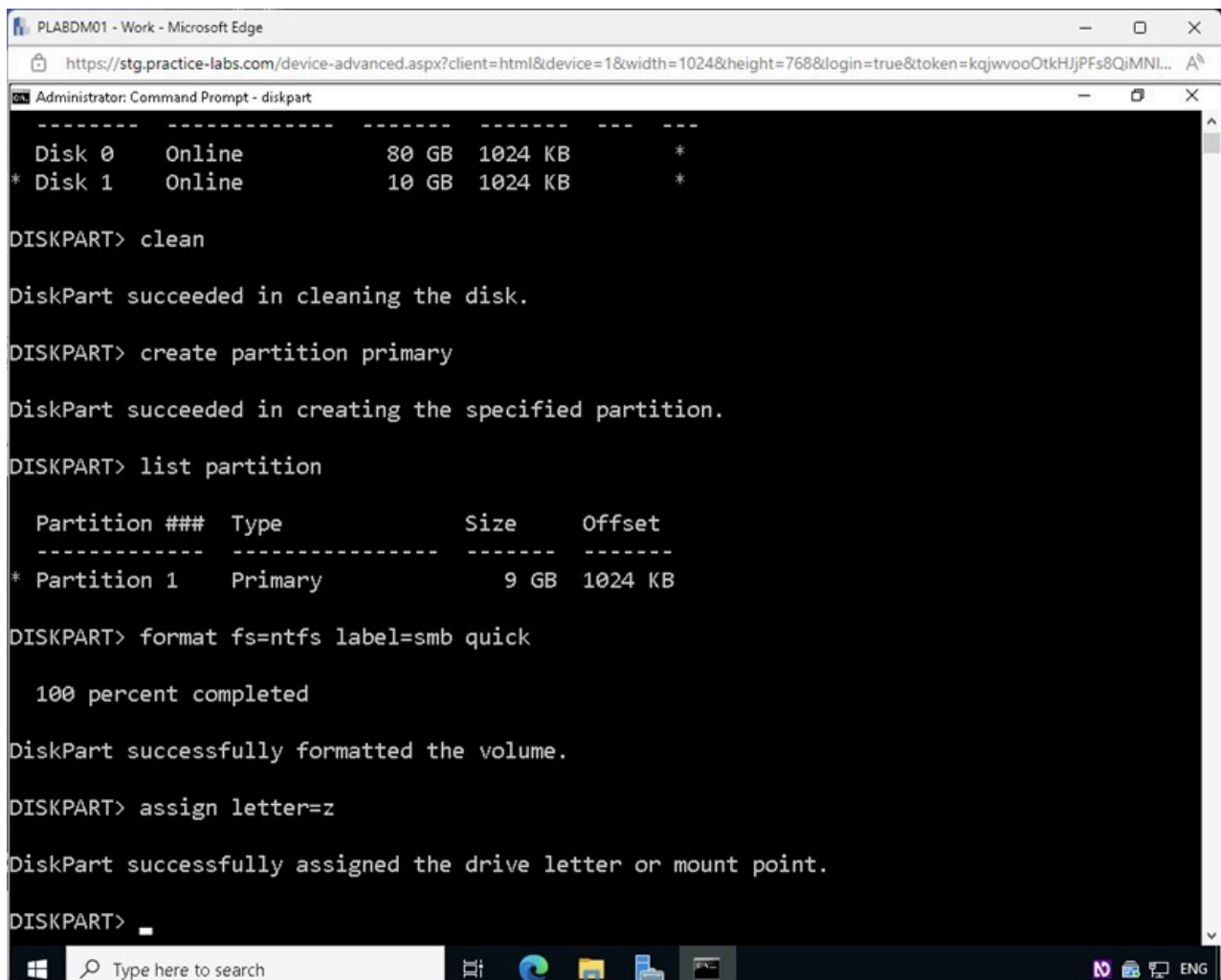
Figure 3.9 Screenshot of PLABDM01: Displaying typing the required command in Command Prompt and pressing enter.

Step 2

Type the following command to create a mount point for the volume:

```
assign letter=z
```

Press **Enter**.



```
-----
Disk 0      Online      80 GB  1024 KB  *
* Disk 1    Online      10 GB  1024 KB  *

DISKPART> clean

DiskPart succeeded in cleaning the disk.

DISKPART> create partition primary

DiskPart succeeded in creating the specified partition.

DISKPART> list partition

   Partition ###  Type              Size      Offset
   -----
* Partition 1     Primary          9 GB      1024 KB

DISKPART> format fs=ntfs label=smb quick

   100 percent completed

DiskPart successfully formatted the volume.

DISKPART> assign letter=z

DiskPart successfully assigned the drive letter or mount point.

DISKPART> _
```

Figure 3.10 Screenshot of PLABDM01: Displaying typing the command in Command Prompt and pressing enter.

Step 3

Confirm the creation of the mount point and filesystem by typing the following:

```
list volume
```

Press **Enter**.

```

DISKPART> list partition

   Partition ###   Type              Size      Offset
   -----
* Partition 1      Primary              9 GB    1024 KB

DISKPART> format fs=ntfs label=smb quick

    100 percent completed

DiskPart successfully formatted the volume.

DISKPART> assign letter=z

DiskPart successfully assigned the drive letter or mount point.

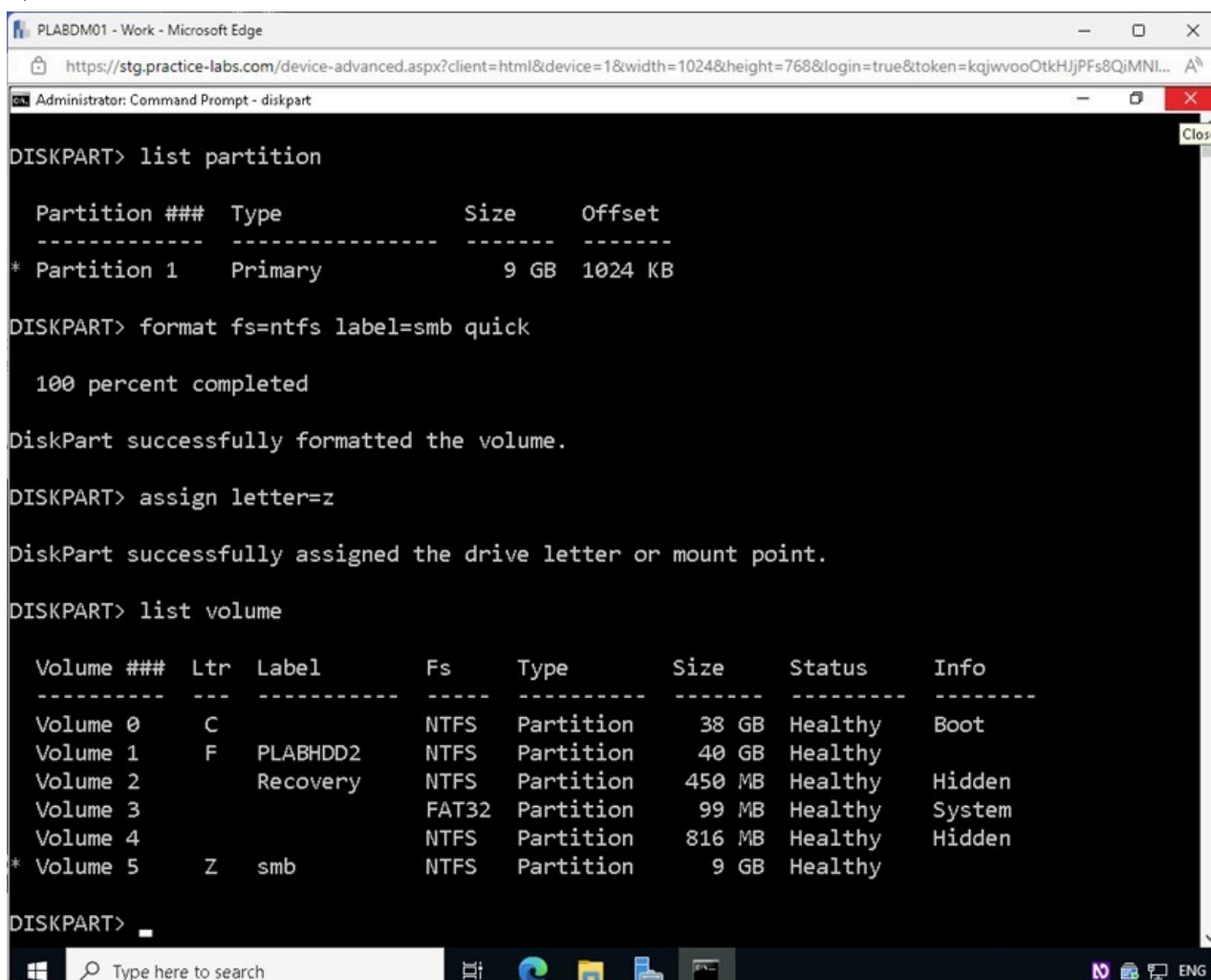
DISKPART> list volume

   Volume ###   Ltr   Label          Fs   Type          Size      Status      Info
   -----
Volume 0        C     Boot            NTFS Partition    38 GB    Healthy      Boot
Volume 1        F     PLABHDD2        NTFS Partition    40 GB    Healthy
Volume 2                Recovery        NTFS Partition    450 MB    Healthy      Hidden
Volume 3                System          FAT32 Partition    99 MB    Healthy      System
Volume 4                Hidden          NTFS Partition    816 MB    Healthy      Hidden
* Volume 5      Z     smb             NTFS Partition     9 GB    Healthy
  
```

Figure 3.11 Screenshot of PLABDM01: Displaying typing the command in Command Prompt and pressing enter.

Step 4

Close the **Command Prompt** window.



```

DISKPART> list partition

   Partition ###   Type              Size      Offset
   -----
* Partition 1      Primary              9 GB    1024 KB

DISKPART> format fs=ntfs label=smb quick

    100 percent completed

DiskPart successfully formatted the volume.

DISKPART> assign letter=z

DiskPart successfully assigned the drive letter or mount point.

DISKPART> list volume

   Volume ###   Ltr   Label          Fs   Type          Size      Status      Info
   -----
Volume 0        C          NTFS   Partition     38 GB   Healthy      Boot
Volume 1        F   PLABHDD2     NTFS   Partition     40 GB   Healthy
Volume 2                Recovery  NTFS   Partition     450 MB   Healthy      Hidden
Volume 3                FAT32   Partition      99 MB   Healthy      System
Volume 4                NTFS   Partition     816 MB   Healthy      Hidden
* Volume 5       Z     smb        NTFS   Partition      9 GB   Healthy

DISKPART>

```

Figure 3.12 Screenshot of PLABDM01: Displaying closing the Command Prompt window.

Exercise 4 - Configuring Shared Storage in Windows using SMB

When creating a shared storage location on a network with Windows Operating systems, the Server Message Block protocol (SMB) is used.

In this exercise, a shared network storage location will be created, and a Windows device will connect to the shared storage location.

Learning Outcomes

After completing this exercise, you should be able to:

- Create a Network Share for Storage
- Connect a Windows 10 Client to the Shared Storage

- Map a Network Drive to the Shared Storage

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)

Task 1 - Create a Network Share for Storage

In this task, you will create a network share for storage using the SMB protocol.

Step 1

Connect to **PLABDM01**.

Launch **File Explorer** by clicking the folder icon on the Windows **Taskbar**.

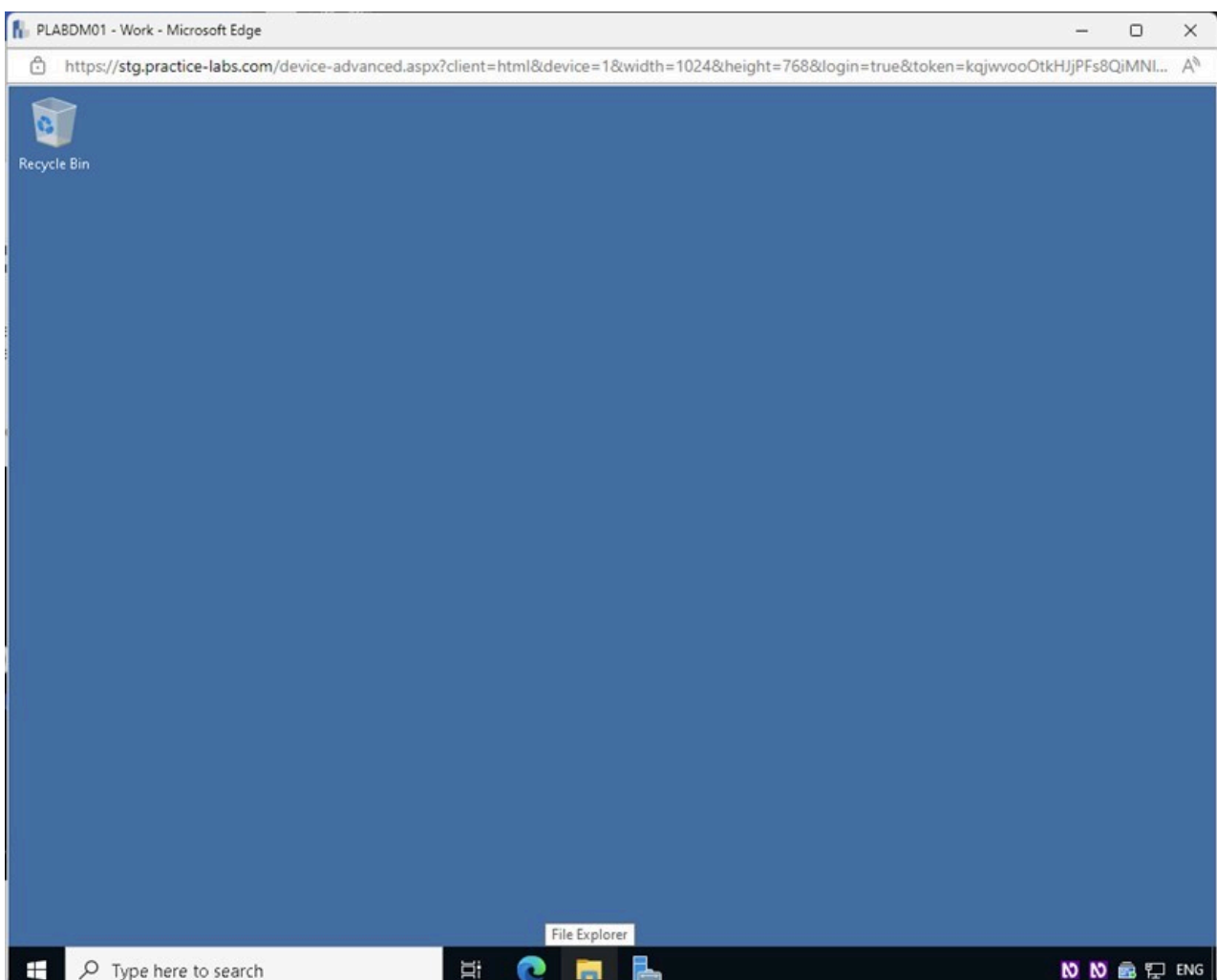


Figure 4.1 Screenshot of PLABDM01: Displaying opening File Explorer from the Taskbar.

Step 2

In the **File Explorer** window, in the left navigation pane, select and right-click on **smb (Z:)**, and choose **Properties**.

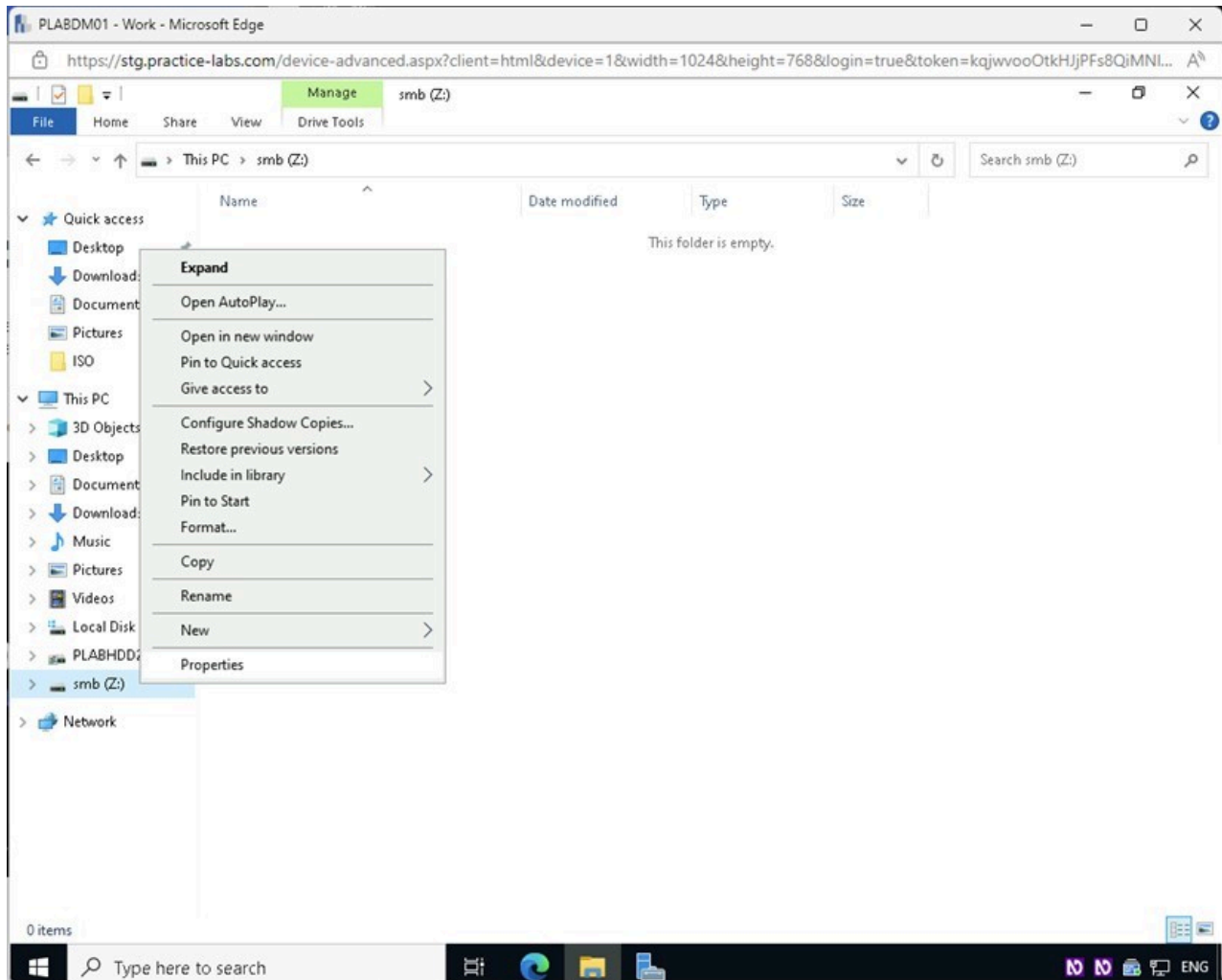


Figure 4.2 Screenshot of PLABDM01: Displaying the File Explorer window. smb is right-clicked and Properties selected.

Step 3

In the **smb (Z:) Properties** window, select the **Sharing** tab.

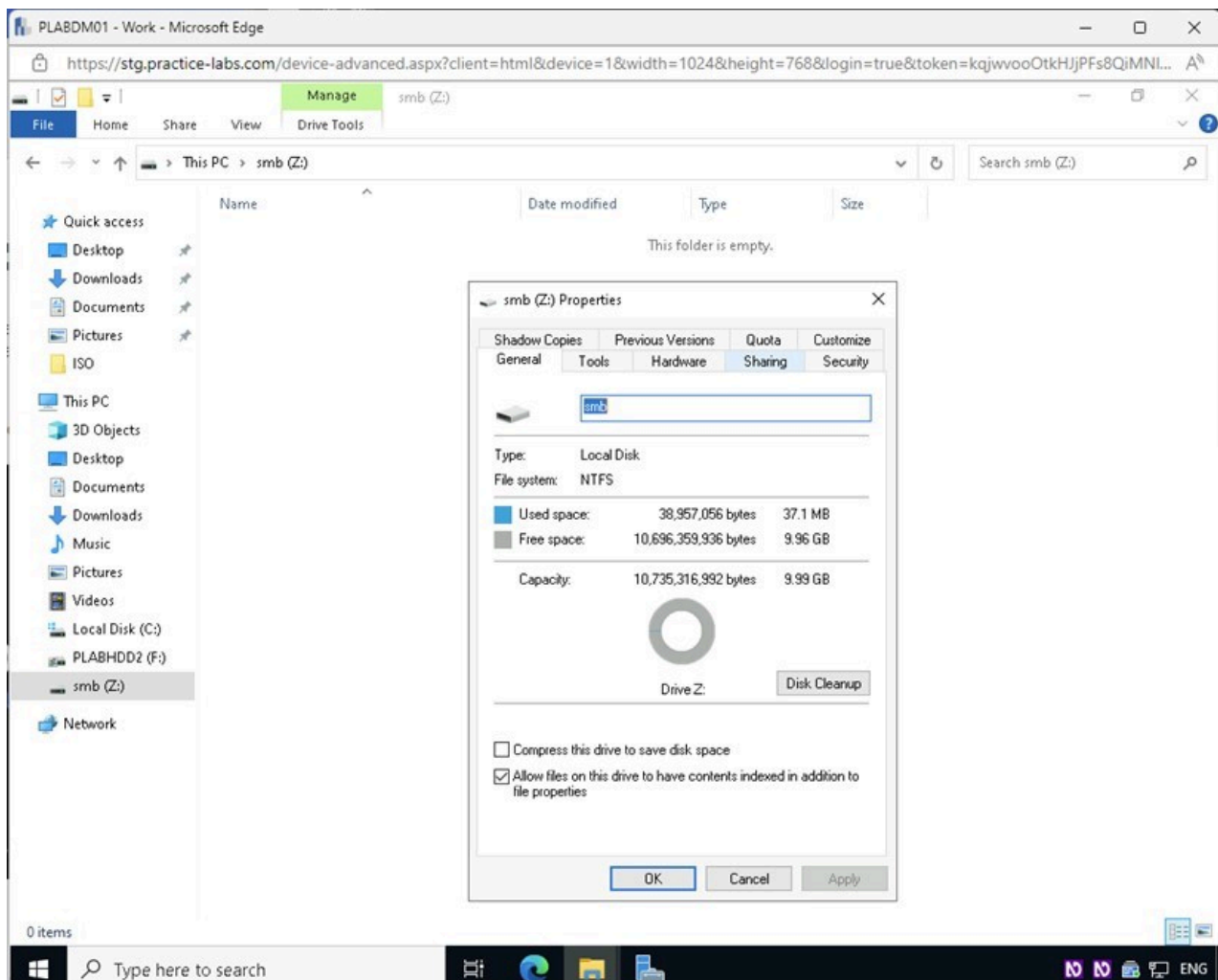


Figure 4.3 Screenshot of PLABDM01: Displaying selecting the Sharing tab on the smb (Z:) Properties window.

Step 4

In the **Sharing** tab, click the **Advanced Sharing** button.

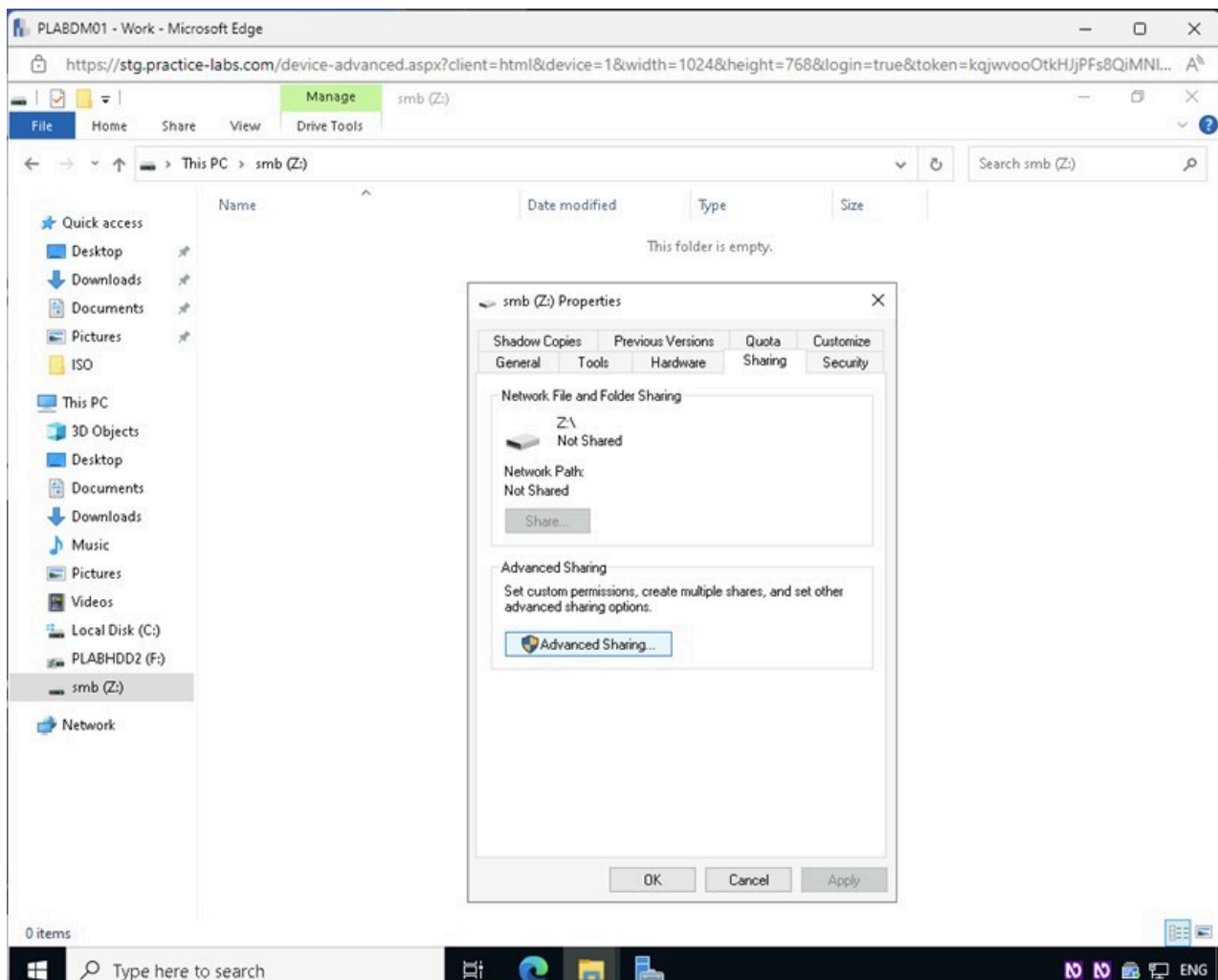


Figure 4.4 Screenshot of PLABDM01: Displaying selecting Advanced Sharing in the Sharing tab.

Step 5

In the **Advanced Sharing** dialog box, tick the **Share this folder** checkbox.

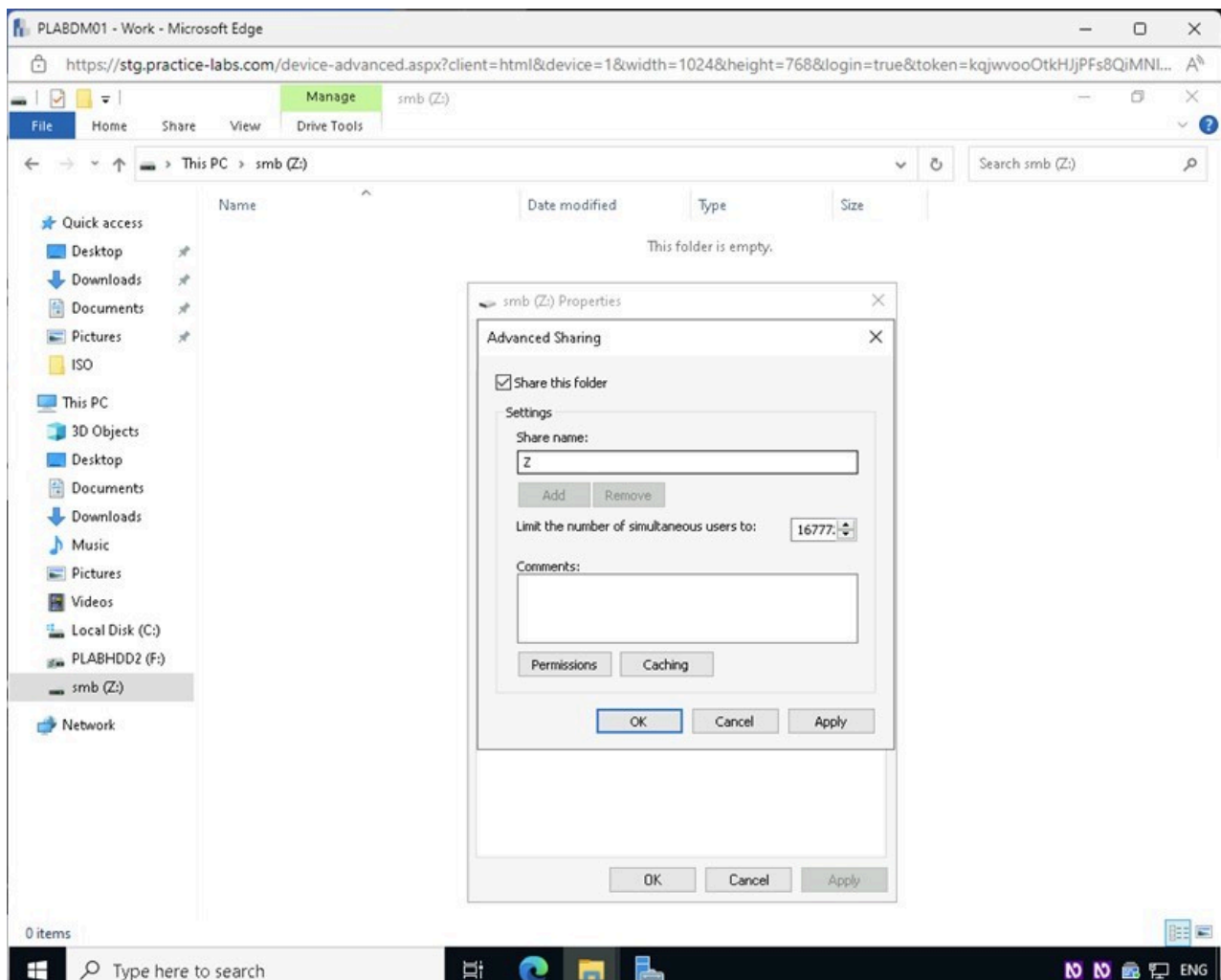


Figure 4.5 Screenshot of PLABDM01: Displaying ticking the Share this folder checkbox in the Advanced Sharing dialog box.

Step 6

Below the **Share name** field, type the following:

smb

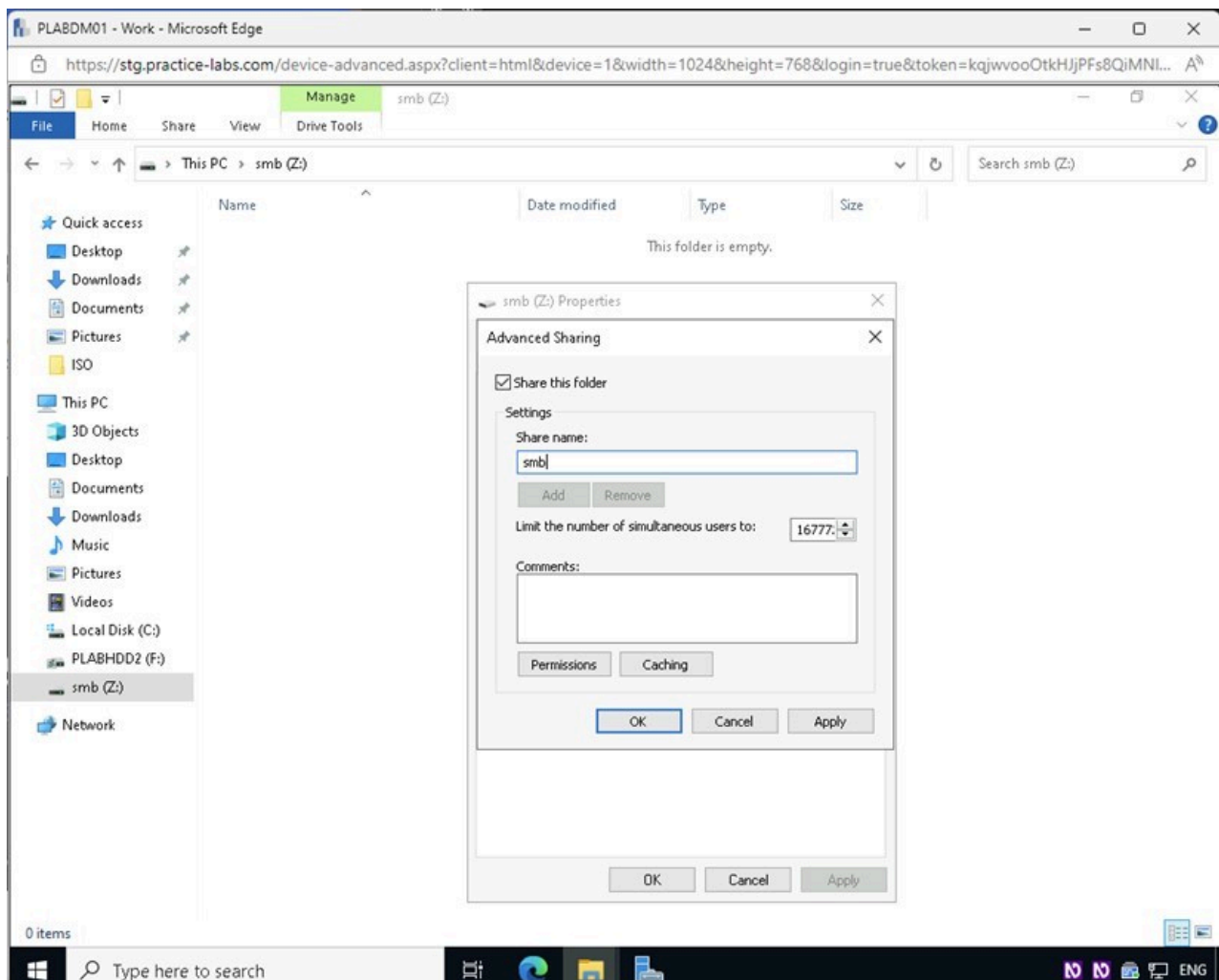


Figure 4.6 Screenshot of PLABDM01: Displaying entering the File share name on the Advanced sharing dialog box.

Step 7

Click the **Permissions** button.

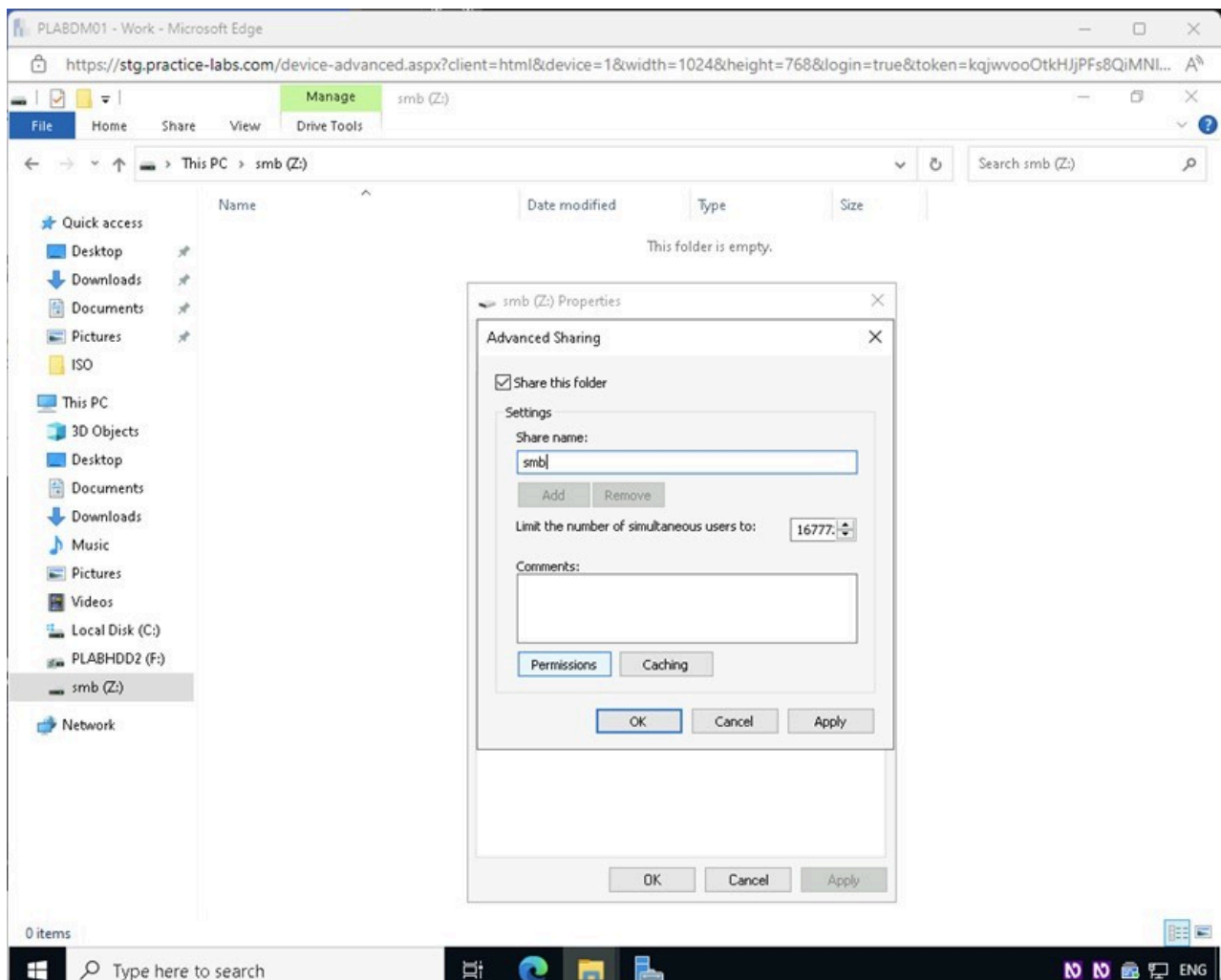


Figure 4.7 Screenshot of PLABDM01: Displaying clicking Permissions on the Advanced Sharing dialog box.

Step 8

In the **Permissions for smb** window, below the **Permissions for Everyone** section, enable the **Allow: Full Control** checkbox.

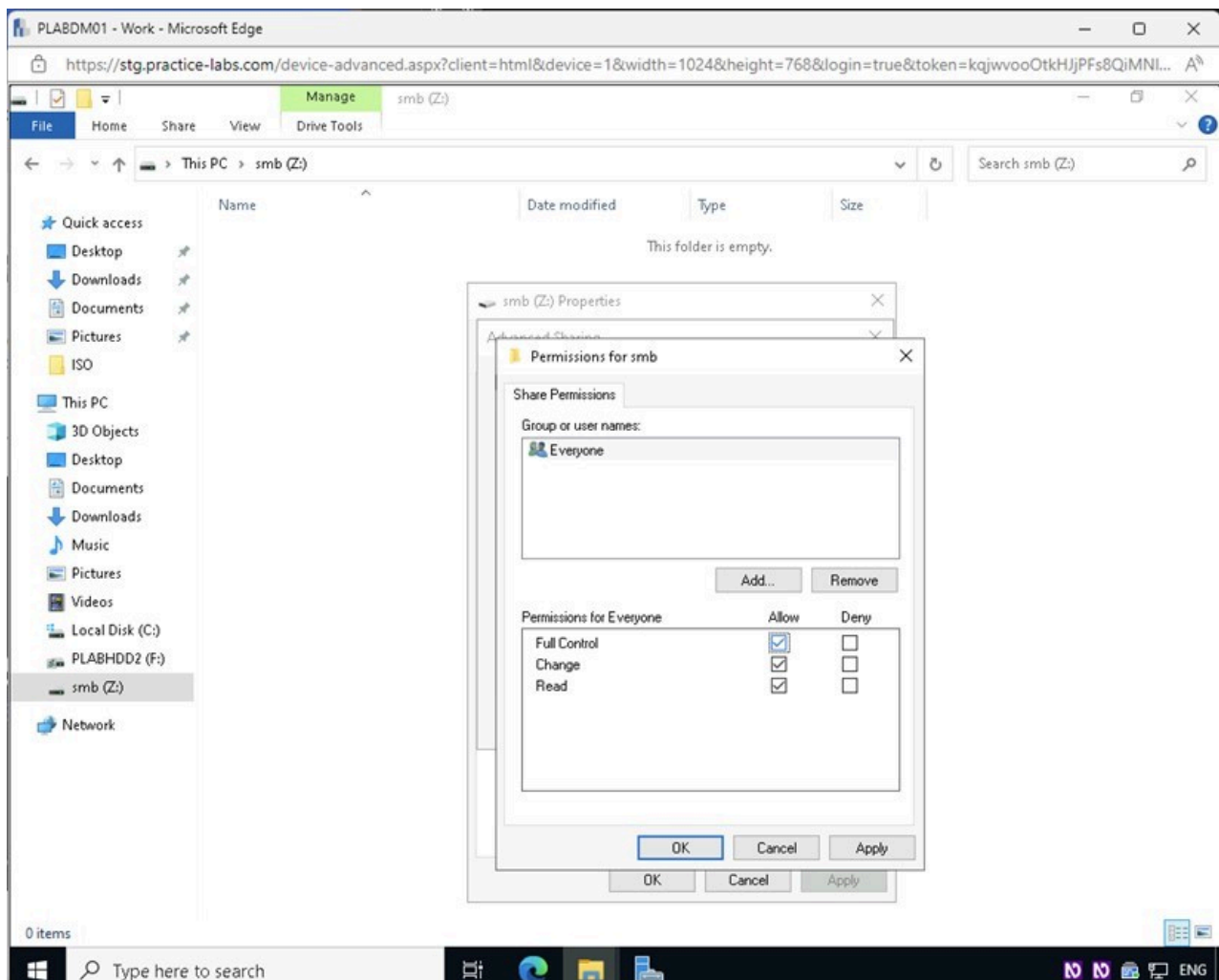


Figure 4.8 Screenshot of PLABDM01: Displaying ticking the Full control checkbox on the Permissions for smb window.

Step 9

In the **Permissions for smb** window, click **OK**.

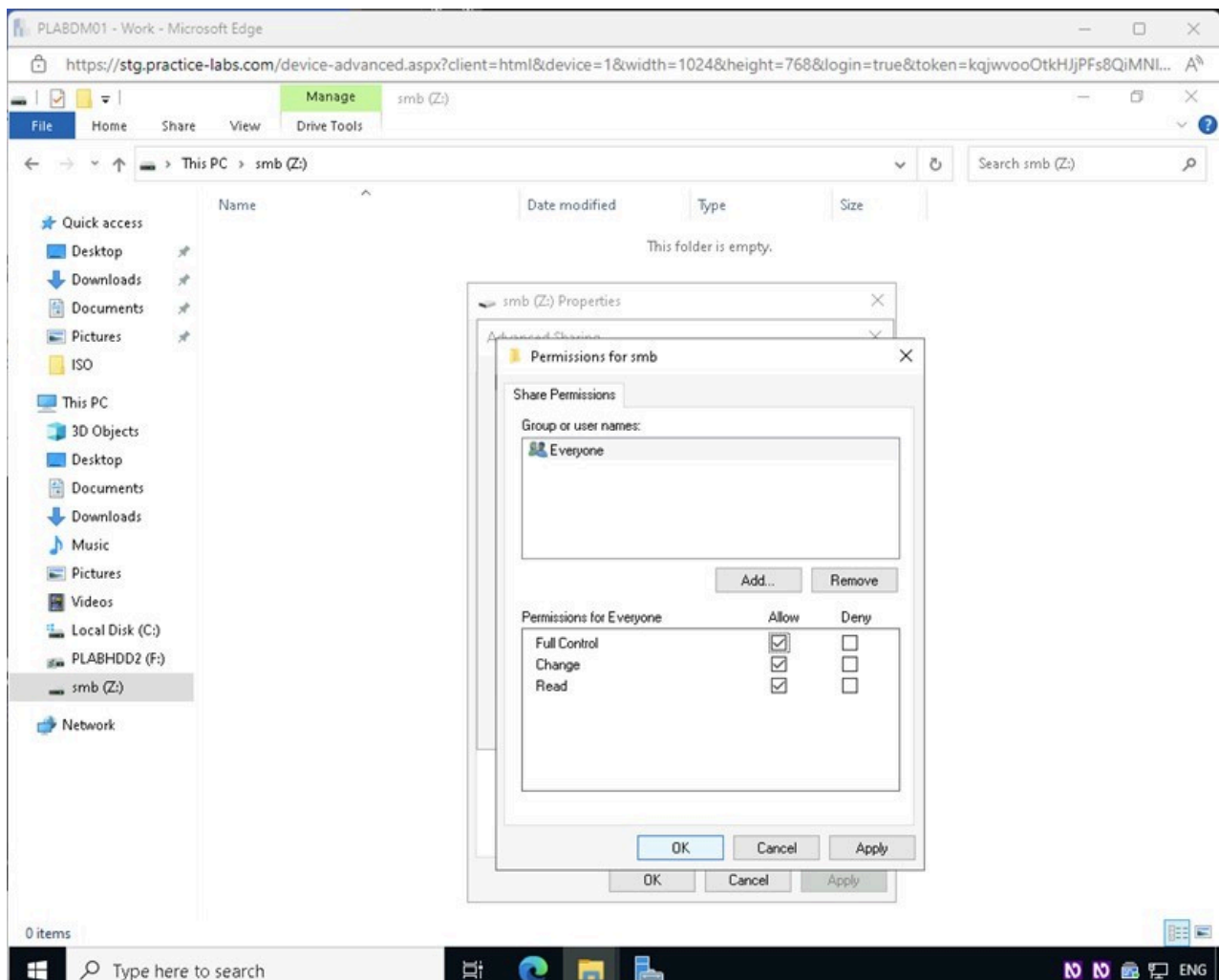


Figure 4.9 Screenshot of PLABDM01: Displaying clicking OK on the Permissions for smb window.

Step 10

In the **Advanced Sharing** dialog box, click **OK**.

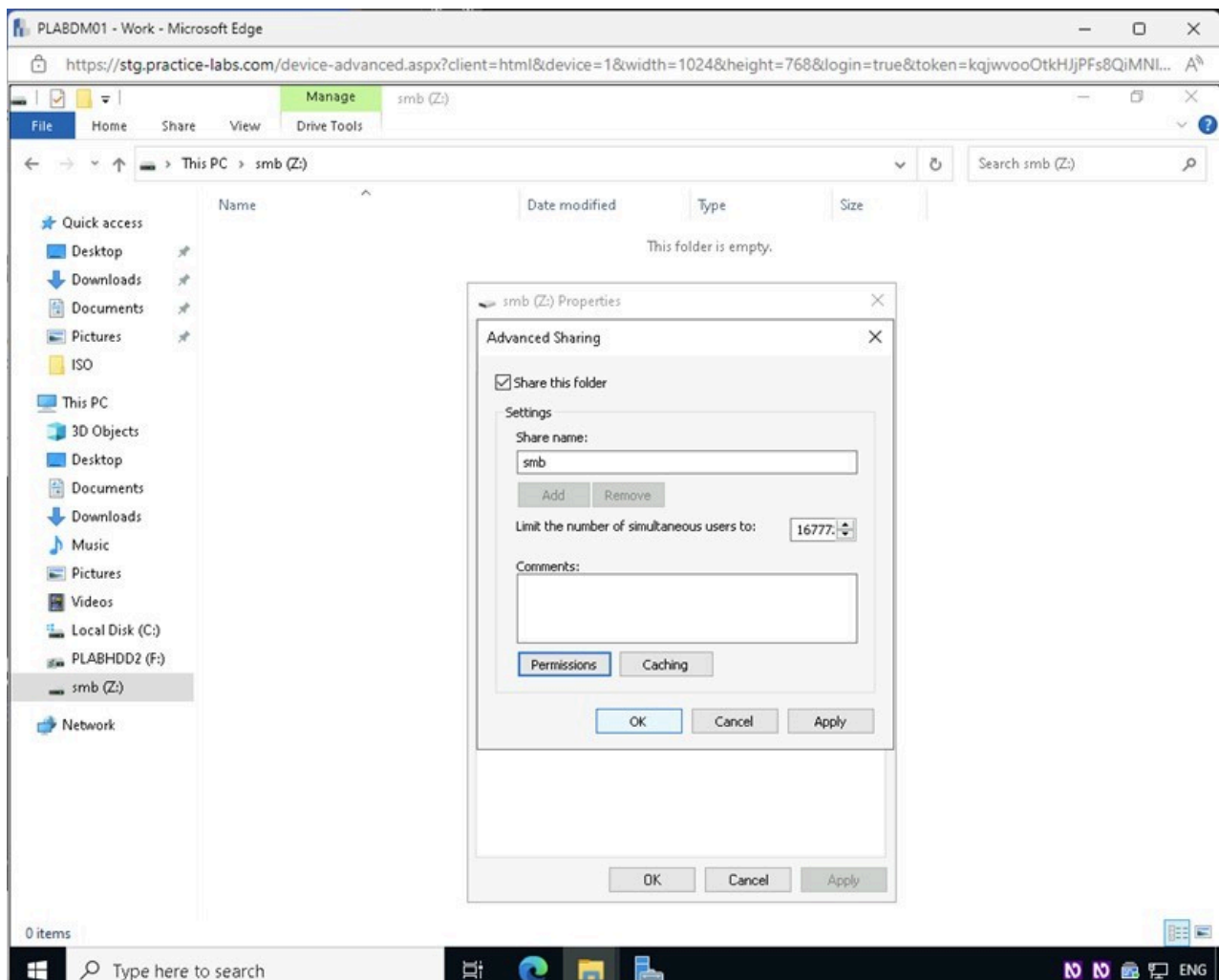


Figure 4.10 Screenshot of PLABDM01: Displaying clicking OK on the Advanced Sharing dialog box.

Step 11

In the **smb (Z:) Properties** window, click **Close**.

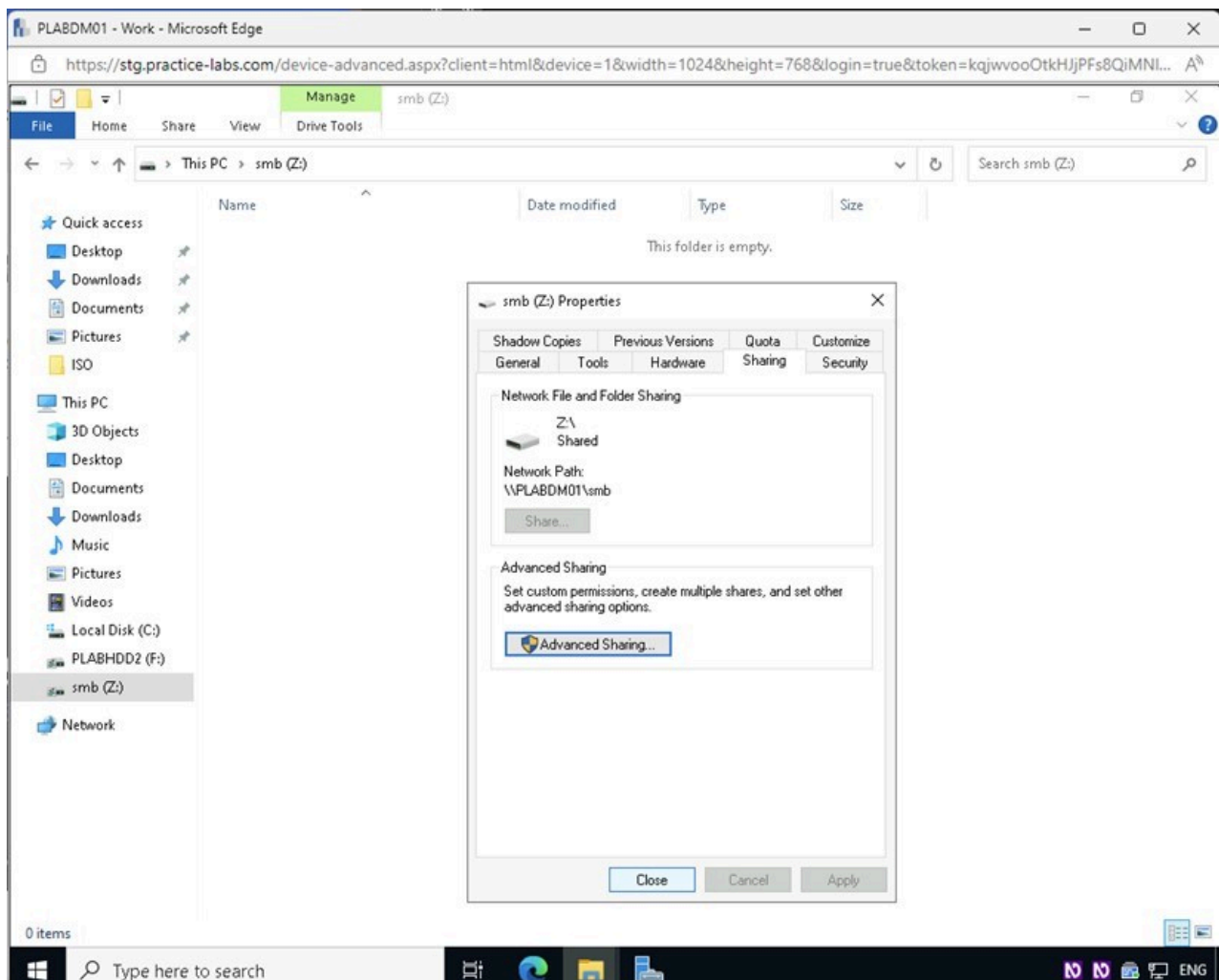


Figure 4.11 Screenshot of PLABDM01: Displaying clicking closing in the smb (Z:) Properties window.

Close the **File Explorer** window.

Task 2 - Connect a Windows 10 Client to the Shared Storage

In this task, you will connect a Windows 10 client to the shared network storage.

Step 1

Connect to **PLABWIN10**.

Launch **File Explorer** by clicking the folder icon on the Windows **Taskbar**.

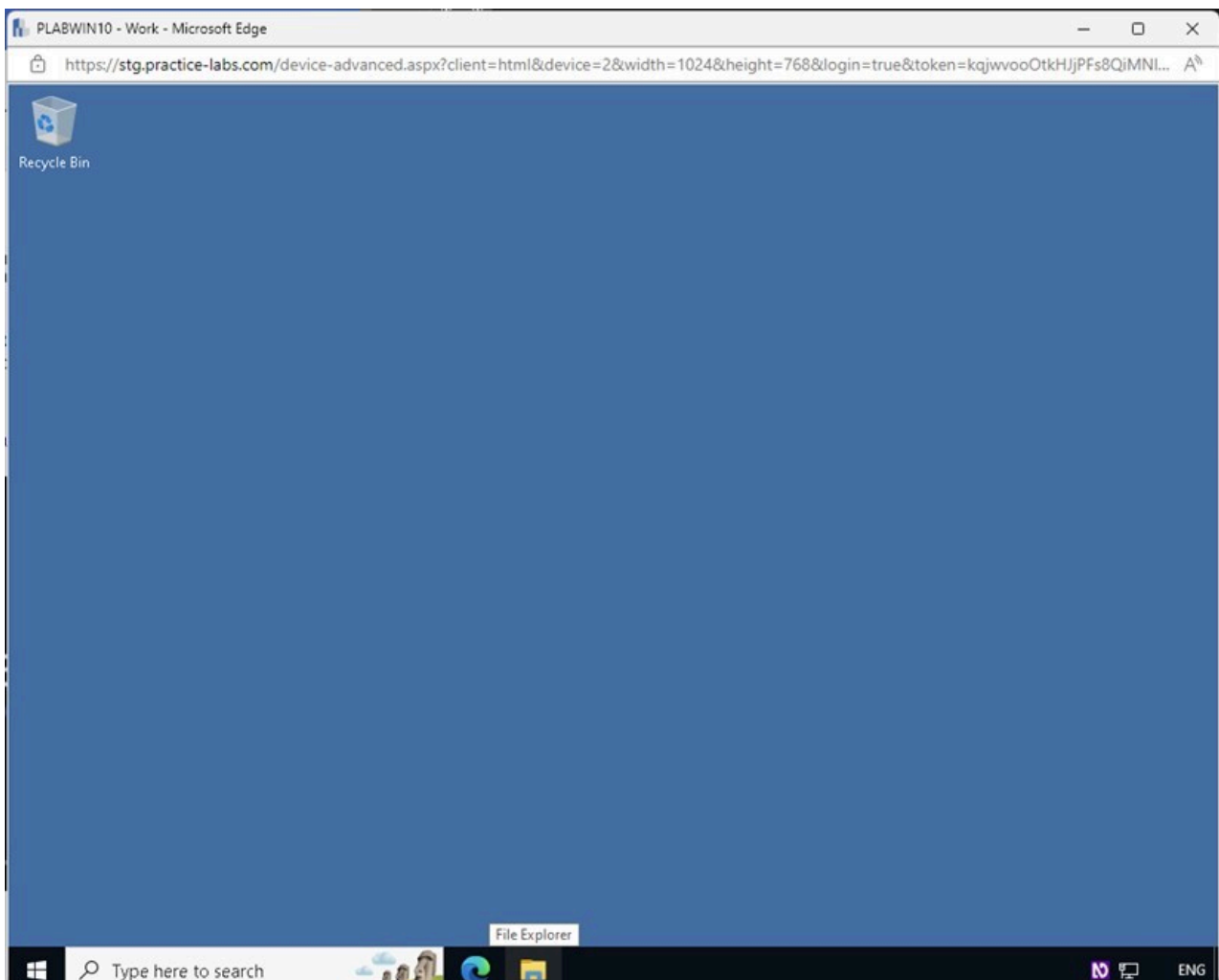


Figure 4.12 Screenshot of PLABWIN10: Displaying opening File Explorer from the Taskbar.

Step 2

In the **File Explorer** window, type the following in the address bar and press **Enter**:

```
\\plabdm01\smb
```

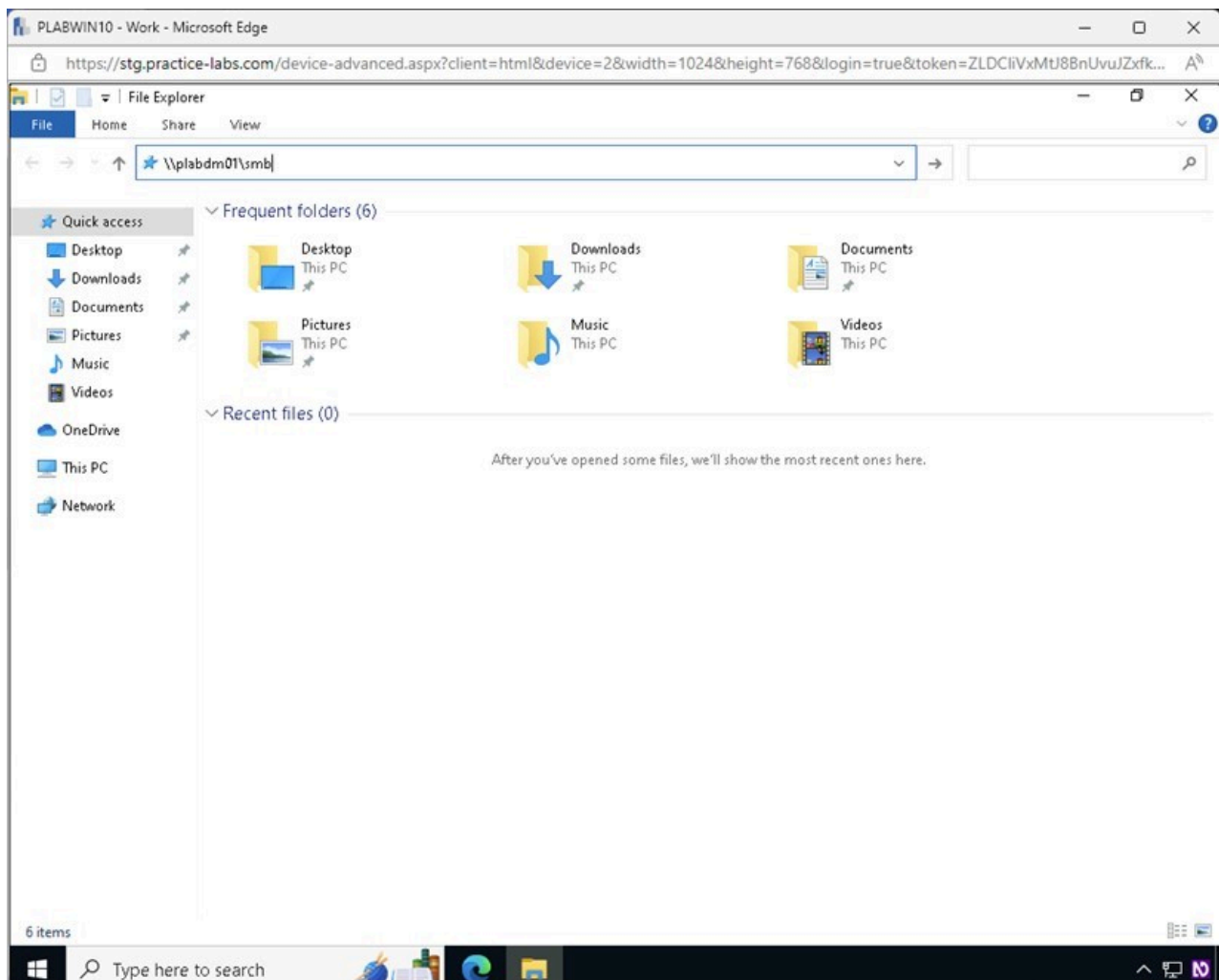


Figure 4.13 Screenshot of PLABWIN10: Displaying the File Explorer window, showing the required path entered in the address bar.

Step 3

Notice that File Explorer connects to the shared storage. In the address bar, you have the path **Network > plabdm01 > smb**. Windows is using Server Message Block(SMB) to connect to a shared resource.

Close the **File Explorer** window.

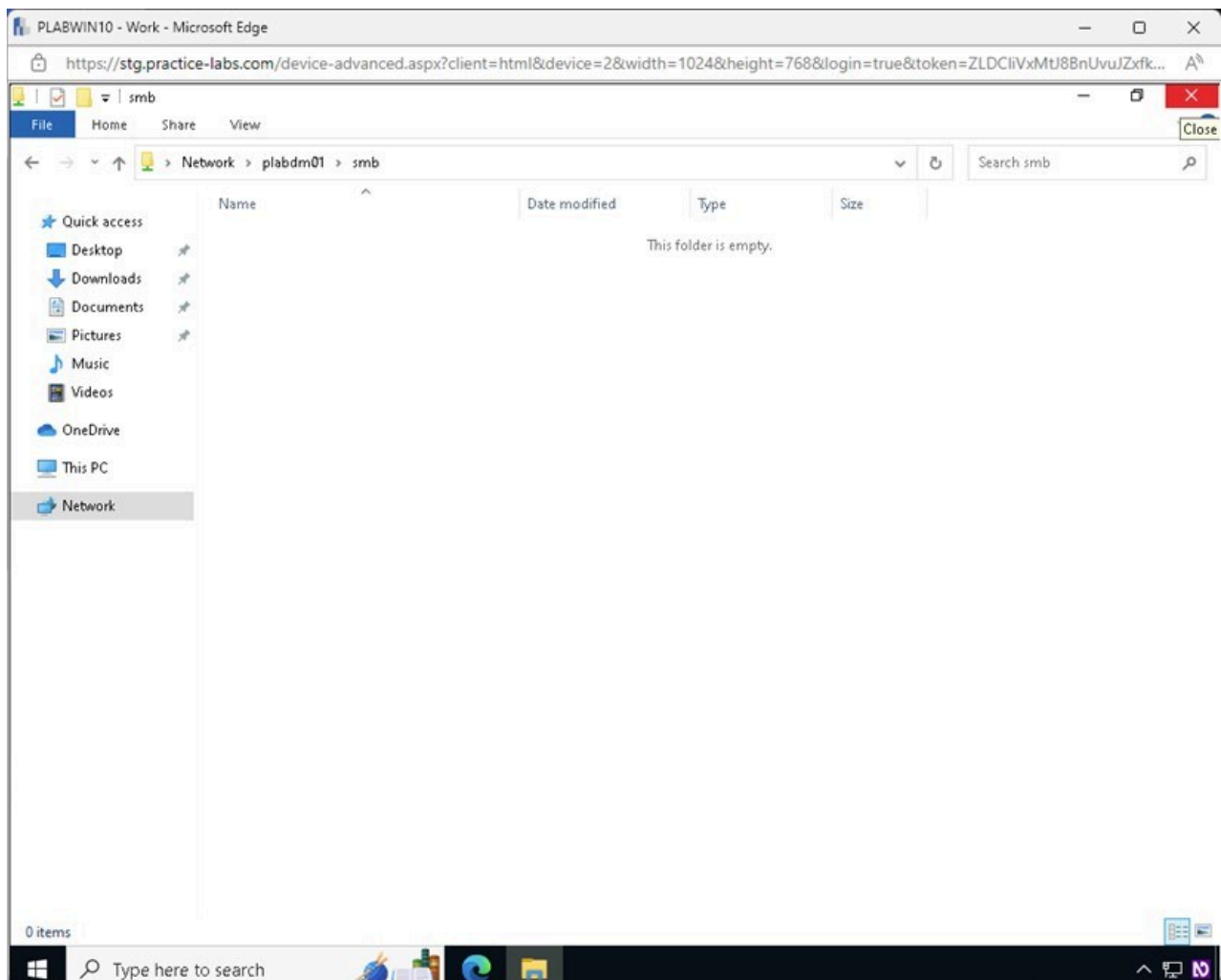


Figure 4.14 Screenshot of PLABWIN10: Displaying typing the required url in the address bar of File Explorer.

Task 3 - Map a Network Drive to the Shared Storage

In this task, you will map a network drive to the shared storage.

Step 1

Ensure you are connected to **PLABWIN10**.

Click the **Start** charm and type the following:

```
cmd
```

In the **Best match**, hover on **Command Prompt** and select **Run as administrator**.

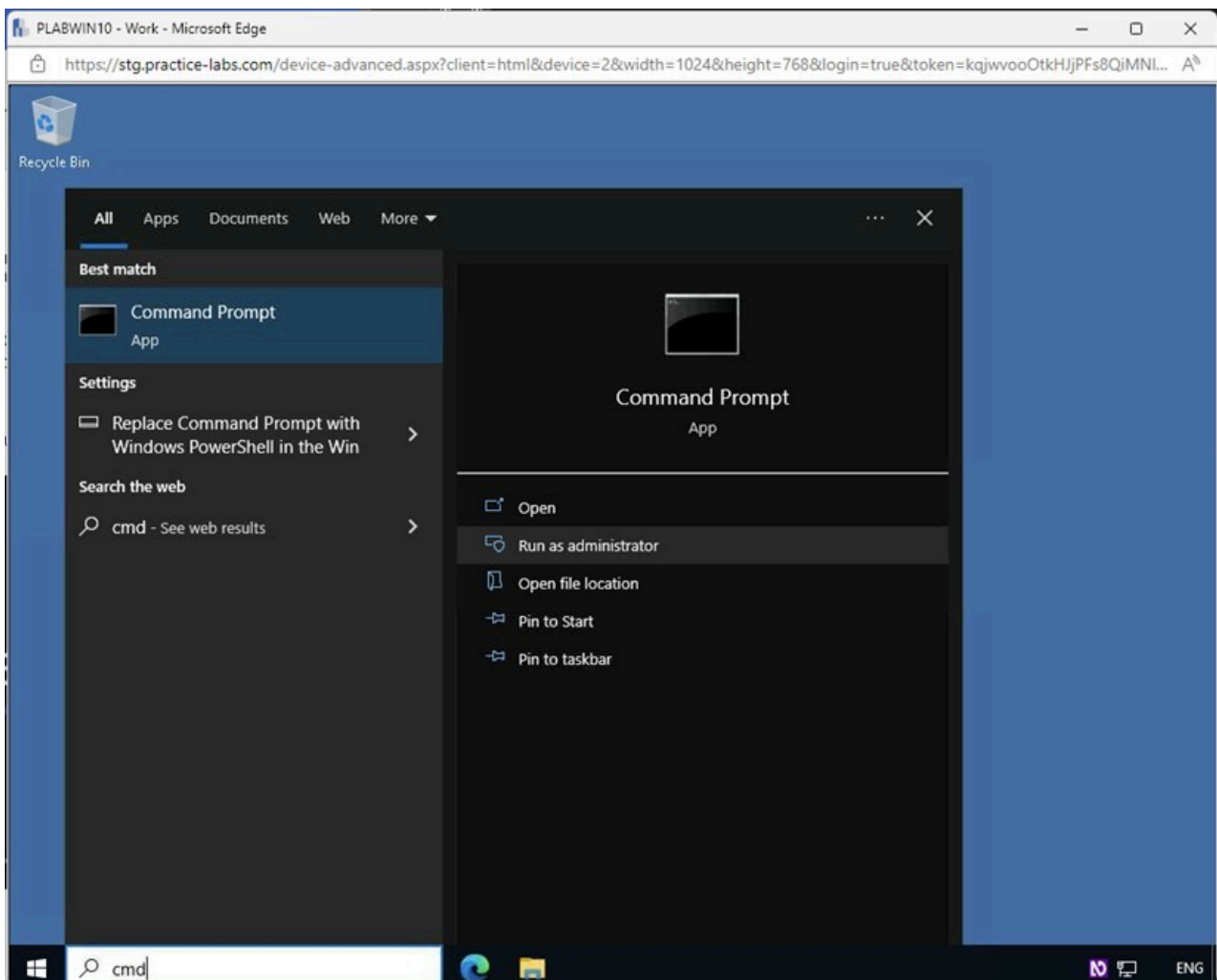


Figure 4.15 Screenshot of PLABWIN10: Displaying clicking Start, typing cmd and selecting Run as administrator for Command Prompt.

Step 2

In the **Command Prompt** window, type the following command:

```
net use Z: \\plabdm01\smb /user:administrator  
/persistent:yes
```

Press **Enter**.

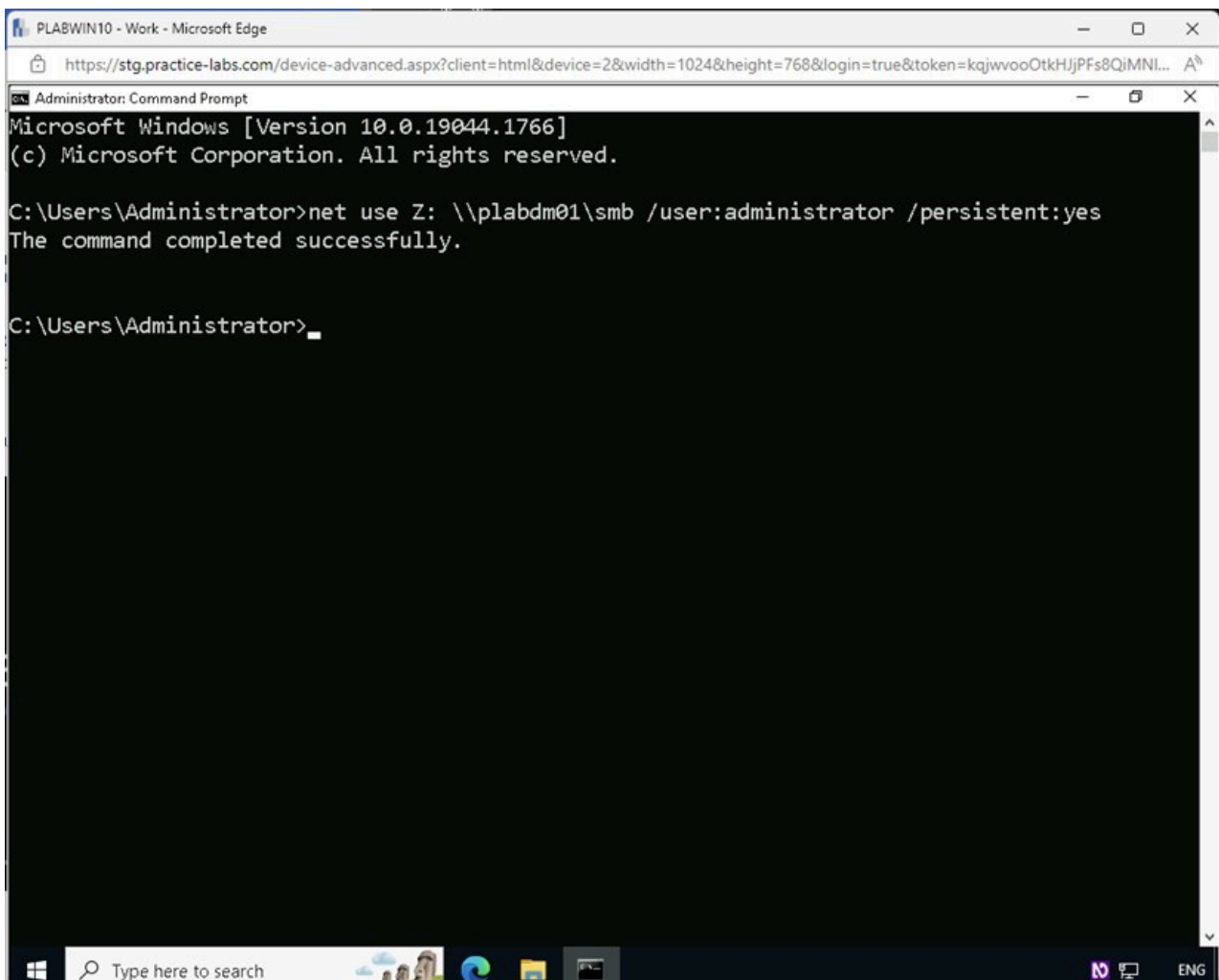


Figure 4.16 Screenshot of PLABWIN10: Displaying typing the required command in the Command Prompt window and pressing Enter.

Note: The *net use* command maps a shared network drive to the Windows device using Command Prompt.

Step 3

Close the **Command Prompt** window.

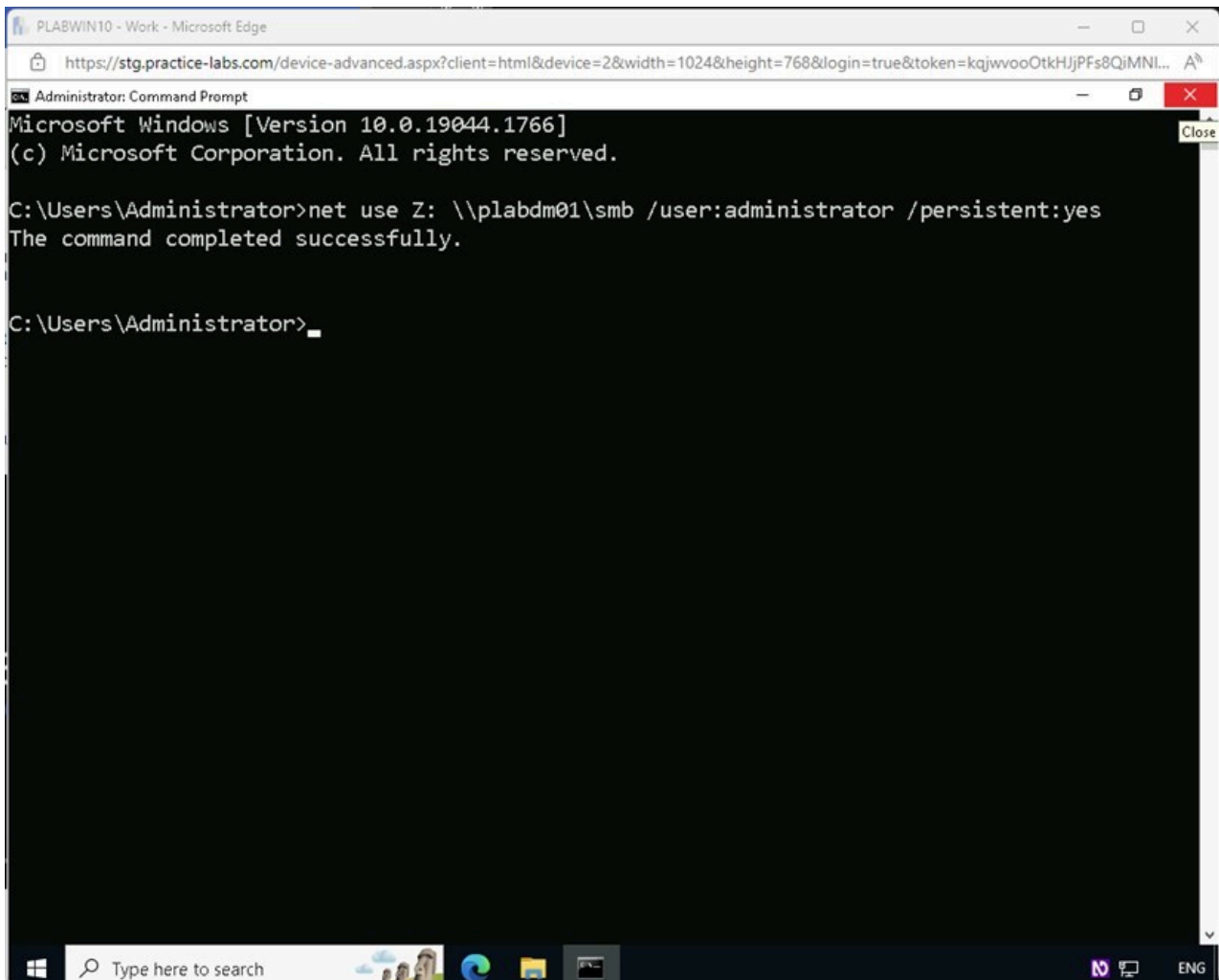


Figure 4.17 Screenshot of PLABWIN10: Displaying closing the Command Prompt window.

Step 4

Click the **File Explorer** icon on the **Taskbar**.

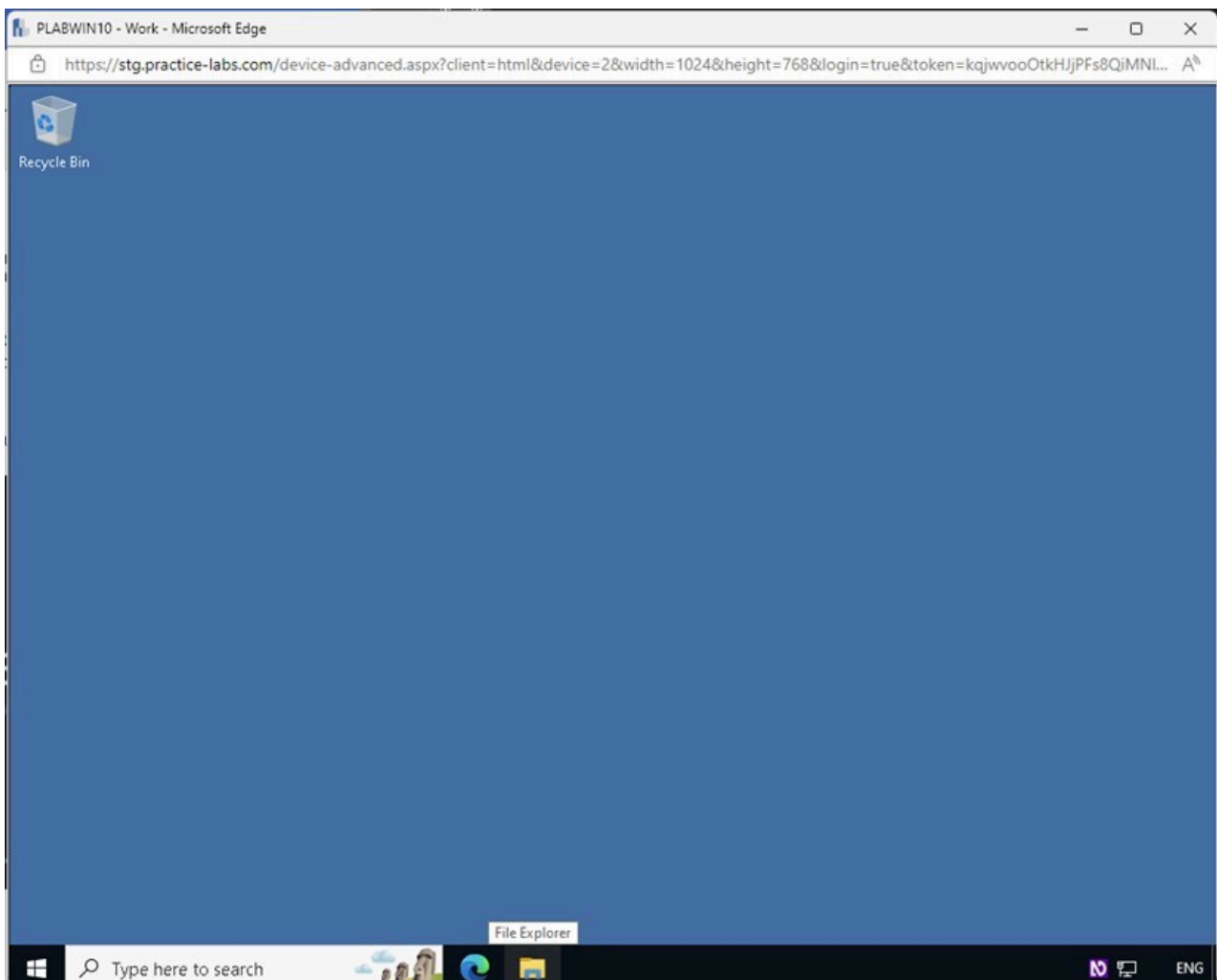


Figure 4.18 Screenshot of PLABWIN10: Displaying opening File Explorer from the Taskbar.

Step 5

In the **File Explorer** window, click **This PC** in the left navigation pane.

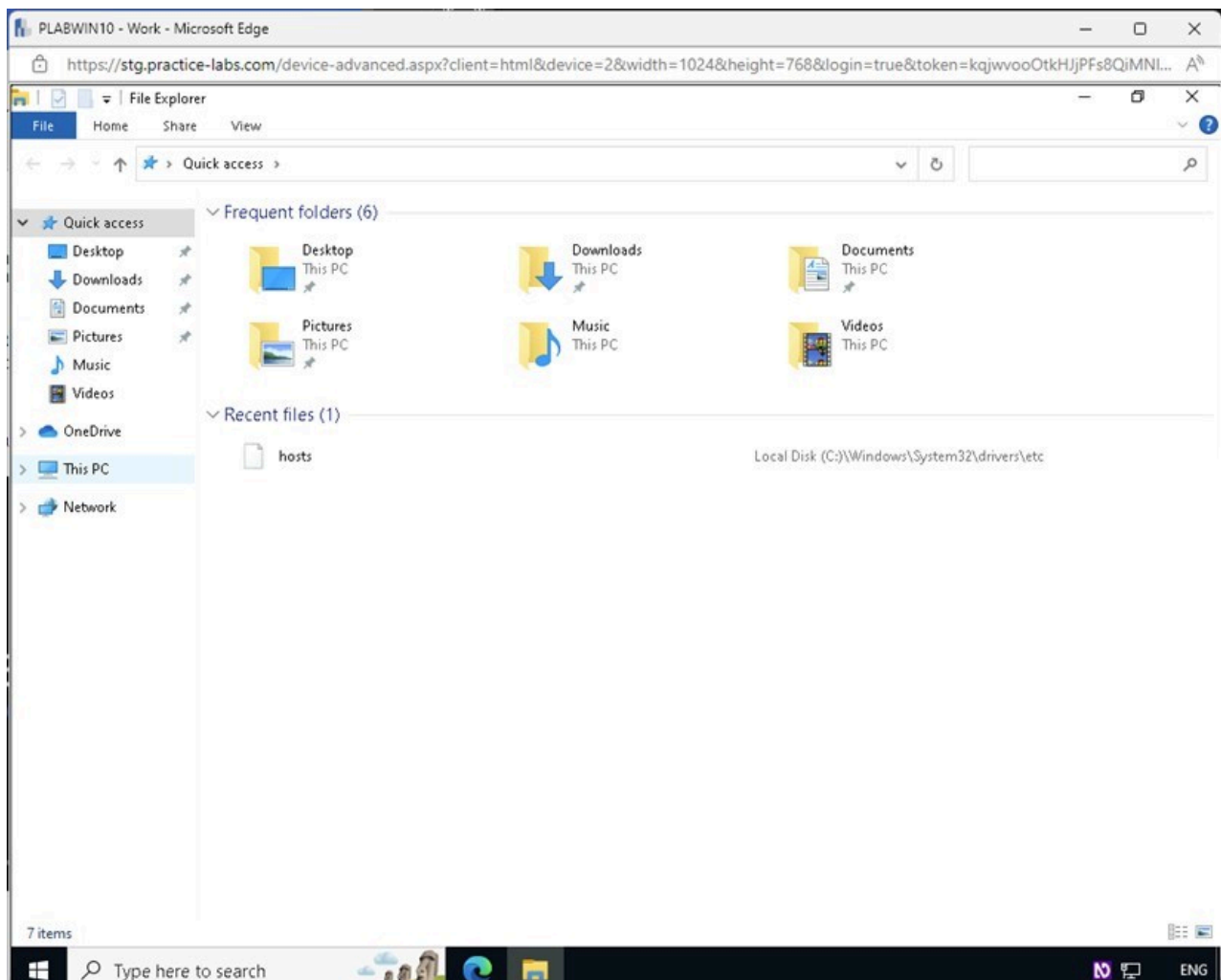


Figure 4.19 Screenshot of PLABWIN10: Displaying selecting This PC in File Explorer.

Step 6

In the right details pane, double-click on **smb (\\plabdm01) (Z:)** below **Network locations**.

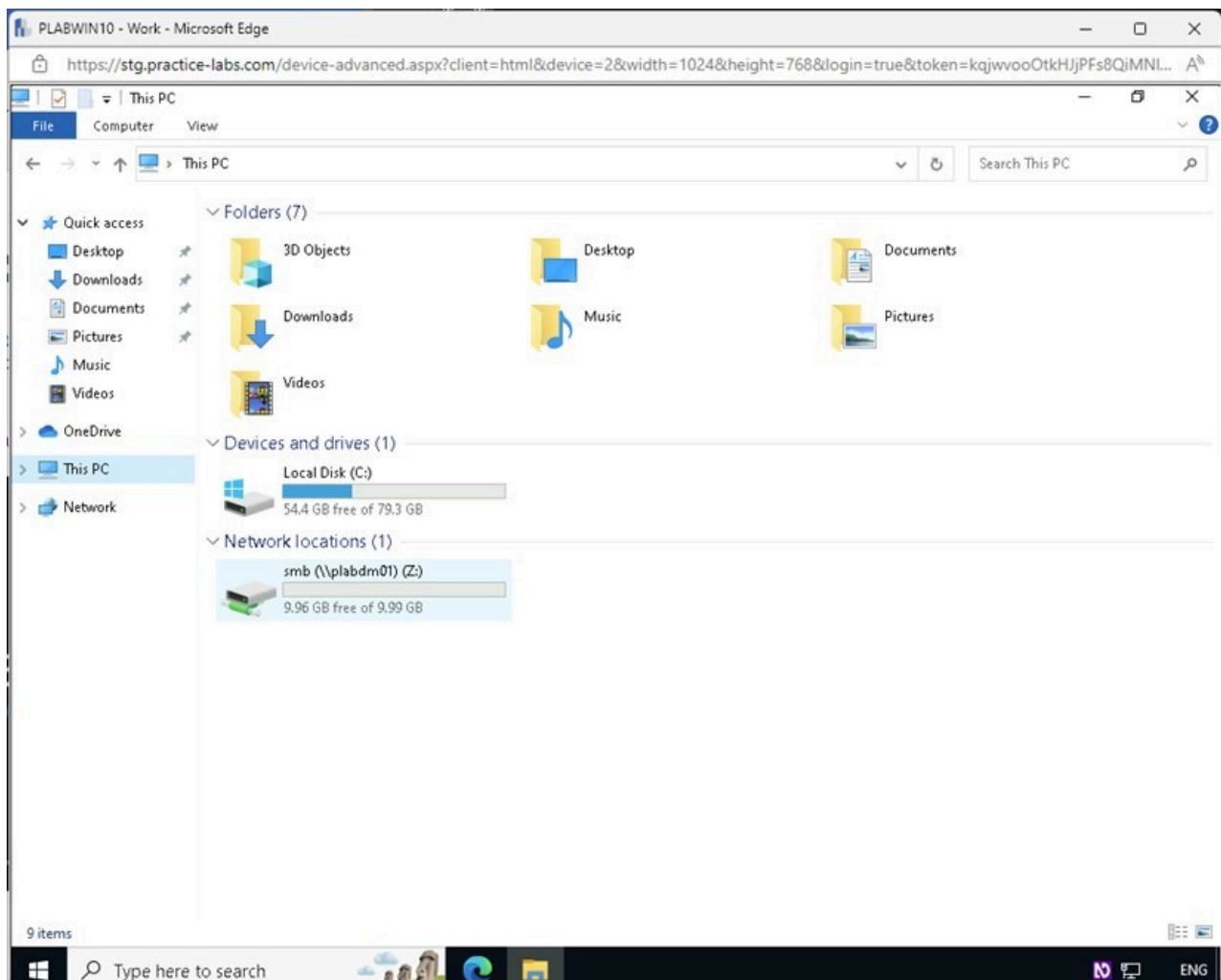


Figure 4.20 Screenshot of PLABWIN10: Displaying double-clicking the smb file share in File Explorer.

Step 7

In the right details pane, right-click anywhere on an empty space and select **New > Text Document**.

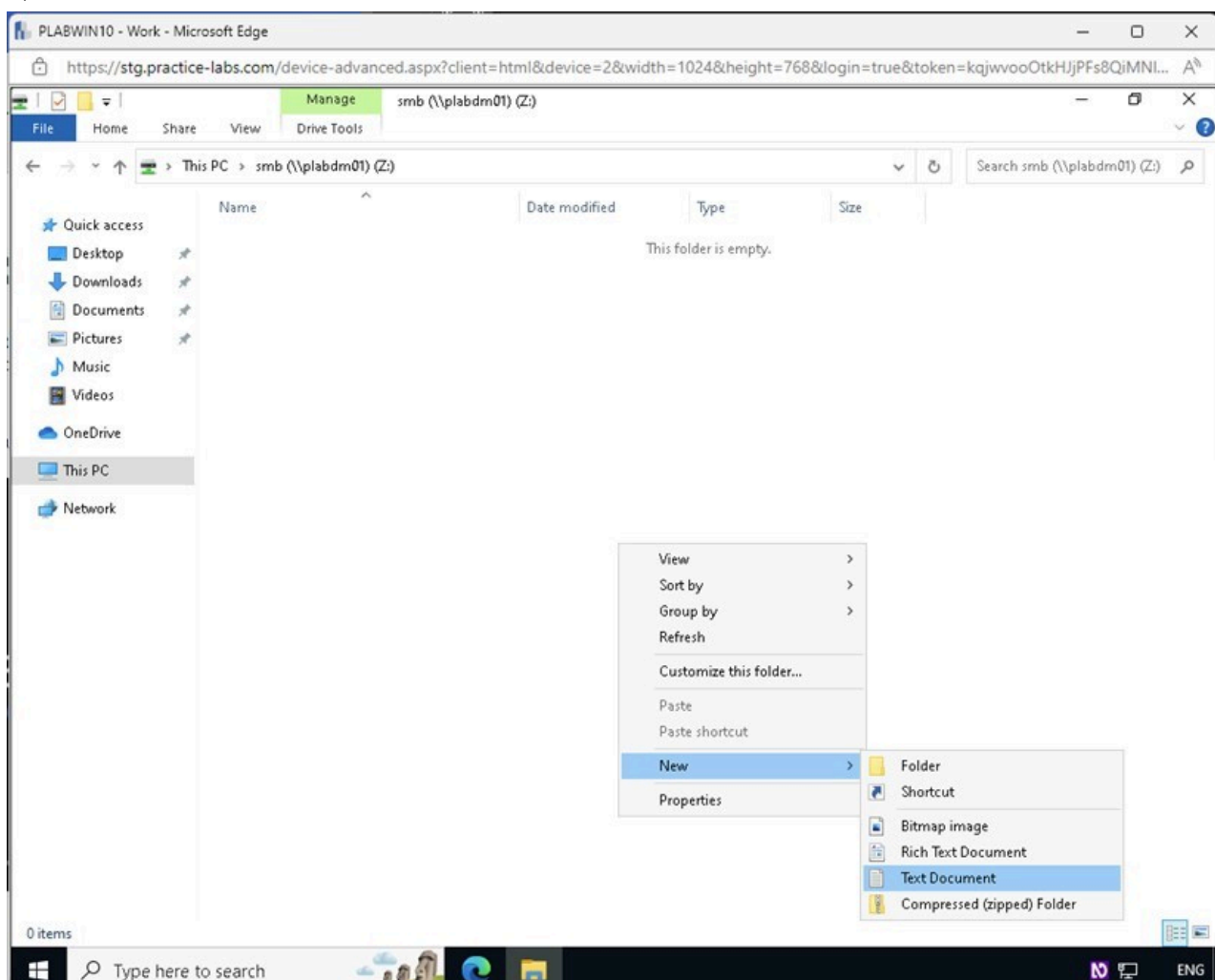


Figure 4.21 Screenshot of PLABWIN10: Displaying right-clicking in the right pane and selecting New > Text Document.

Step 8

Rename the **New Text Document** to:

Shared Storage

Press **Enter**.

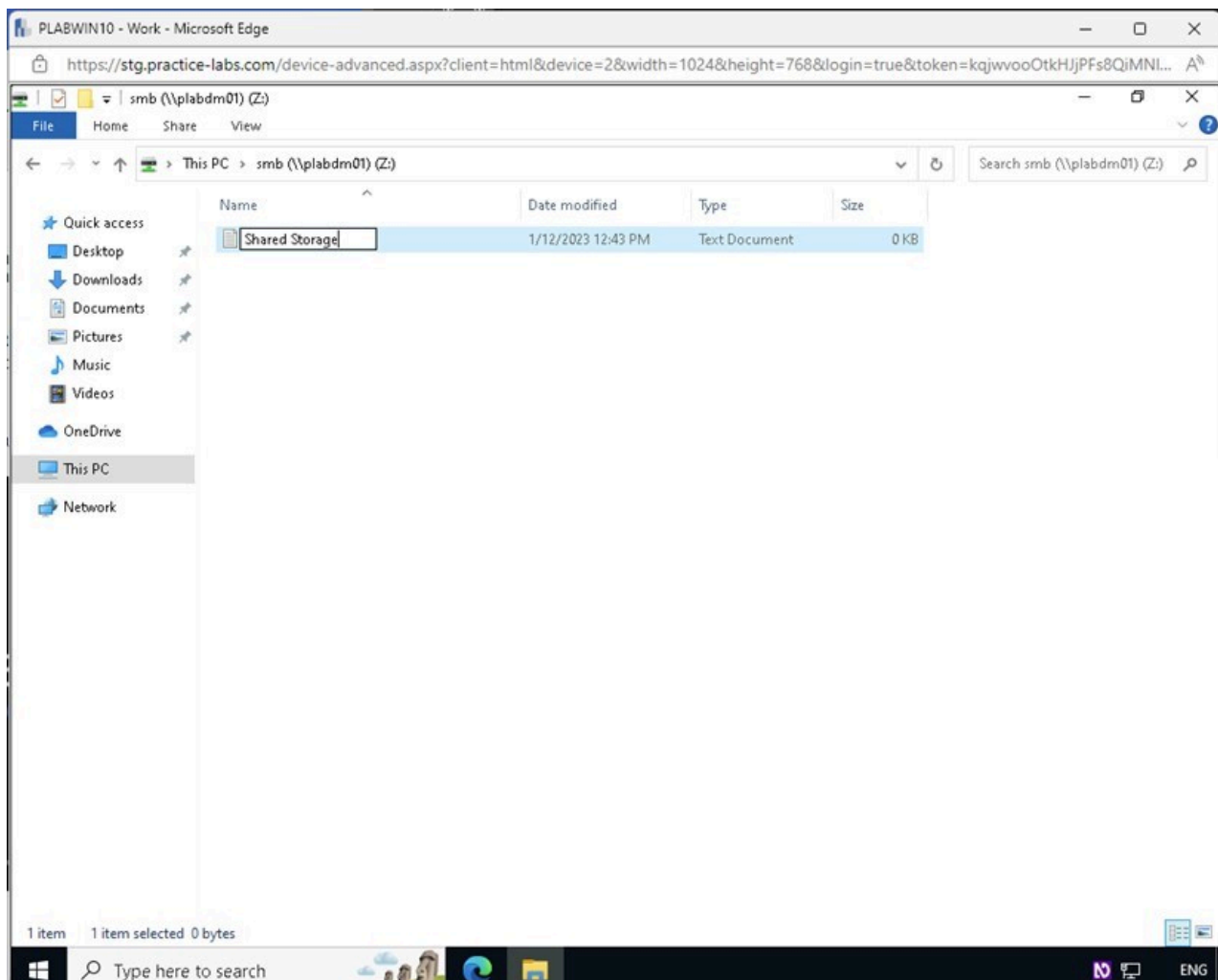


Figure 4.22 Screenshot of PLABWIN10: Displaying renaming the text document and pressing enter.

Step 9

Close the **File Explorer** window.

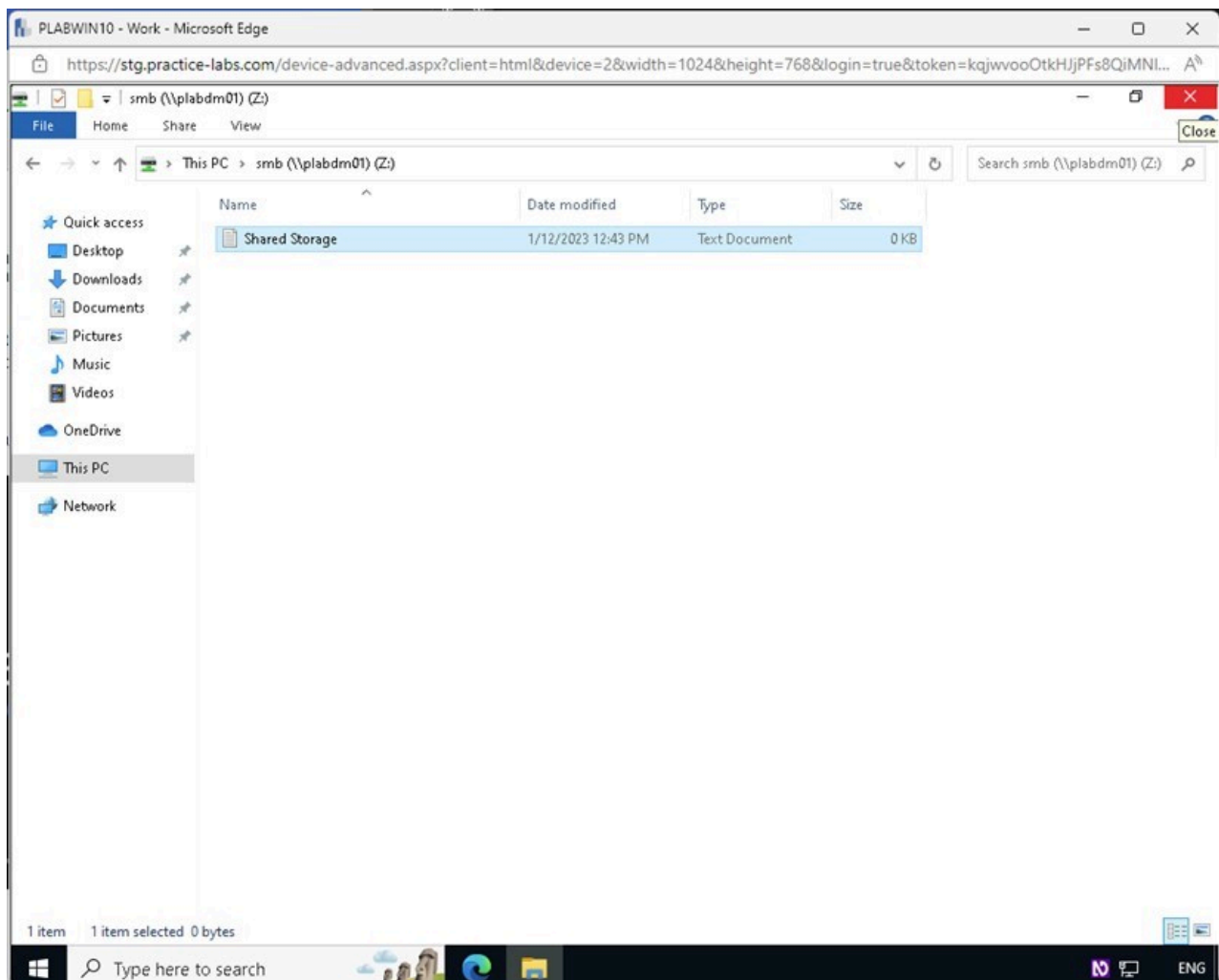


Figure 4.23 Screenshot of PLABWIN10: Displaying closing the File Explorer window.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Deploying and Managing Server Storage** Practice Lab.

In this module, the configuration of storage for different server operating systems was investigated and implemented.

Summary

You completed the following exercises:

- Exercise 1 - Configuring Storage in Linux
- Exercise 2 - Installing and Configuring Shared Storage in Linux using NFS
- Exercise 3 - Configuring Storage in Windows
- Exercise 4 - Configuring Shared Storage in Windows using SMB

You should now be able to:

- Partition a Storage Device in Linux
- Format a Partition with a Filesystem in Linux
- Mount Storage in Linux
- Configure the Permissions on the NFS Directory
- Install and Configure an NFS Server
- Install and Configure an NFS Client
- Partition a Storage Device
- Format the Storage Device with the NTFS Filesystem
- Create a Network Share for Storage
- Connect a Windows 10 Client to the Shared Storage
- Map a Network Drive to the Shared Storage

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.